BABU BANARSI DAS UNIVERSITY

LUCKNOW

SESSION: 2024-25

SCHOOL OF COMPUTER APPLICATION

Project on

DESCRIPTIVE ANALYTICS (BCADSN13201)

Submitted by
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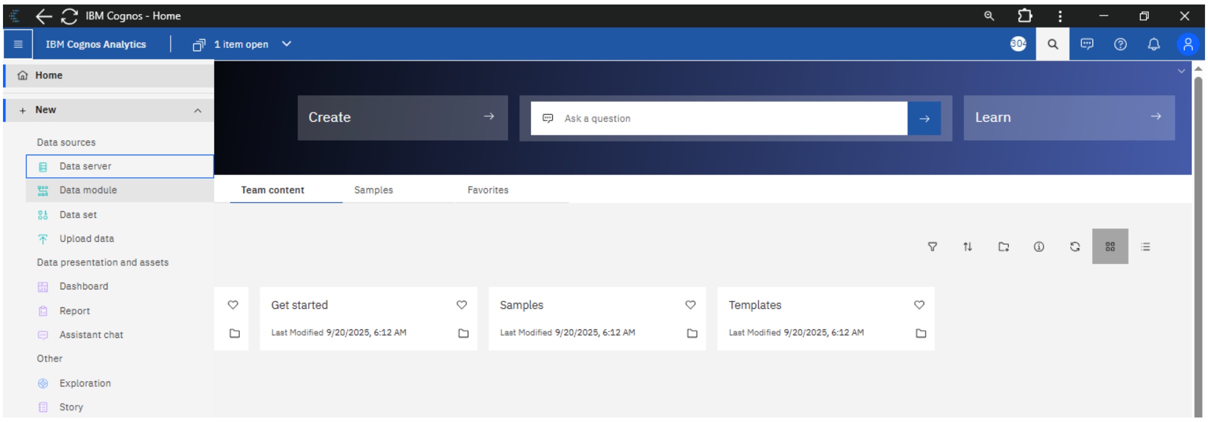
Submitted to
Mr. Ayush Chand

Project

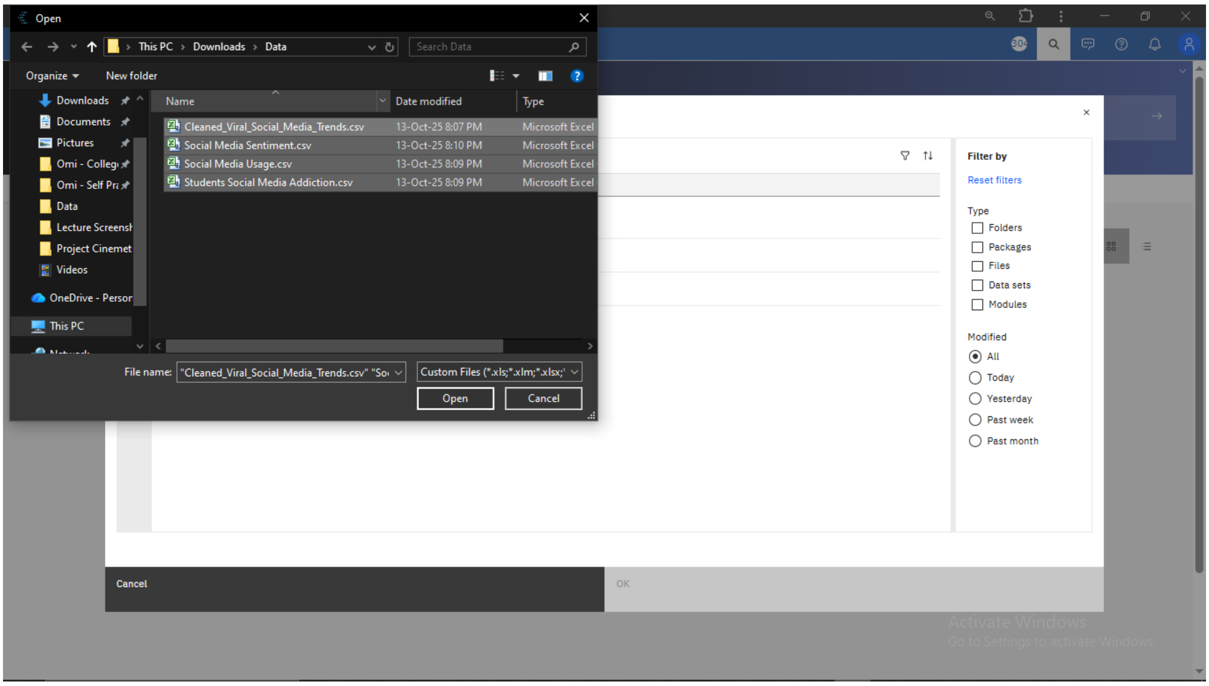
Problem Statement: Analyse the average screen time per platform across user

Solution : Create a Bar Chart using Data Module

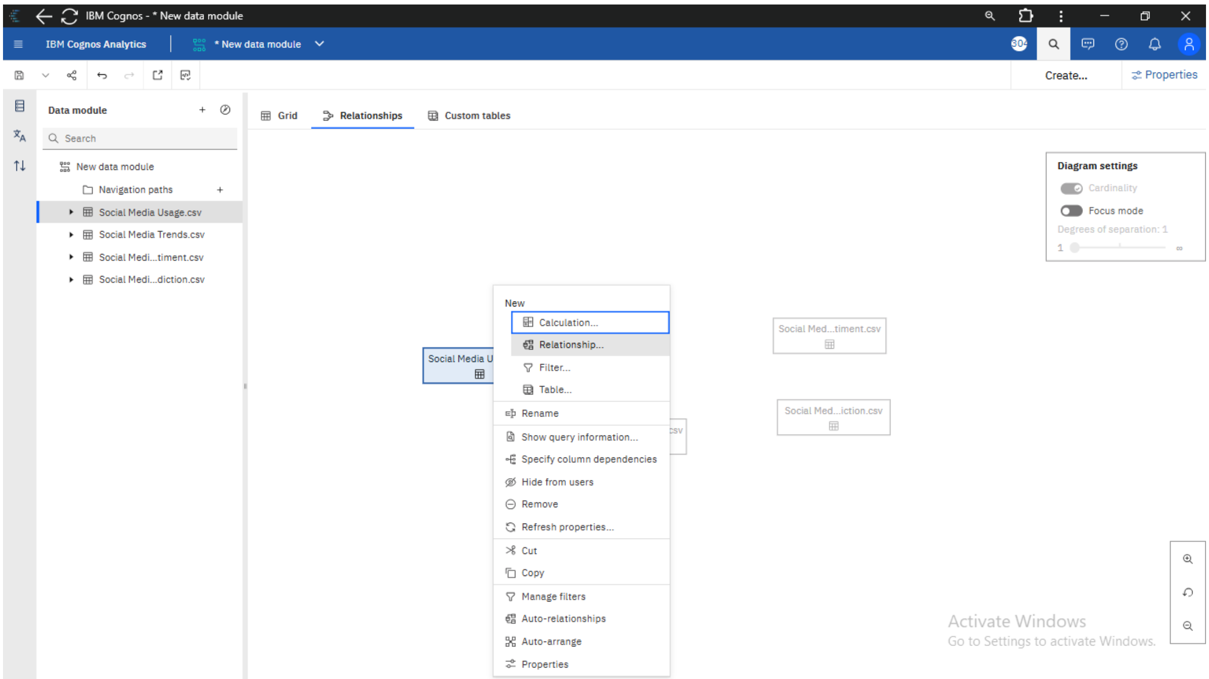
Step 1: Login to IBM Cognos and following interface will be opened. Click Three-Line Icon and Select Data Module option.



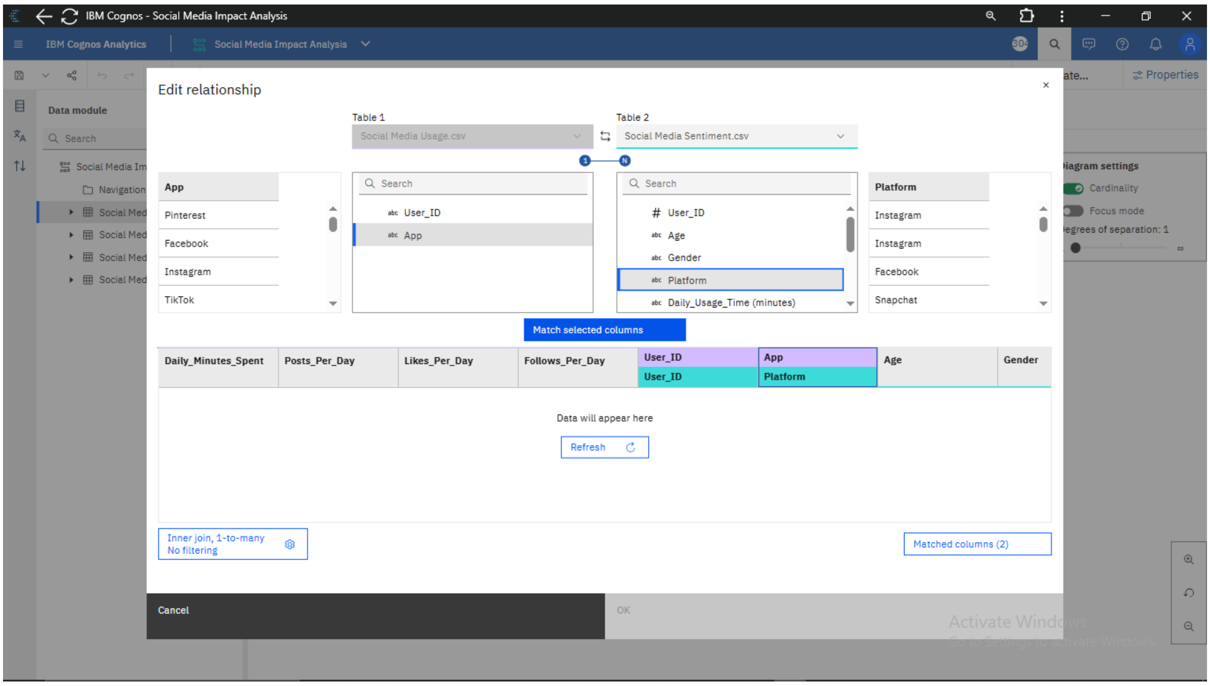
Step 2: Click on upload icon and Select all the files you want to use.



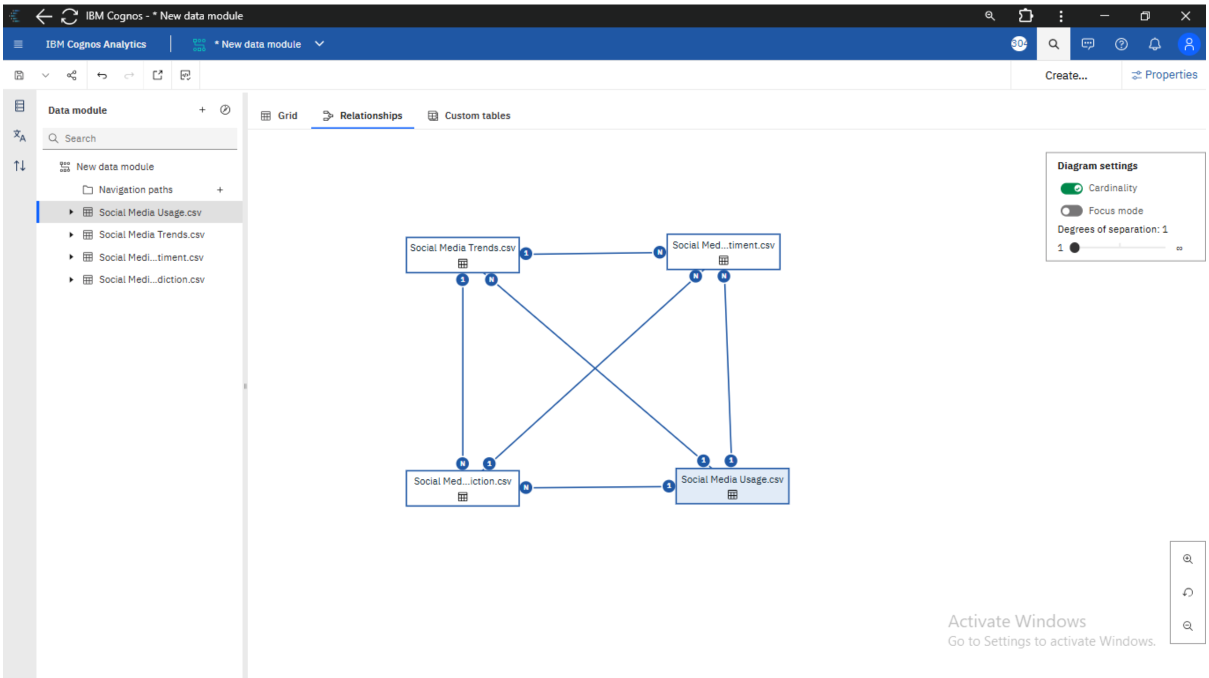
Step 3: Click on the Relationship Option on the canvas, Select the desired dataset and left-click and select Relationships from given options



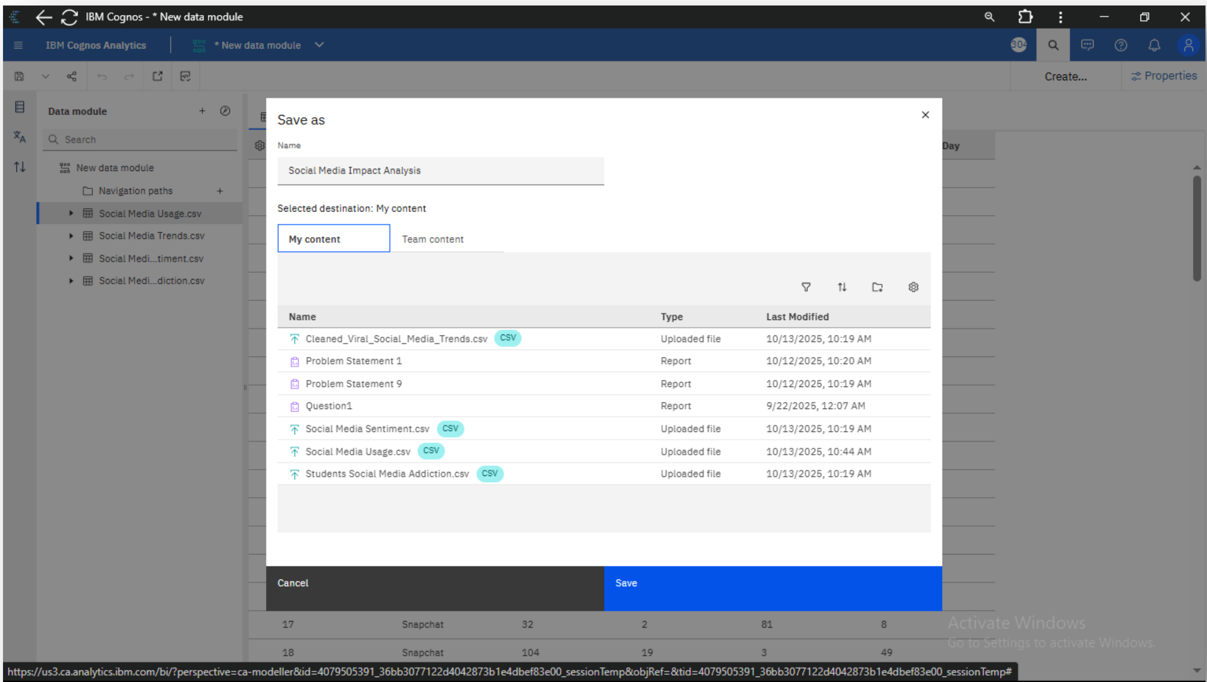
Step 4: Select Table-2 from drop-down option and click on match selected columns and Do this all the datasets



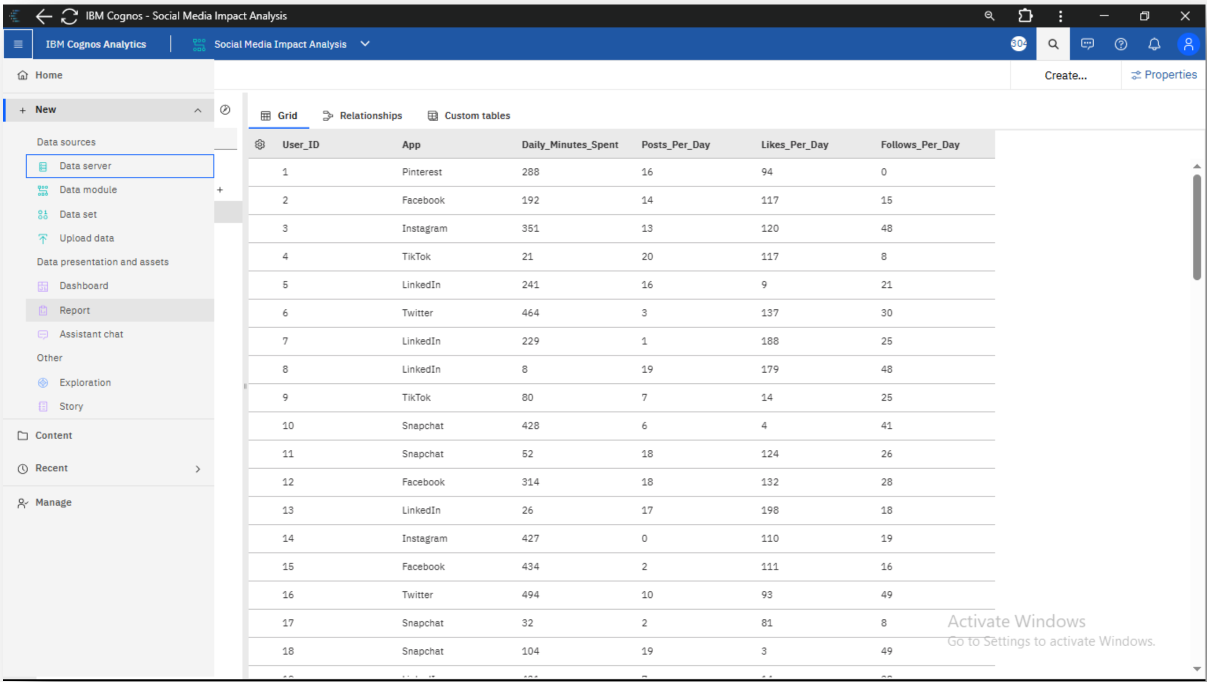
Step 5: Establish Relationship between Datasets and following visuals will appear



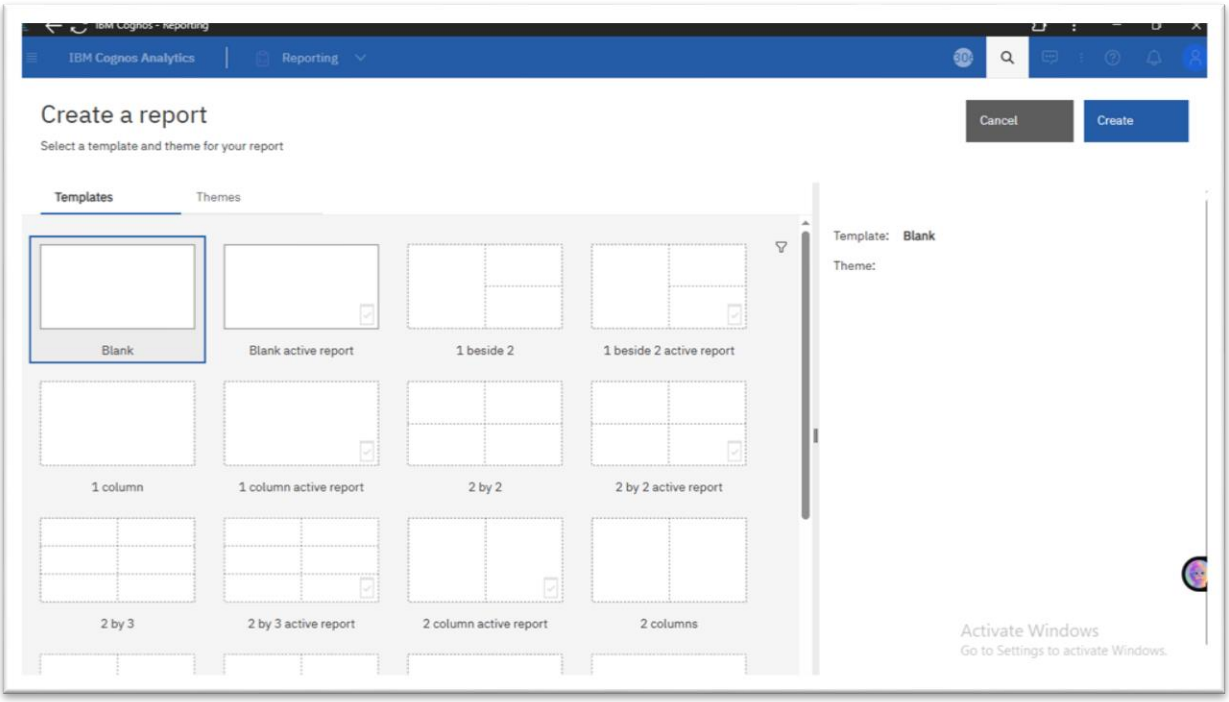
Step 6: Now Click on save option and Give Module an appropriate name, Select My content and click on Save



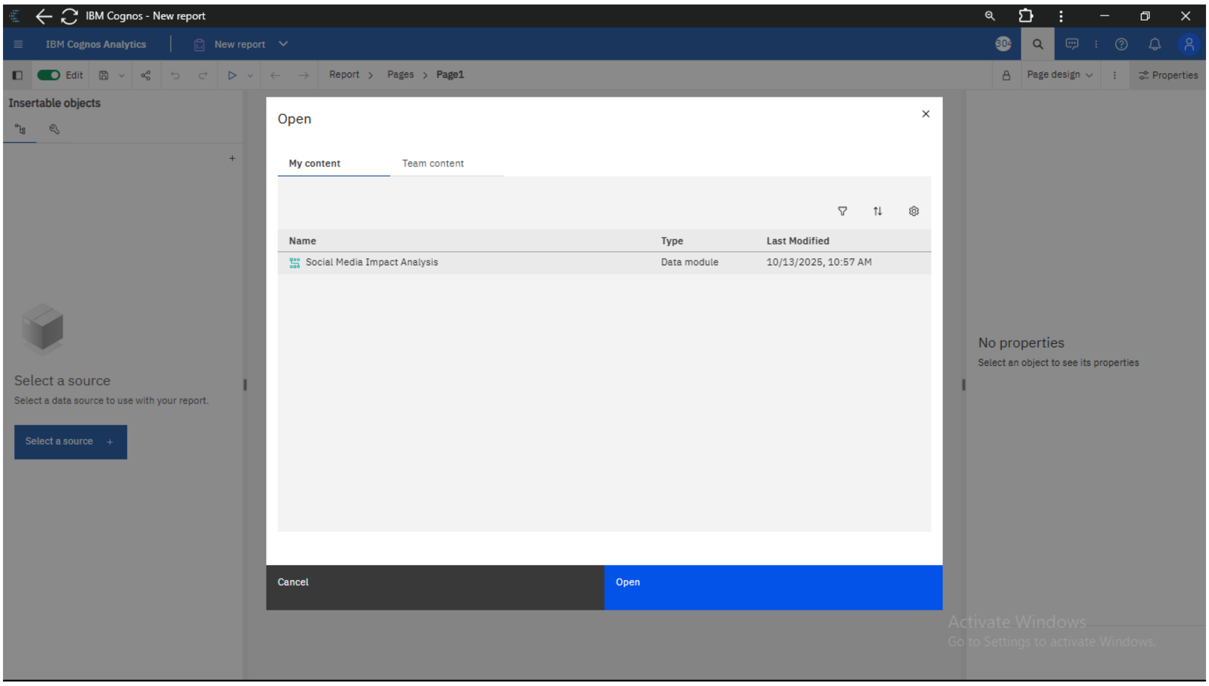
Step 7: Click on Three-Line icon again and select New-> Report option



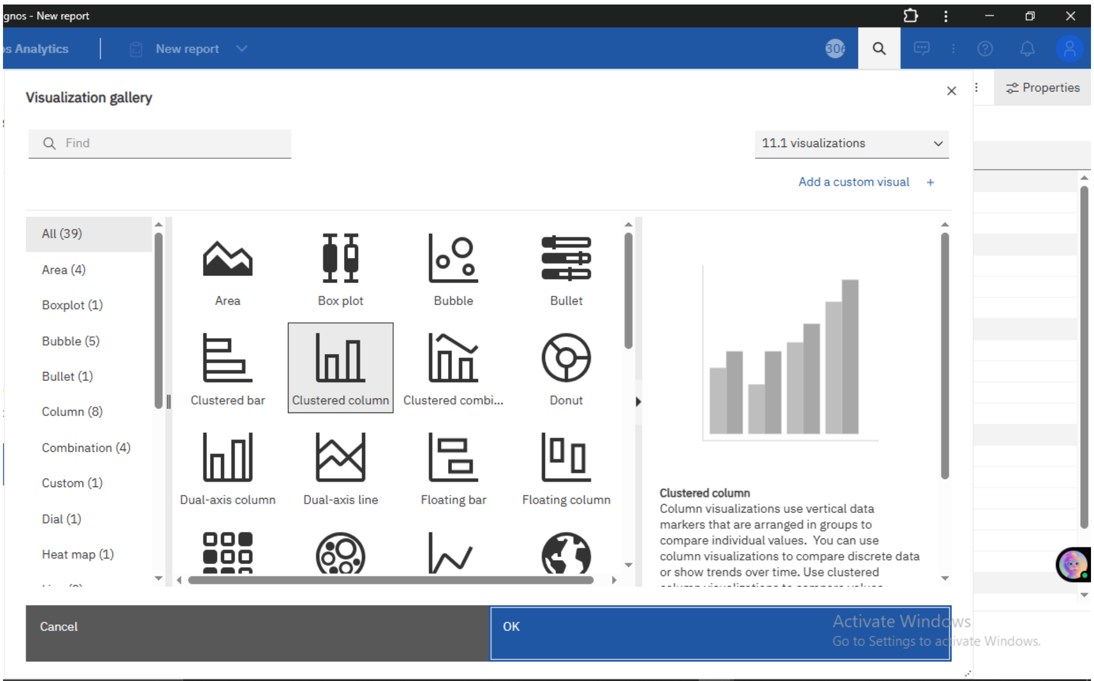
Step 8: Click on Blank Report Option and Then Click on Create Option



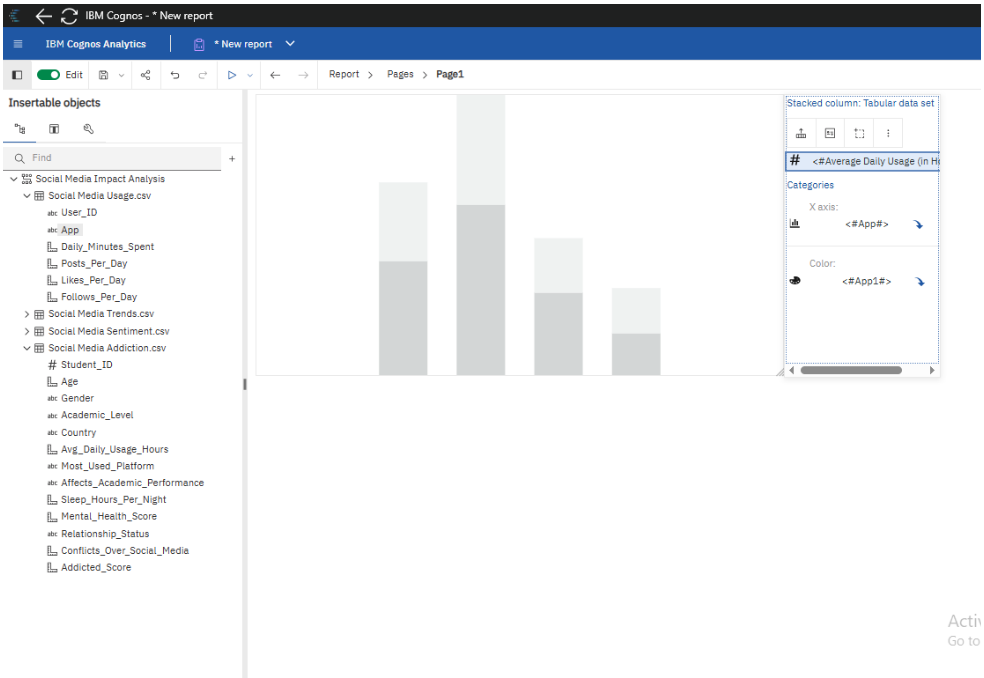
Step 9: Select Data Module from My Content Option and Click on Open.



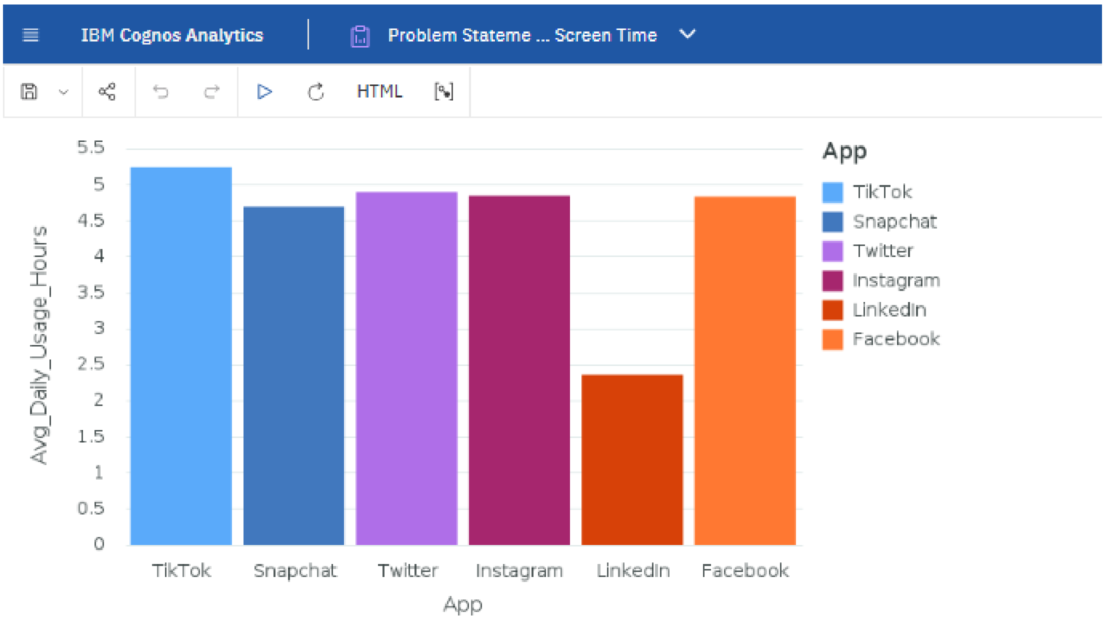
Step 10: Select Visualization Widget from the given options and choose Clustered columns



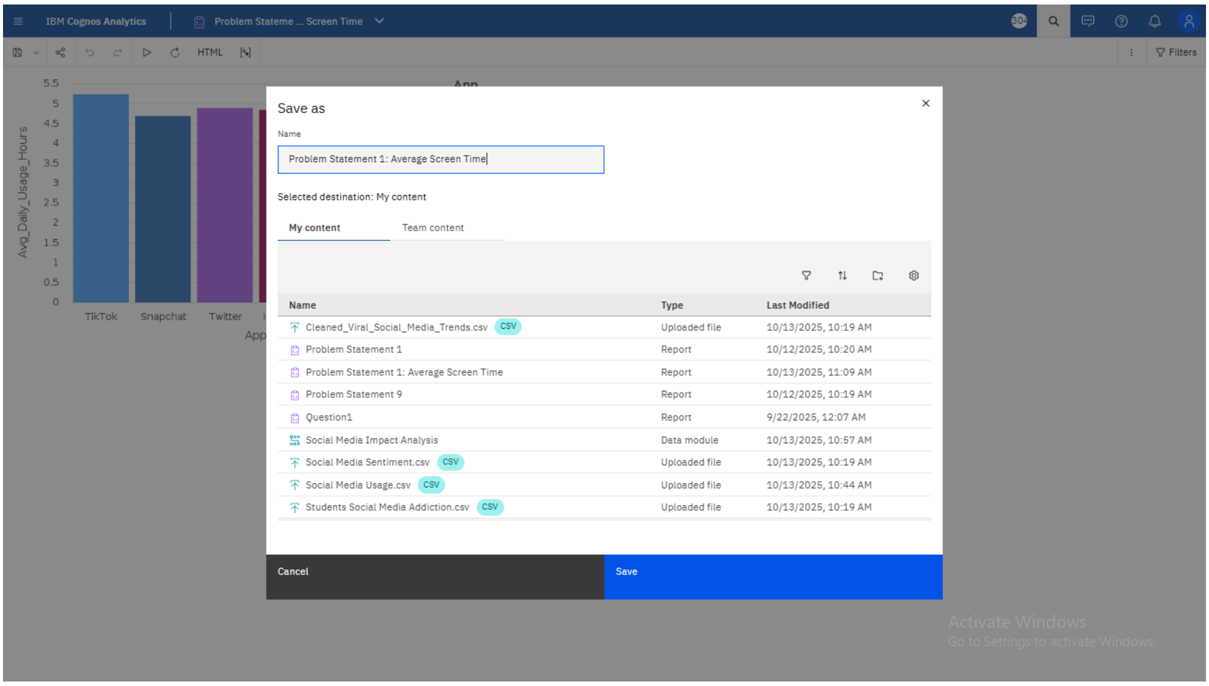
Step 11: Drag 'App' from Social Media Usage into X-axis and Colour, and Average Daily Usage in Y-axis options



Step 12: Run the report and following output will be generated



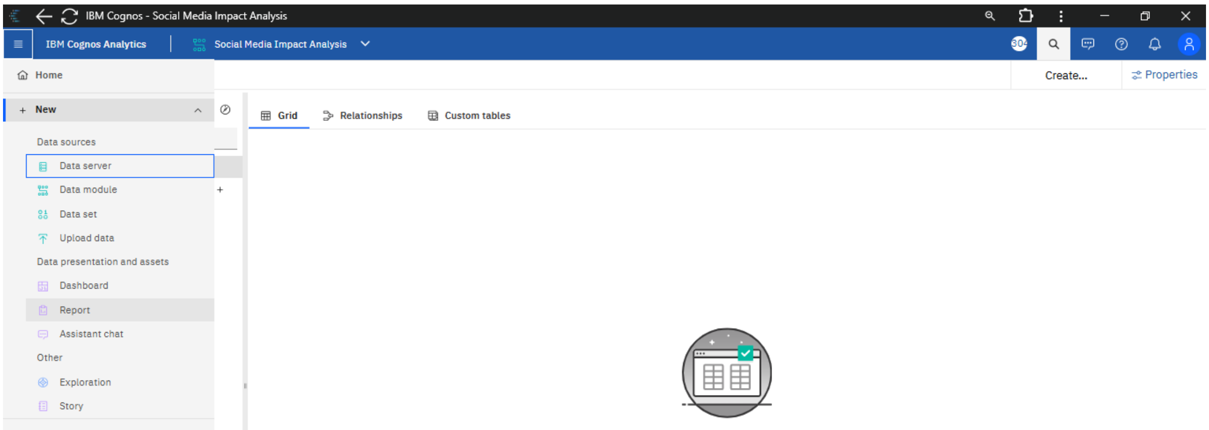
Step 13: Save the Report with the Desired name.



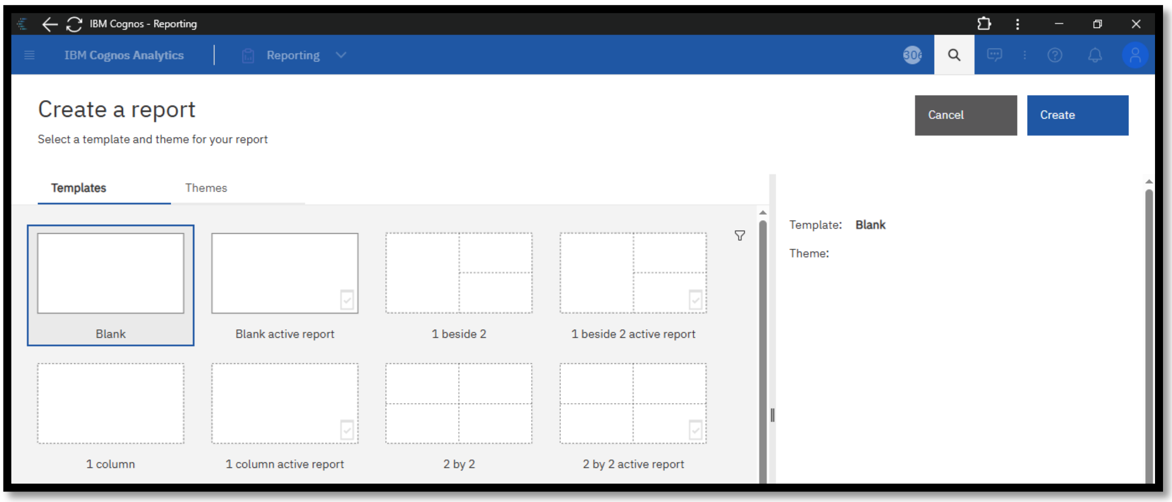
Problem Statement: Rank countries or regions by their average mental health score and usage level

Solution: Create a Map Report using Data Module

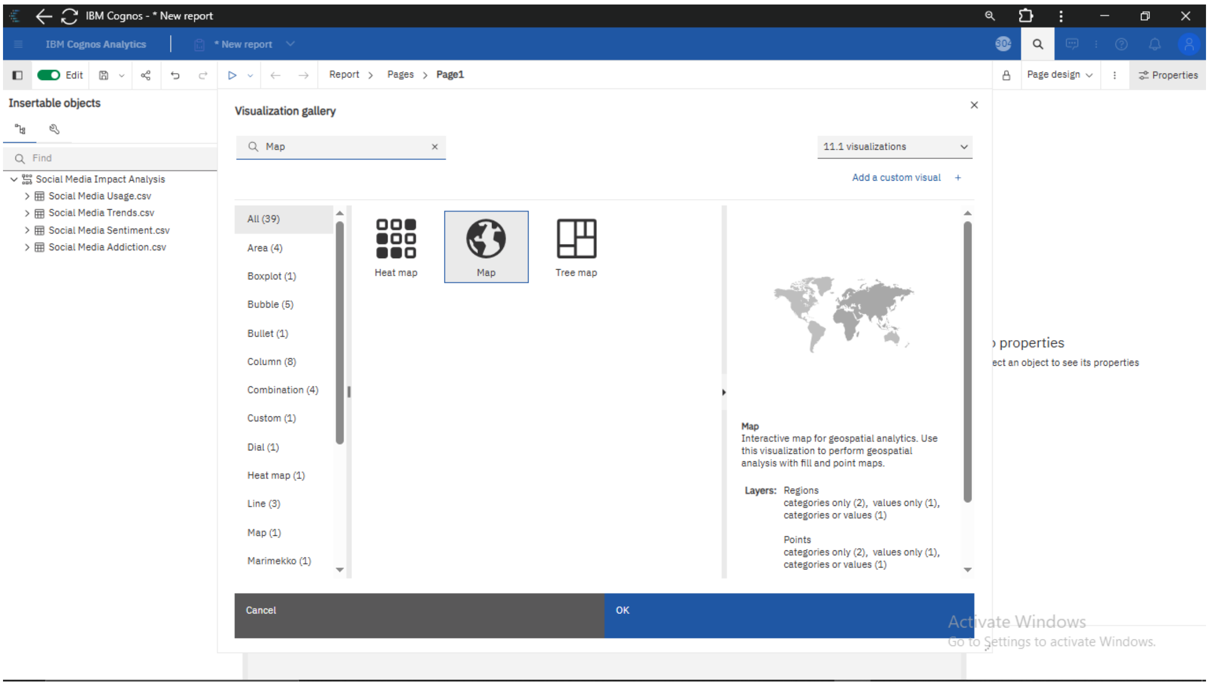
Step 1: Click on Three-Line Icon and Select New -> Report option.



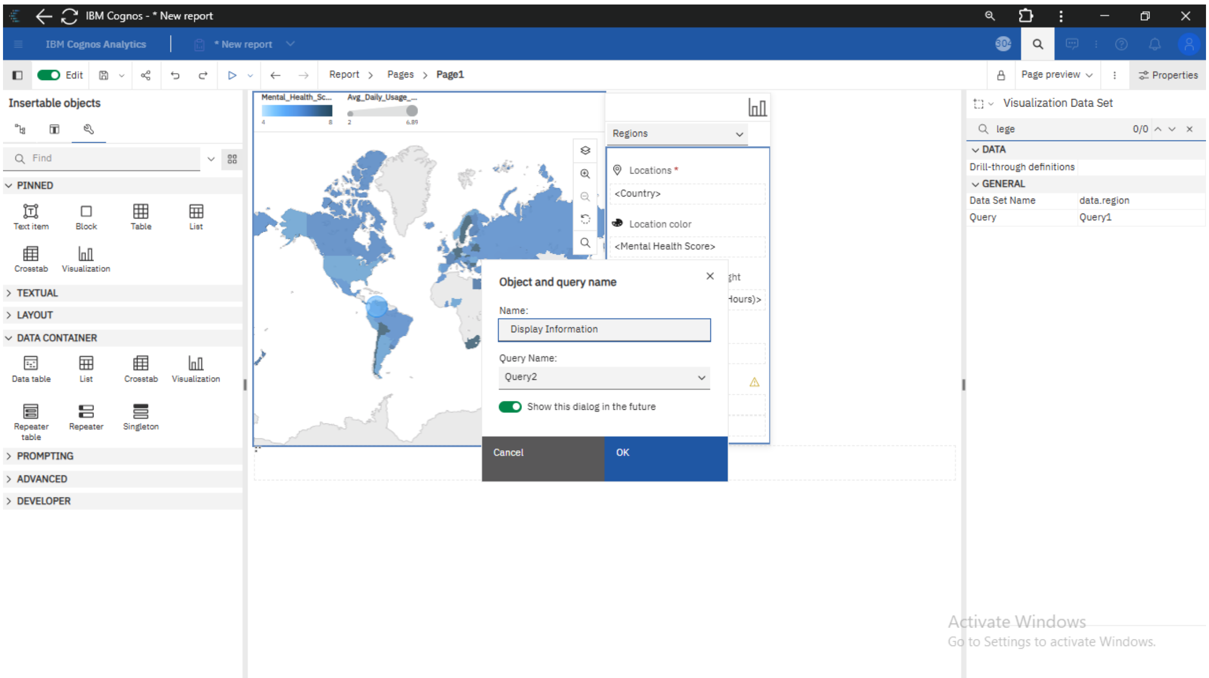
Step 2: Select Blank Report Option and Click on Create



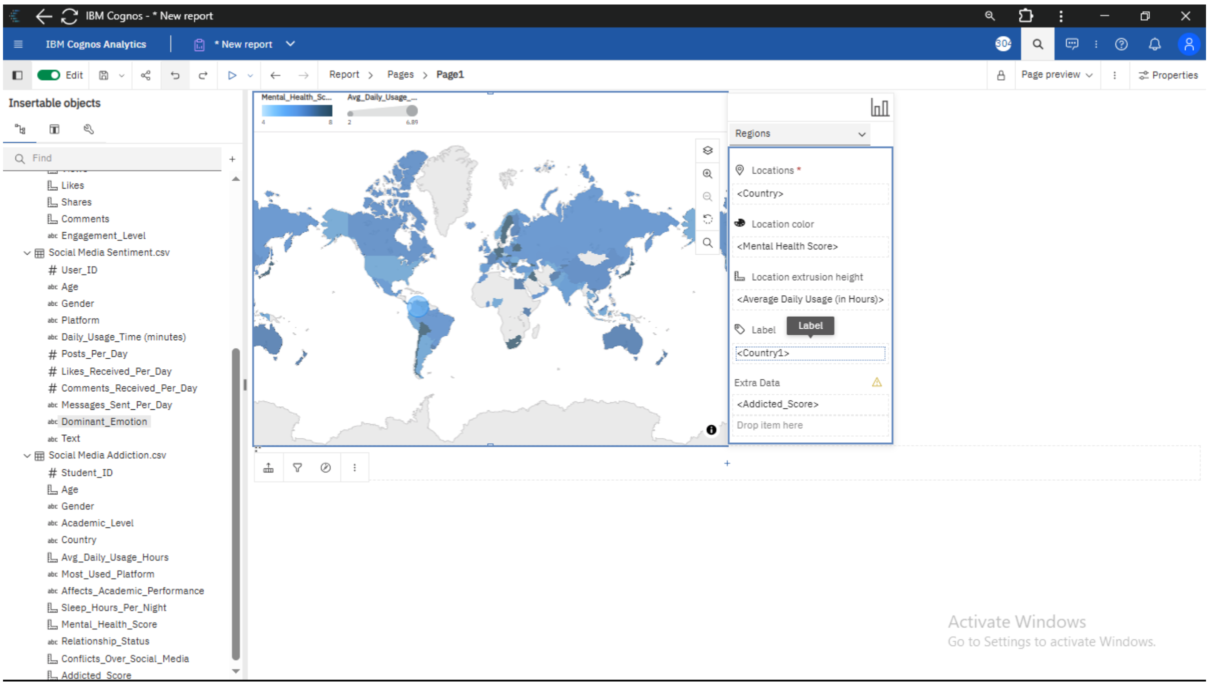
Step 3: Select Visualization Widget and Click on Map option



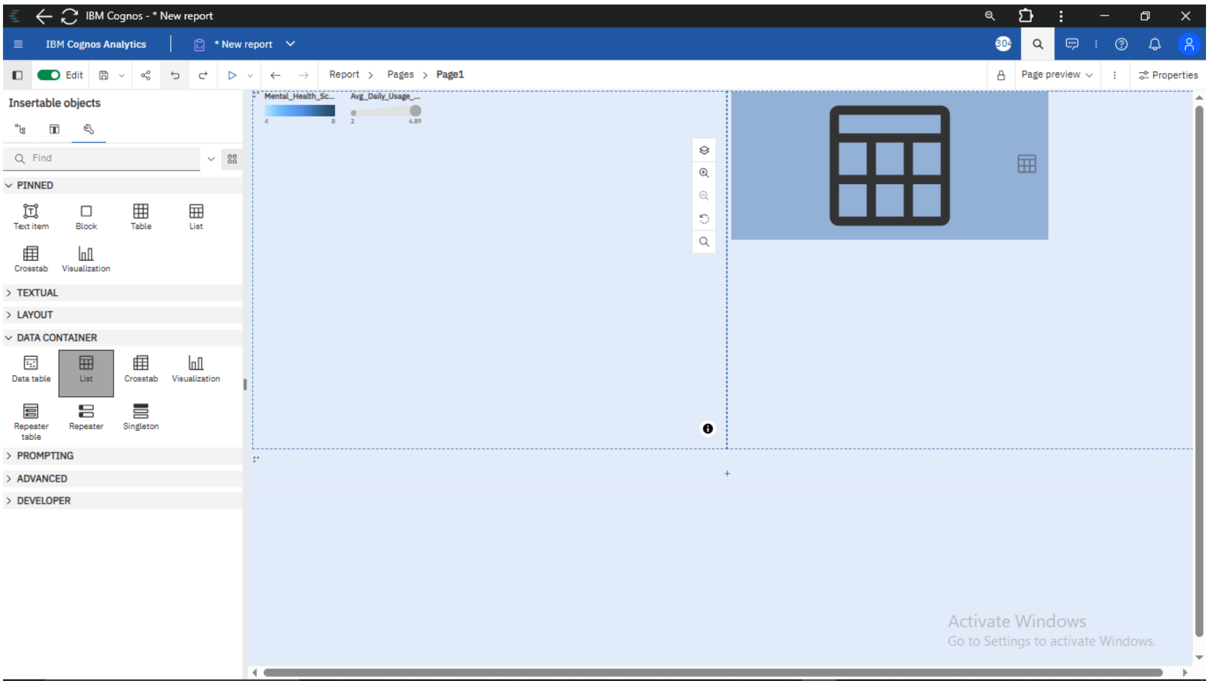
Step 4: Give a suitable name to it.



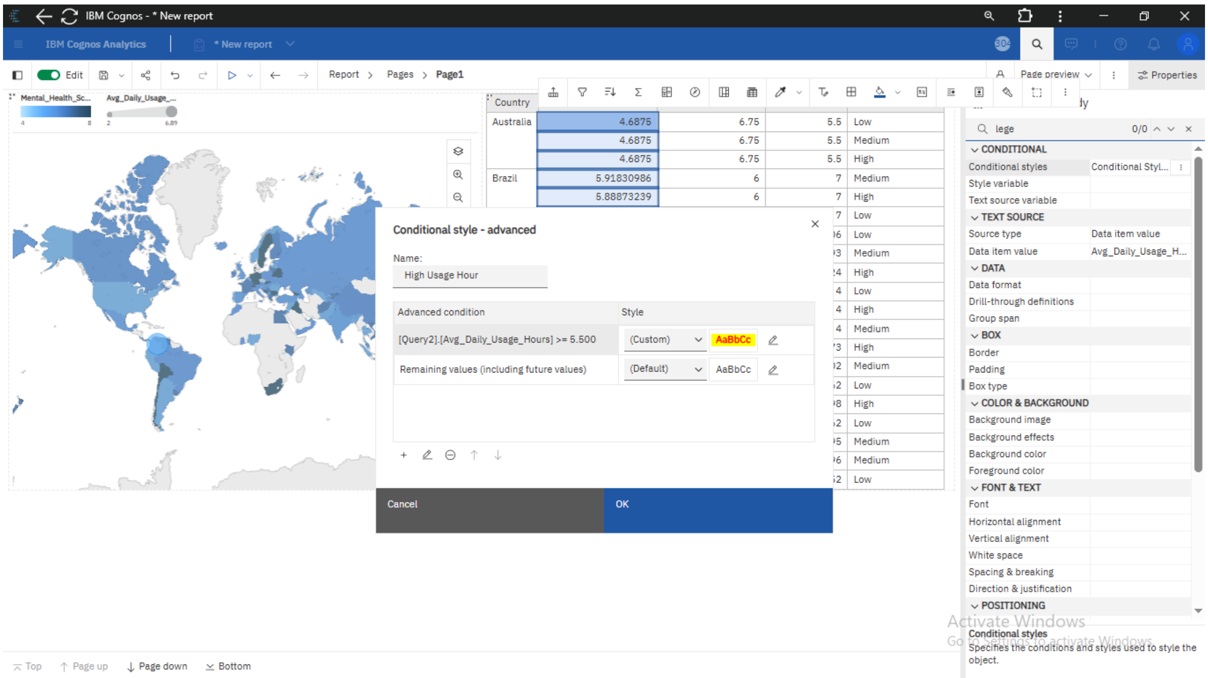
Step 6: Drag and Drop Country in Location, Mental Health Score in Location colour, Average daily usage in Location height and Country in labels option



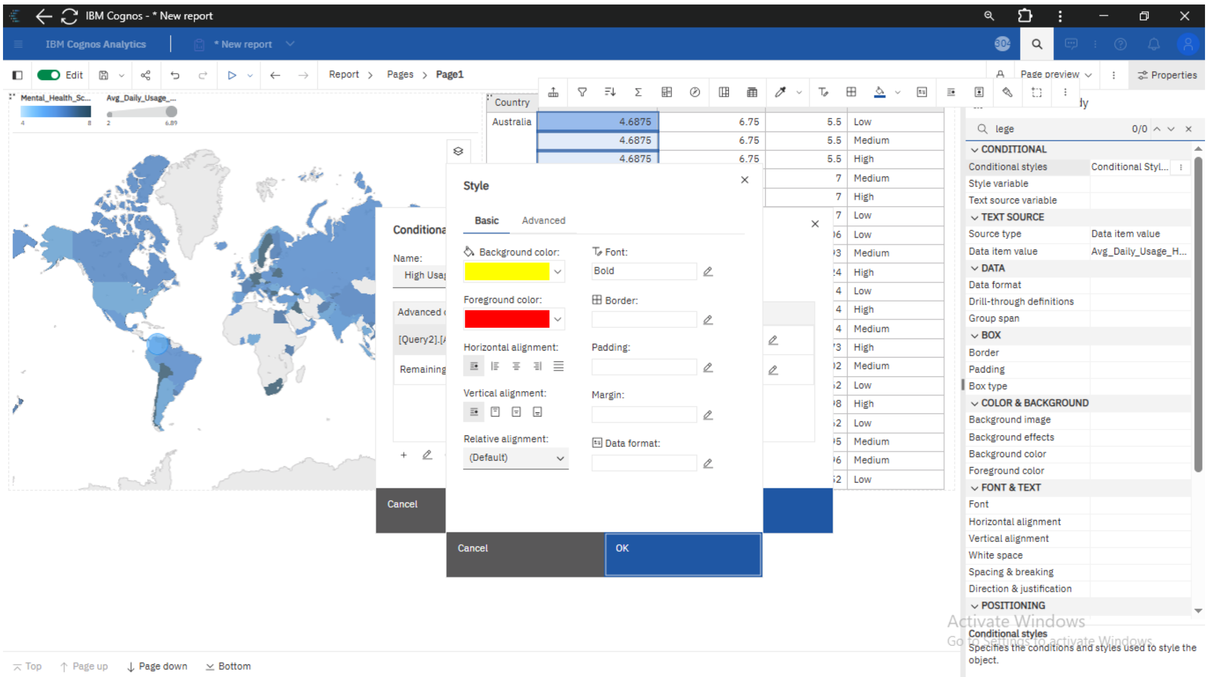
Step 7: Now in Click on Tool-icon in Insertable objects and Drag and Drop List from Data container option



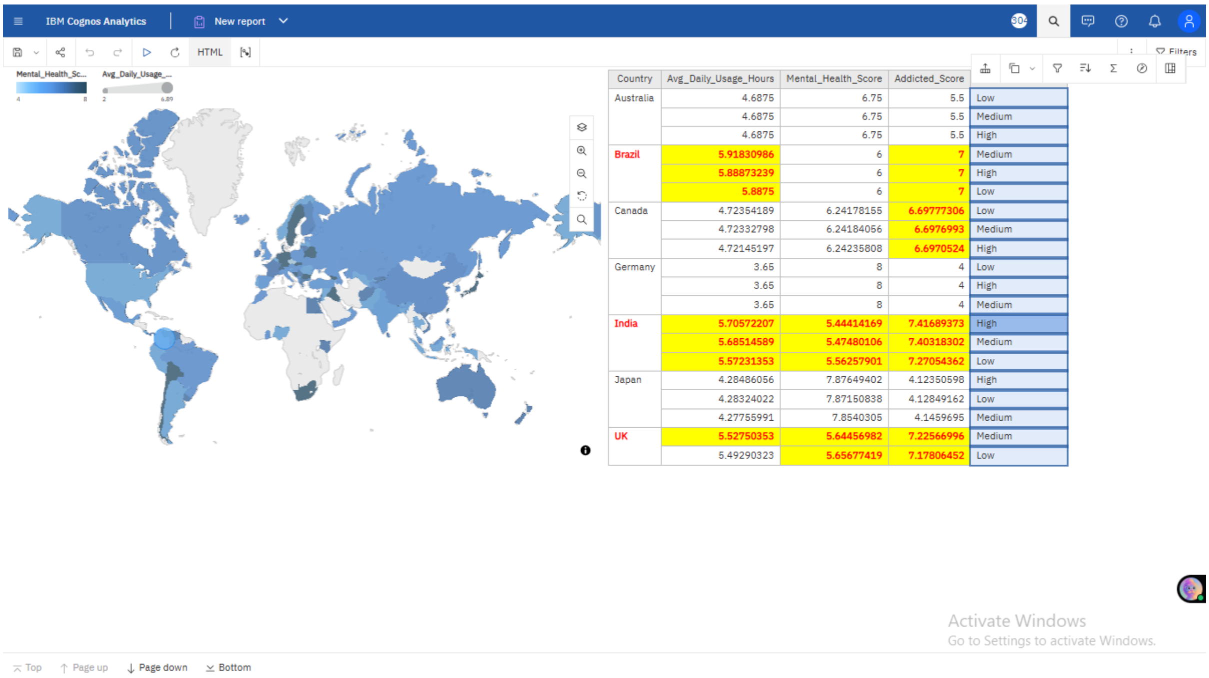
Step 8: Select Desired Columns then Click on Conditional Styles in Properties Panel, Create desired conditions by clicking on "+" icon



Step 9: Select formatting options by clicking on Edit button and Click on OK. Do the same for all the desired columns



Step 10: Run the Report and Save it with suitable name.



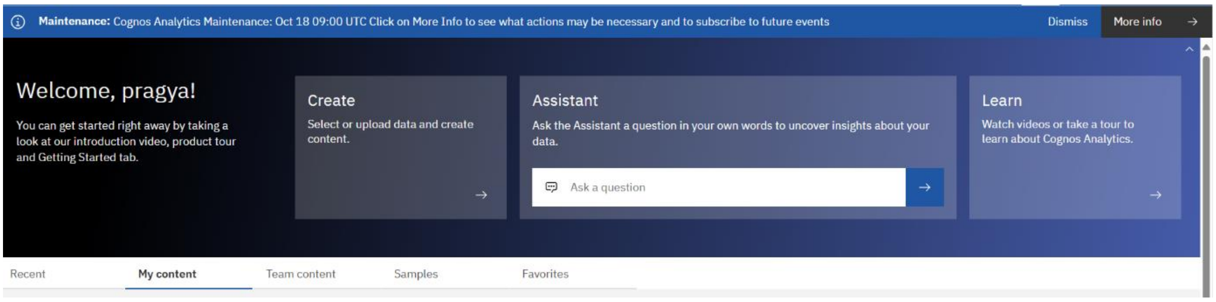
Country	Avg_Daily_Usage_Hours	Mental_Health_Score	Addicted_Score	
Australia	4.6875	6.75	5.5	Low
	4.6875	6.75	5.5	Medium
	4.6875	6.75	5.5	High
Brazil	5.91830986	6	7	Medium
	5.88873239	6	7	High
	5.8875	6	7	Low
Canada	4.72354189	6.24178155	6.69777306	Low
	4.72332798	6.24184056	6.6976993	Medium
	4.72145197	6.24235808	6.6970524	High
Germany	3.65	8	4	Low
	3.65	8	4	High
	3.65	8	4	Medium
India	5.70572207	5.44414169	7.41689373	High
	5.68514589	5.47480106	7.40318302	Medium
	5.57231353	5.56257901	7.27054362	Low
Japan	4.28486056	7.87649402	4.12350598	High
	4.28324022	7.87150838	4.12849162	Low
	4.27755991	7.8540305	4.1459695	Medium
UK	5.52750353	5.64456982	7.22566996	Medium
	5.49290323	5.65677419	7.17806452	Low

Project

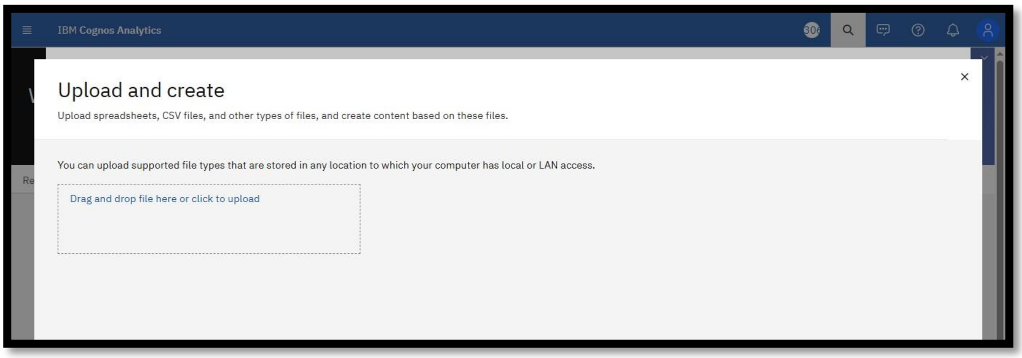
Problem Statement 1: Evaluate gender-wise differences in usage addiction

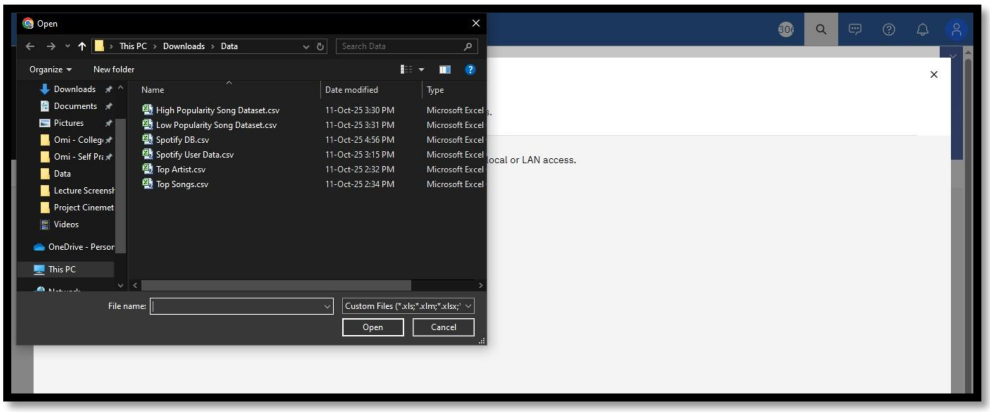
Solution : Create a Bar Chart using Data set

Step 1: Login to IBM Cognos and following interface will be opened. Click on Create option**.**

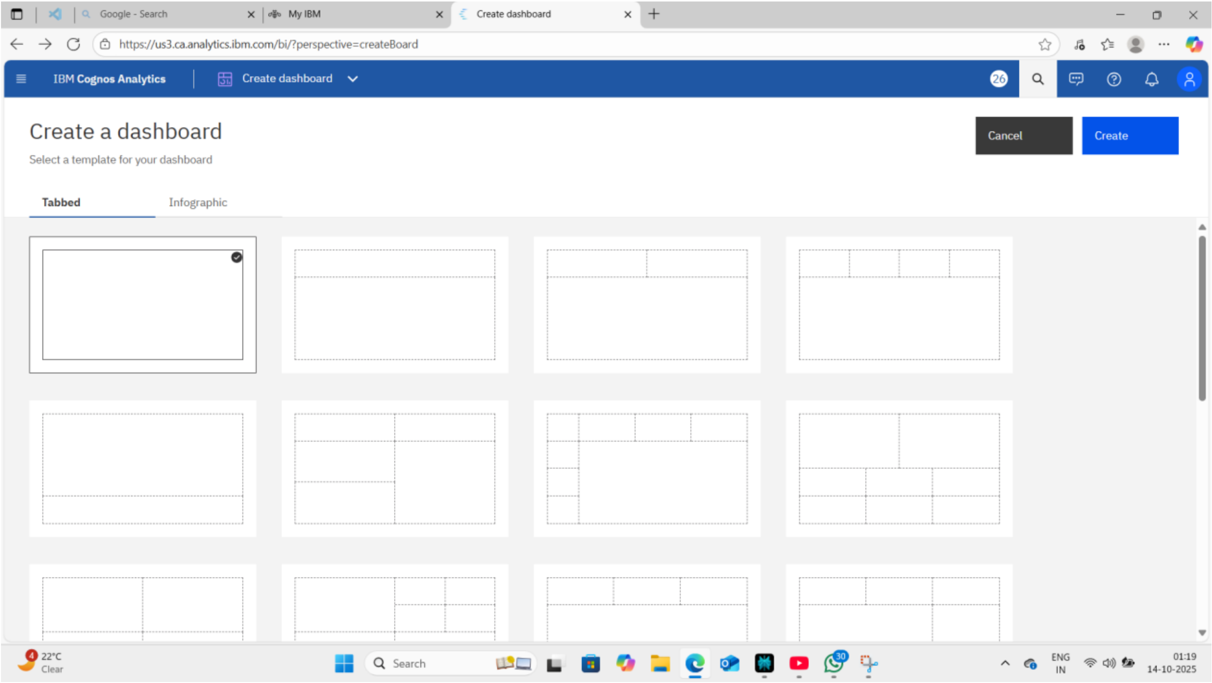


Step 2: Go to Click to Upload option and Select Files you want to Upload and Click on next

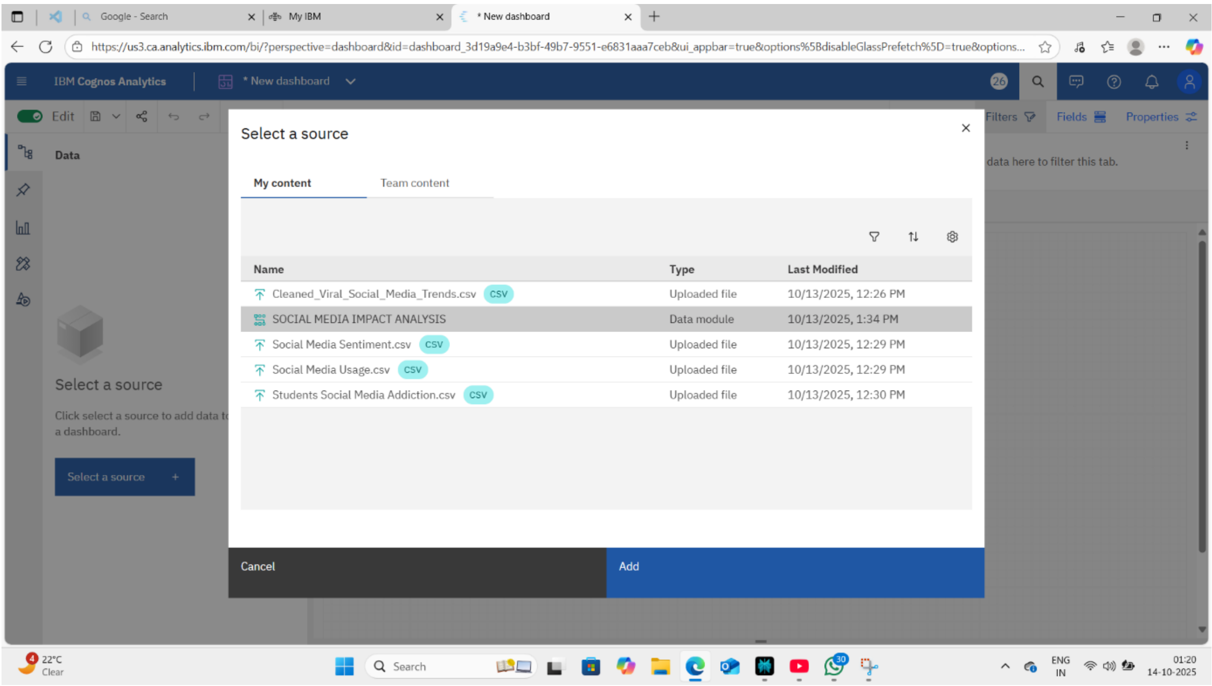




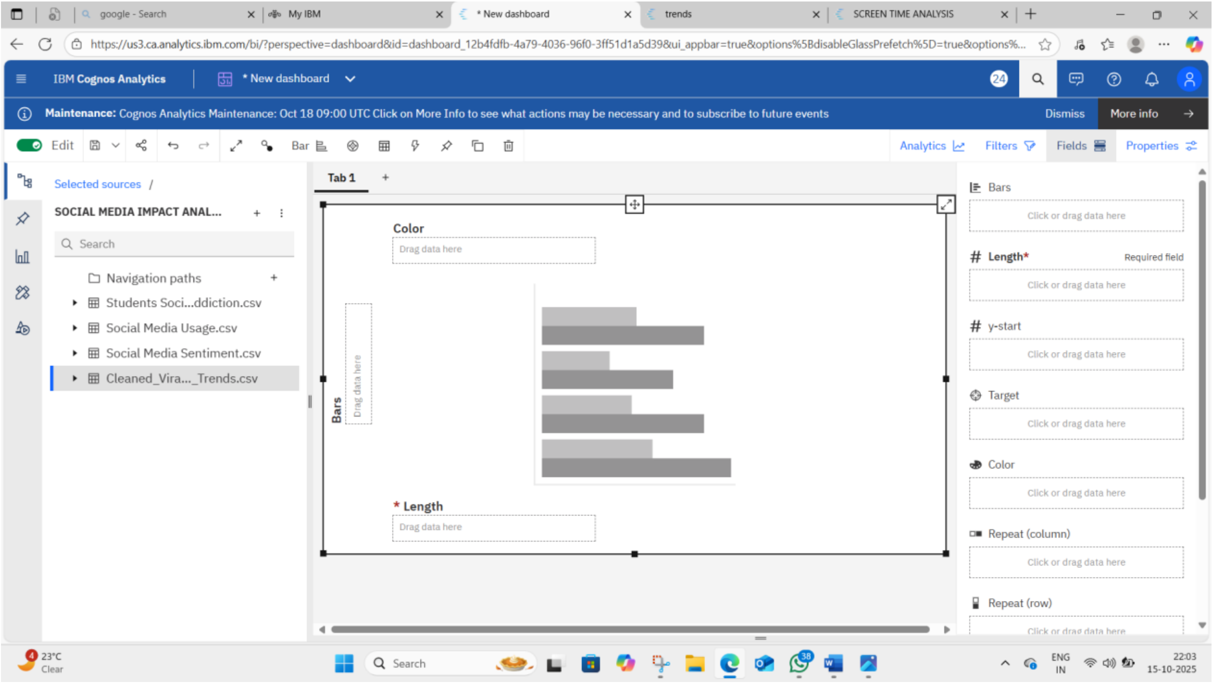
Step 3: Click on the DASHBOARD Option on next screen and Click on Create option.



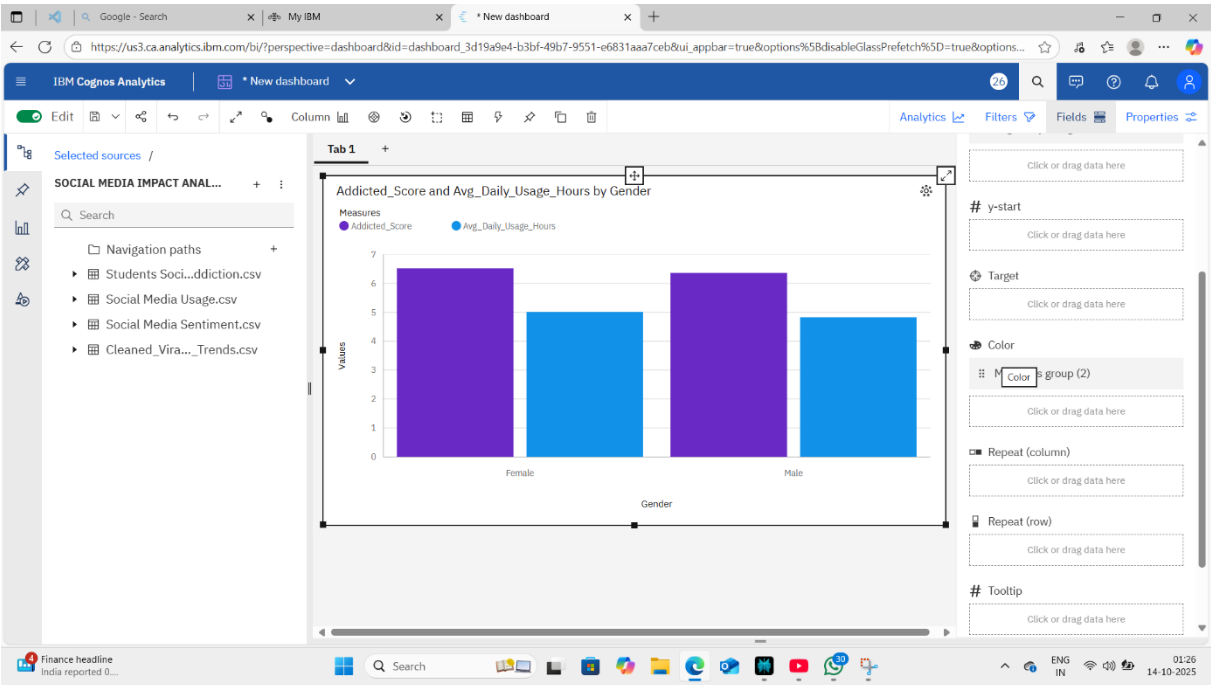
Step 4: Select Datasets to make a Dashboard



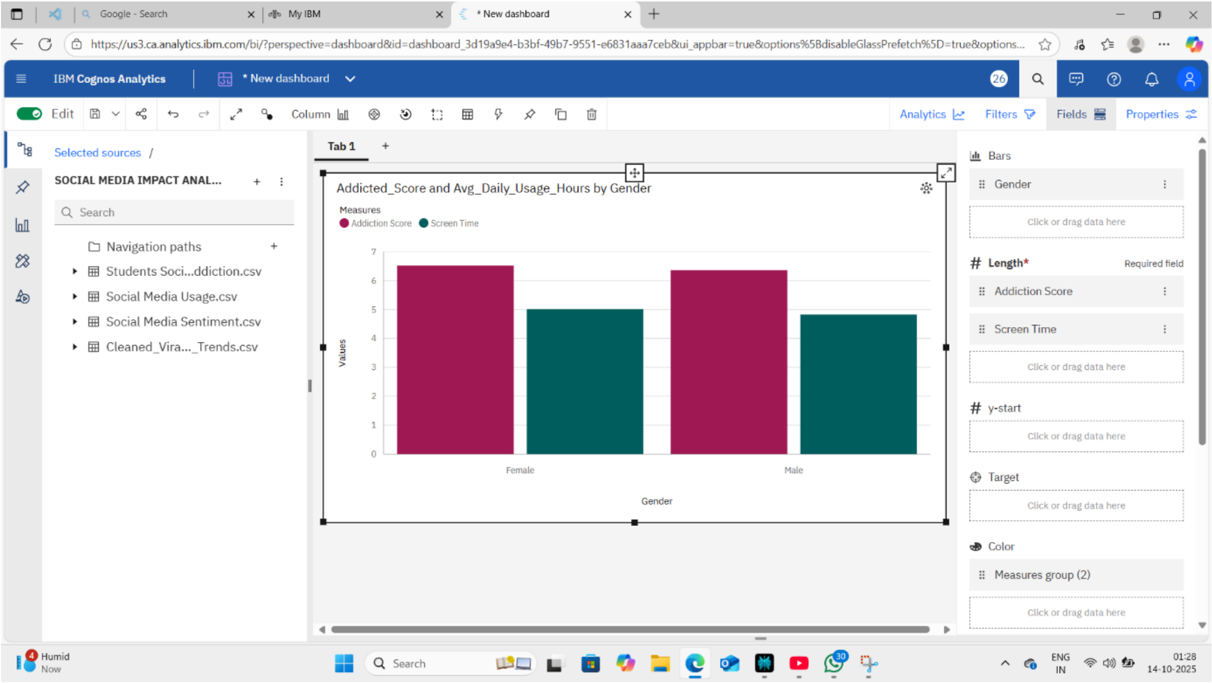
Step 5: Insert a Bar Chart Visualisation



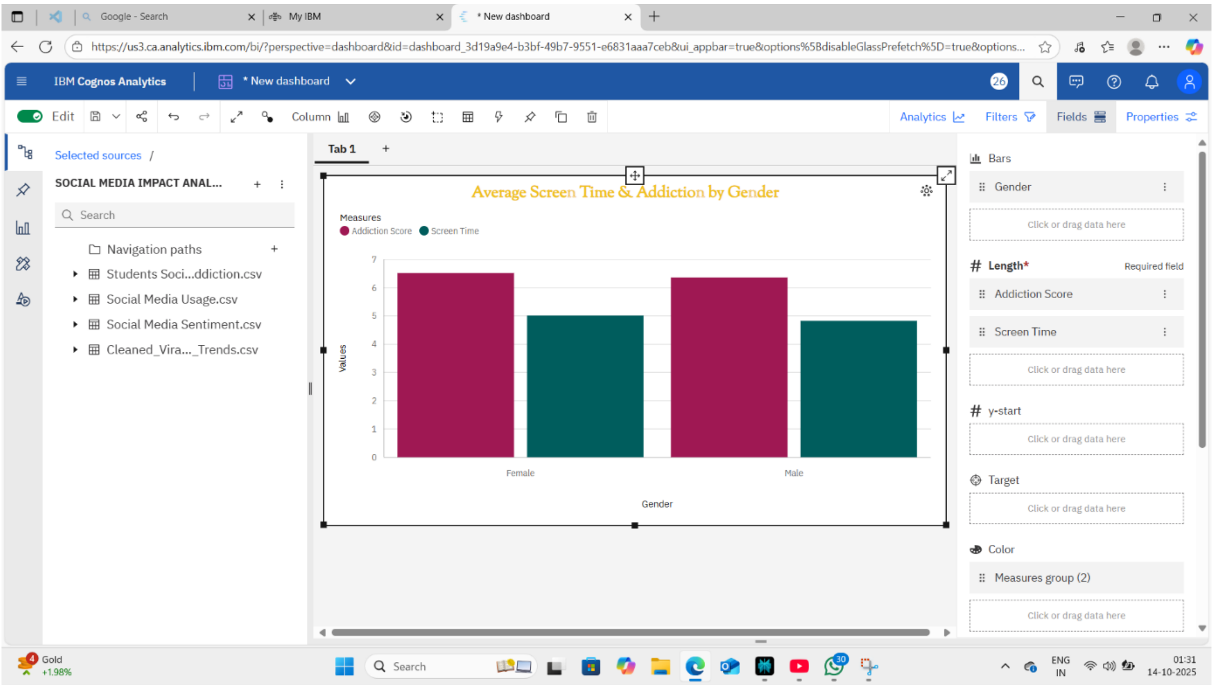
Step 6: ADD data to Bar Chart



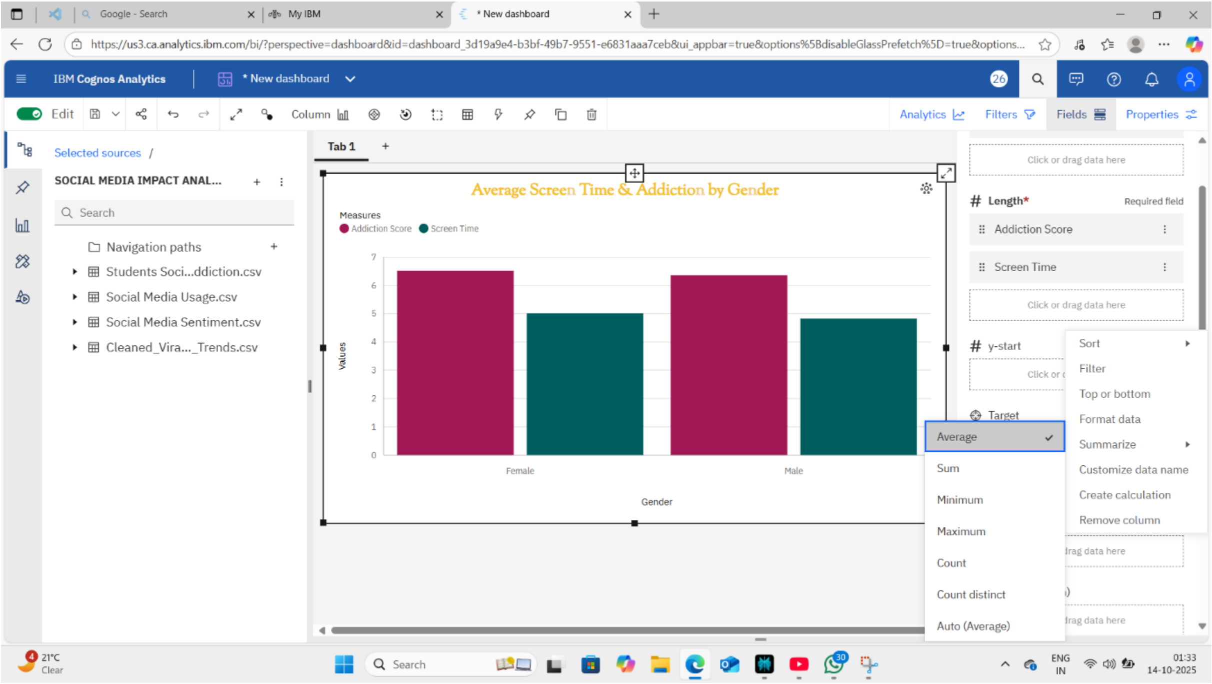
Step 7: Rename columns(for clarity) CHANGE avg_daily_usage to Screen Time



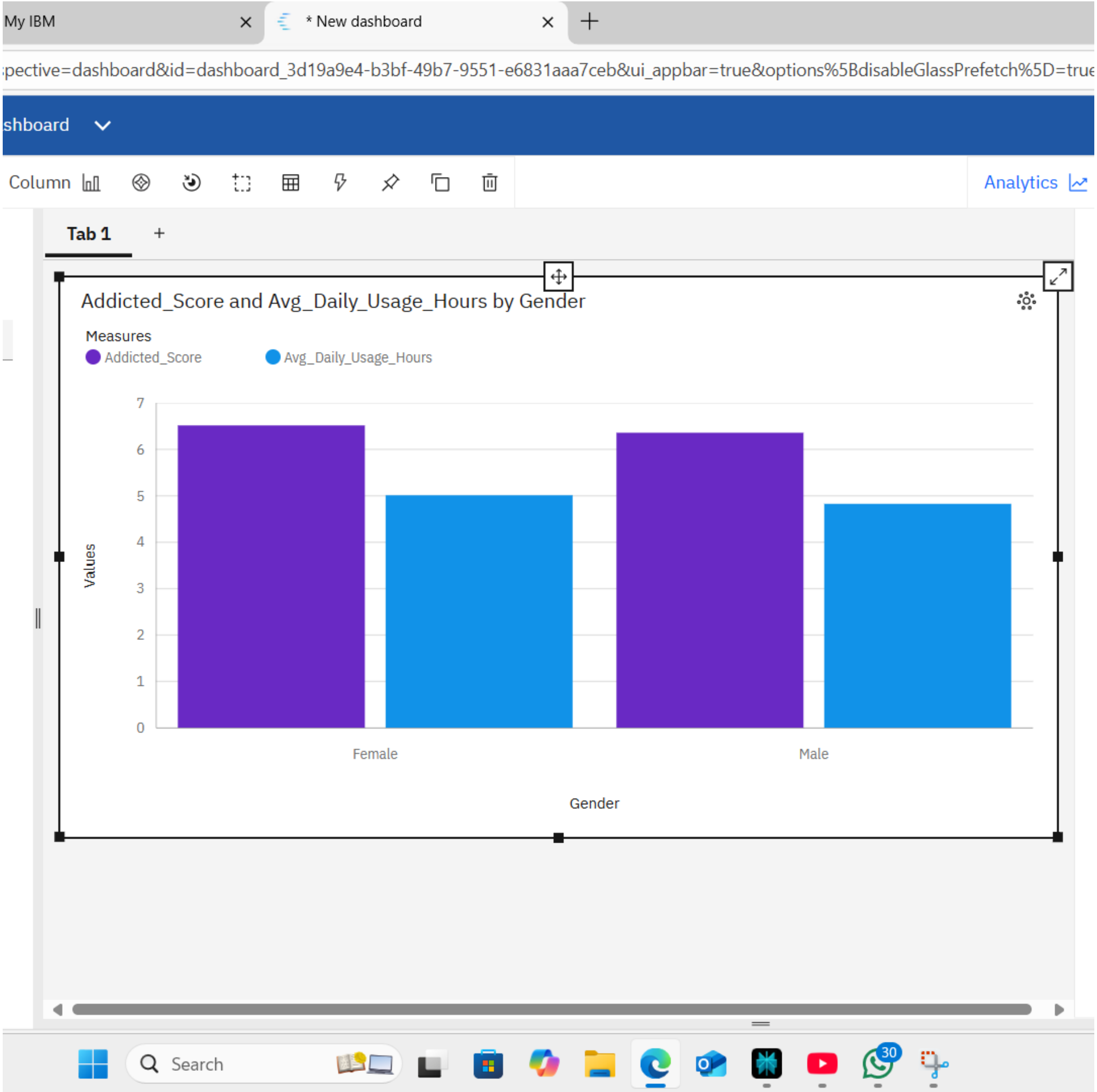
Step 8: ADD additional formatting to make it look good



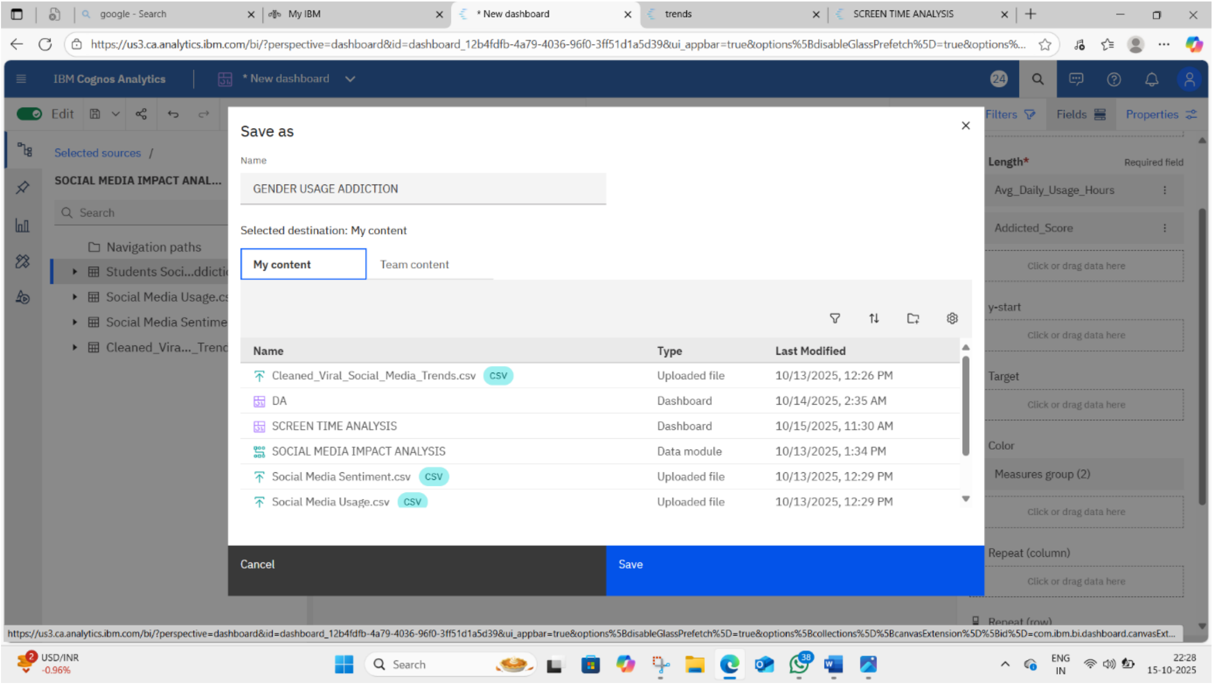
Step 9: Chane the value type to “Average”.



Step 10: ADD COLOR customisation



Step 11: Save your work

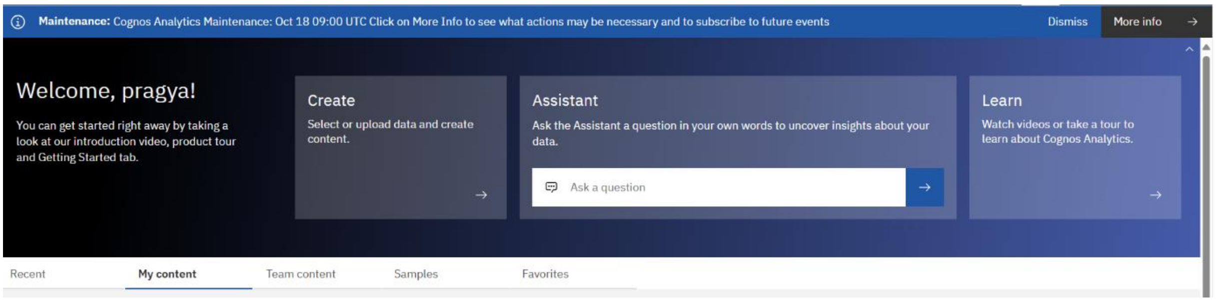


Project

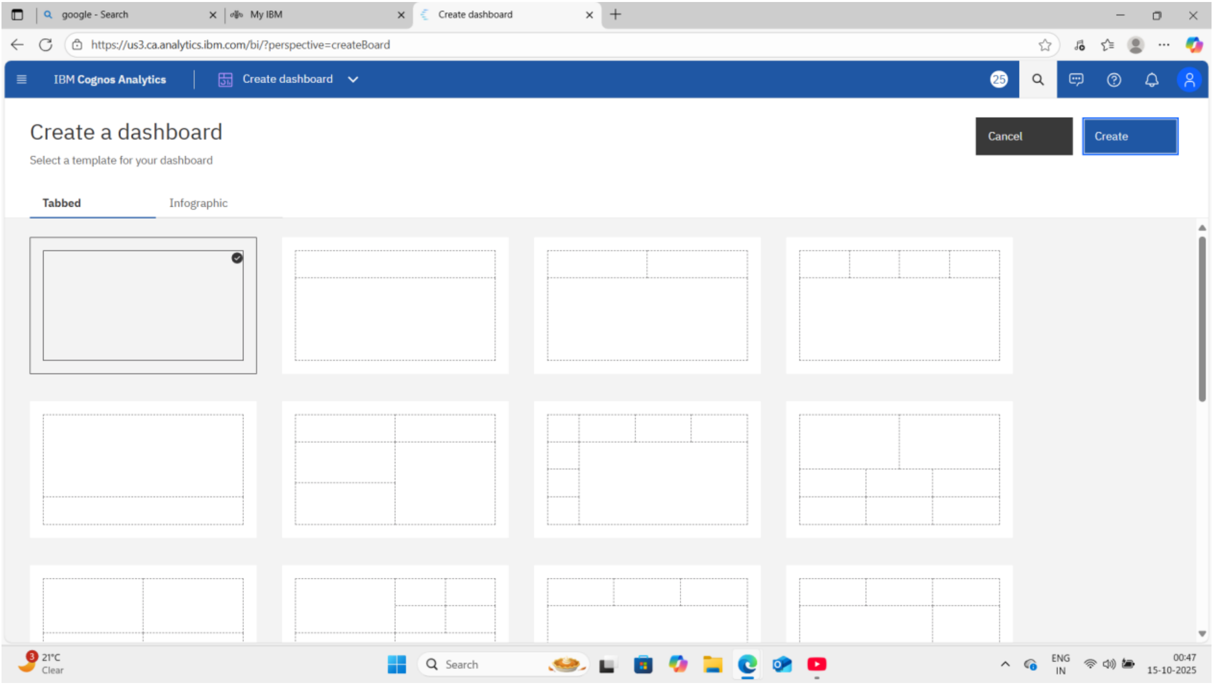
Problem Statement: Identify users with excessive screen time but low engagement rates

Solution: Make a table that shows those users with excessive screen time and low engagement rates.

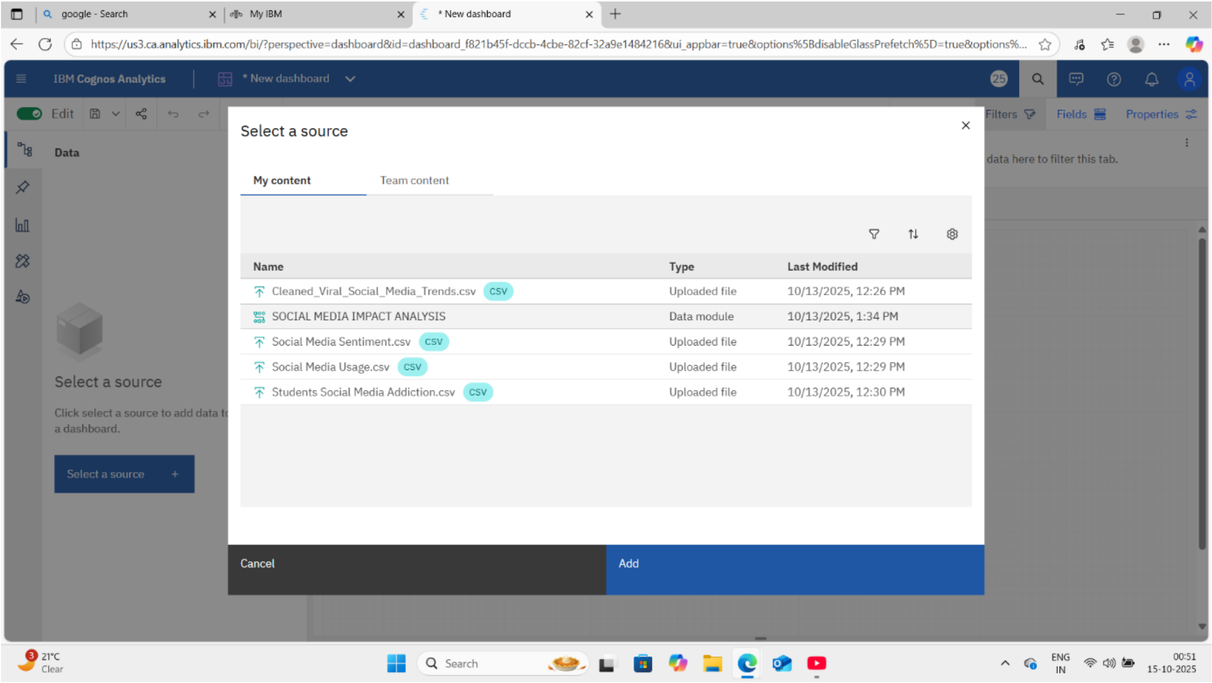
Step 1: loging to IBM following interface will appear



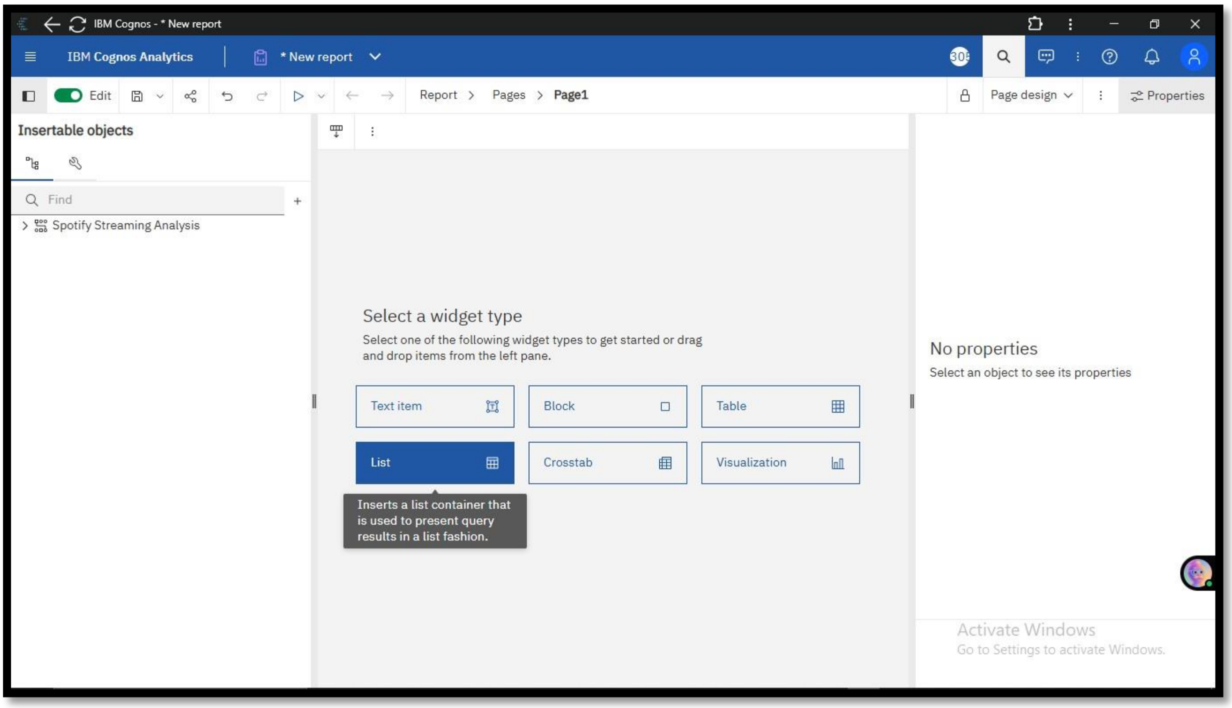
Step 2: Select Blank dashboard Option and Click on Create



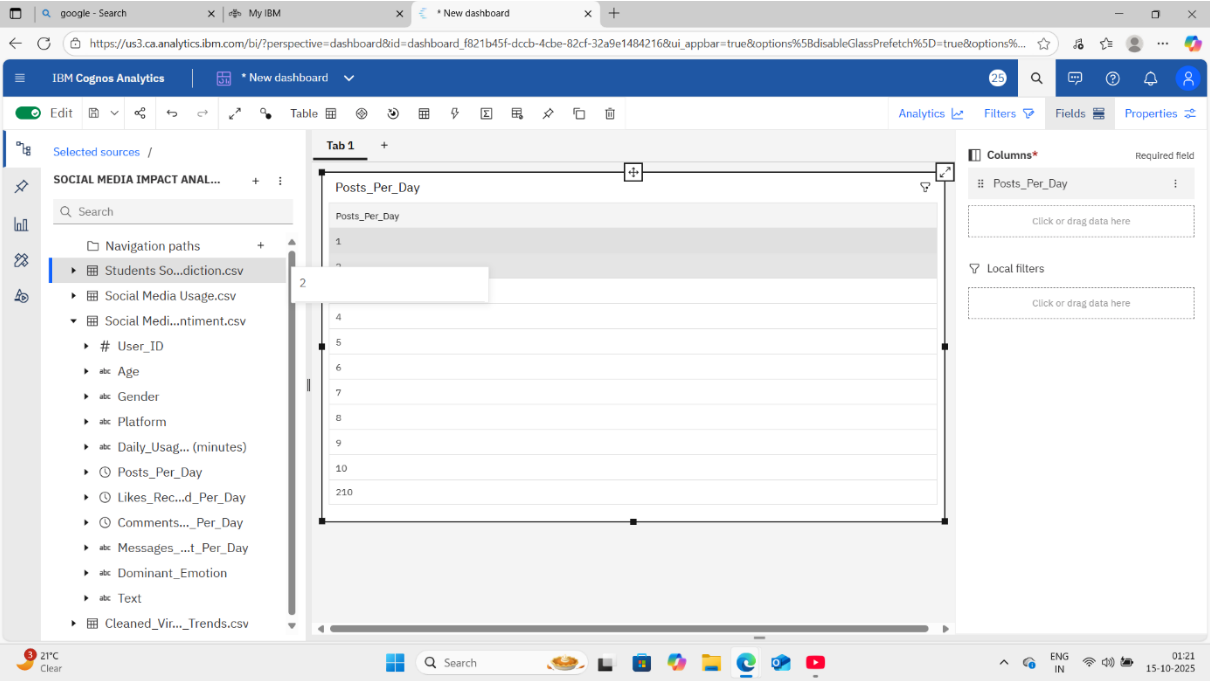
Step 3: Select Data Module from My Content option and Click on Open Option.

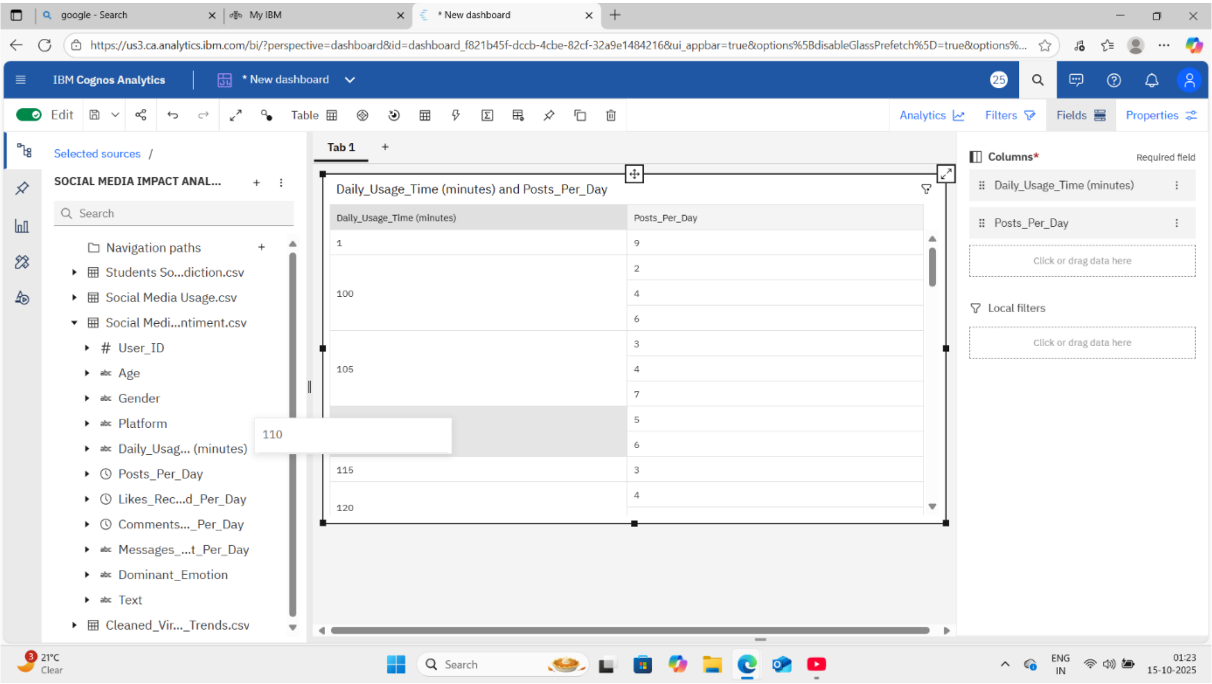


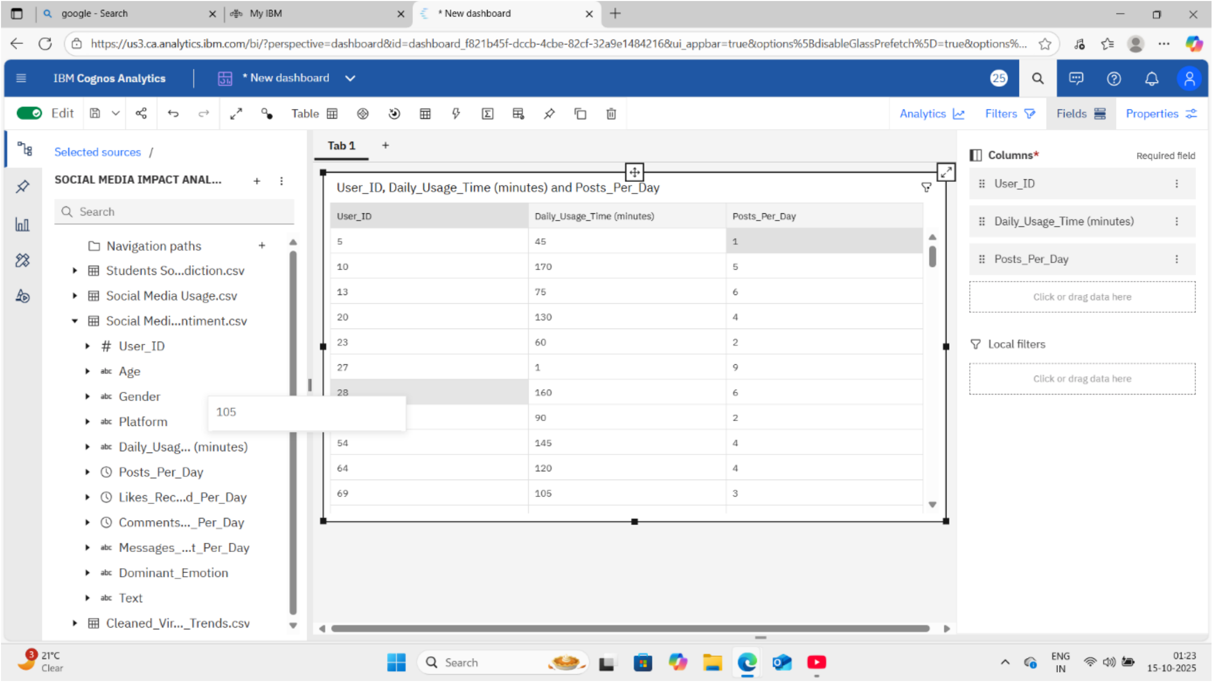
Step 4: Select table Widget Type from the given Options



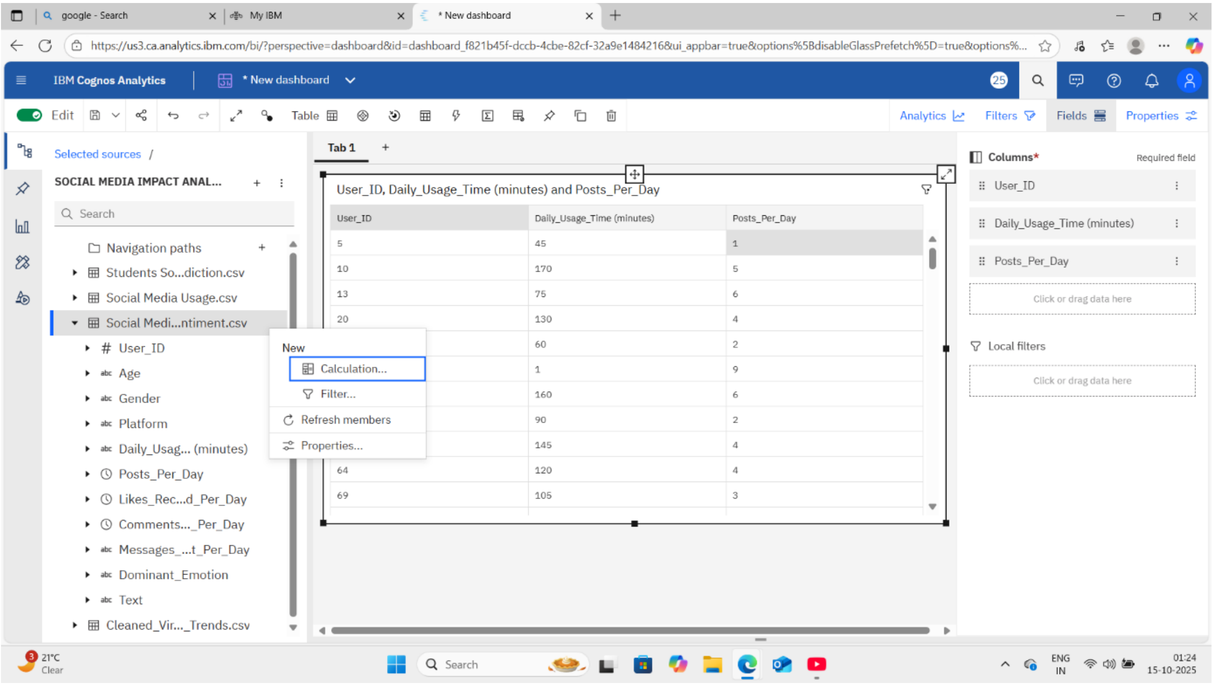
Step 5: Add data from data set one by one into columns fields


IBM Cognos Analytics
* New dashboard
Selected sources /
SOCIAL MEDIA IMPACT ANAL...
Navigation paths
Students So...diction.csv
Social Media Usage.csv
Social Medi...ntiment.csv
User_ID
Age
Gender
Platform
Daily_Usag... (minutes)
Posts_Per_Day
Likes_Rec...d_Per_Day
Comments..._Per_Day
Messages_...t_Per_Day
Dominant_Emotion
Text
Cleaned_Vir..._Trends.csv
Tab 1
Posts_Per_Day
1
2
4
5
6
7
8
9
10
210
Columns*
Required field
Click or drag data here
Local filters


IBM Cognos Analytics
* New dashboard
Selected sources /
SOCIAL MEDIA IMPACT ANAL...
Tab 1
Daily_Usage_Time (minutes) and Posts_Per_Day
Daily_Usage_Time (minutes)
Posts_Per_Day
1
100
105
110
115
120
Columns*
Required field
Daily_Usage_Time (minutes)
Posts_Per_Day
Click or drag data here
Local filters

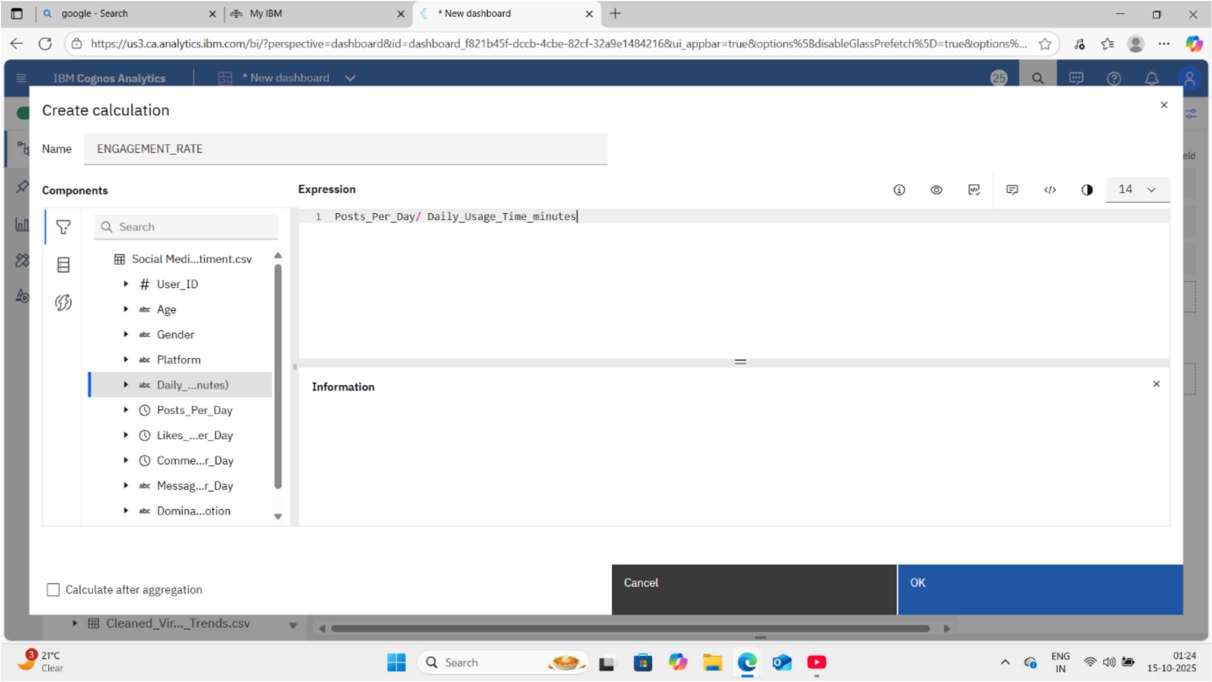


Step 6: ADD a calculation for measuring the engagement rate

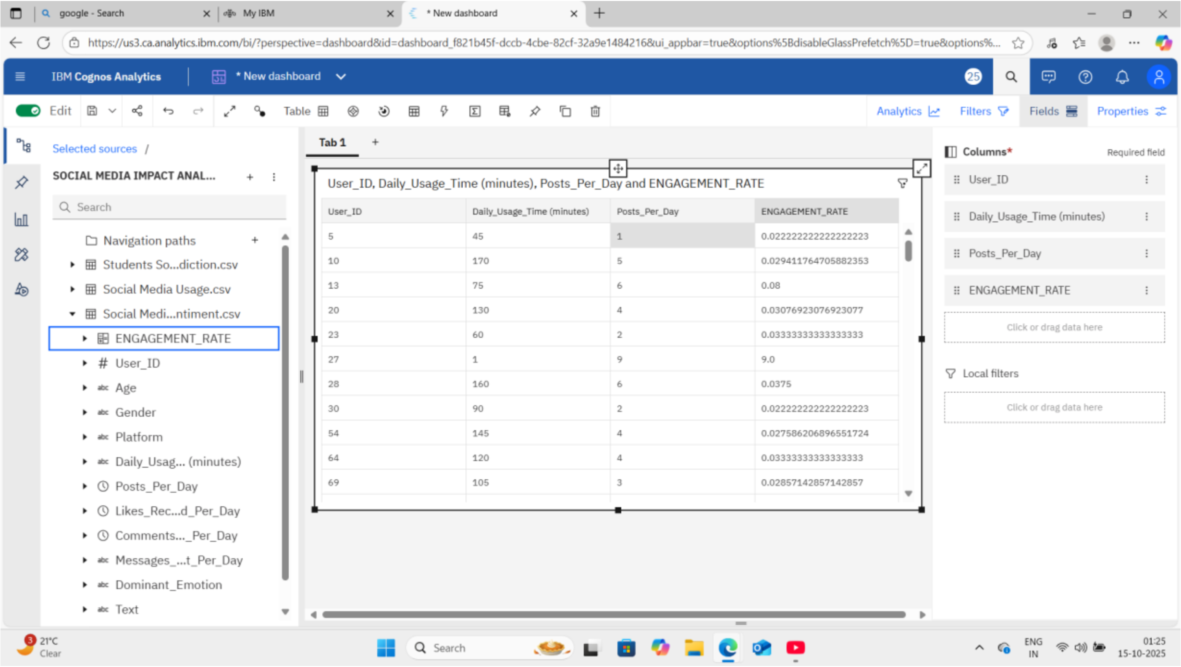


Step 7: IN FORMULA BOX write:

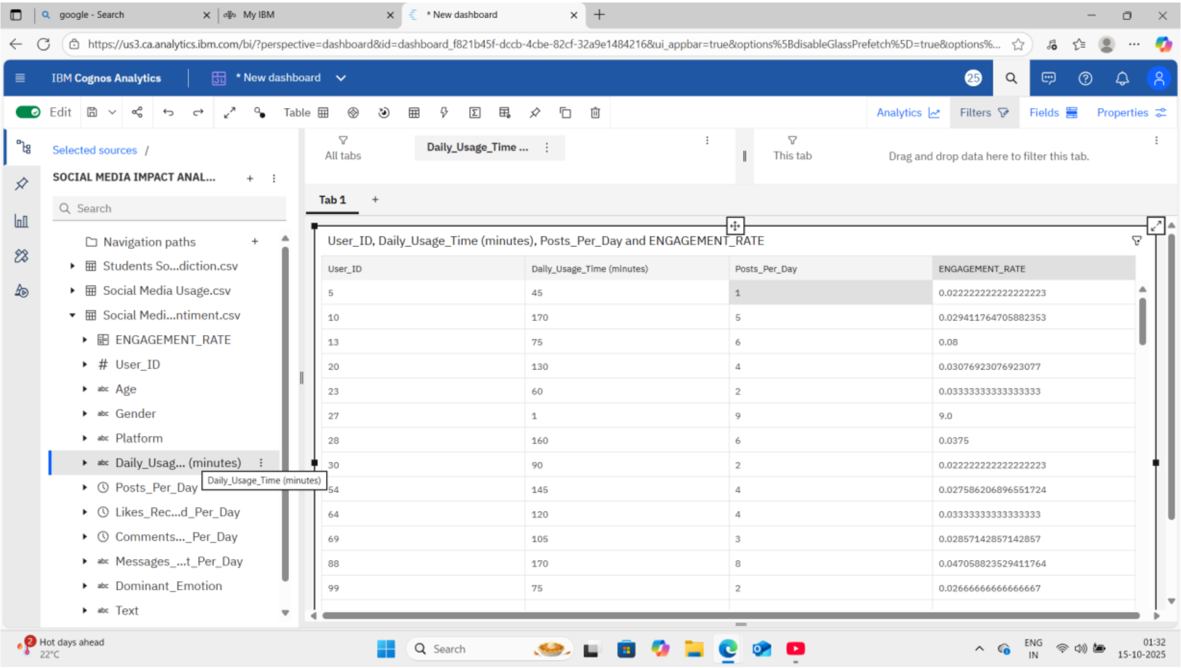
[Engagement_count]/[screen_time]


Create calculation
Name
ENGAGEMENT_RATE
Components
Expression
Posts_Per_Day/ Daily_Usage_Time_minutes
Social Medi...timent.csv
User_ID
Age
Gender
Platform
Daily_...nutes)
Posts_Per_Day
Likes_...er_Day
Comme...r_Day
Messag...r_Day
Domina...otion
Information
Calculate after aggregation
Cancel
OK

Step 8: Make a new column(engagement rate)

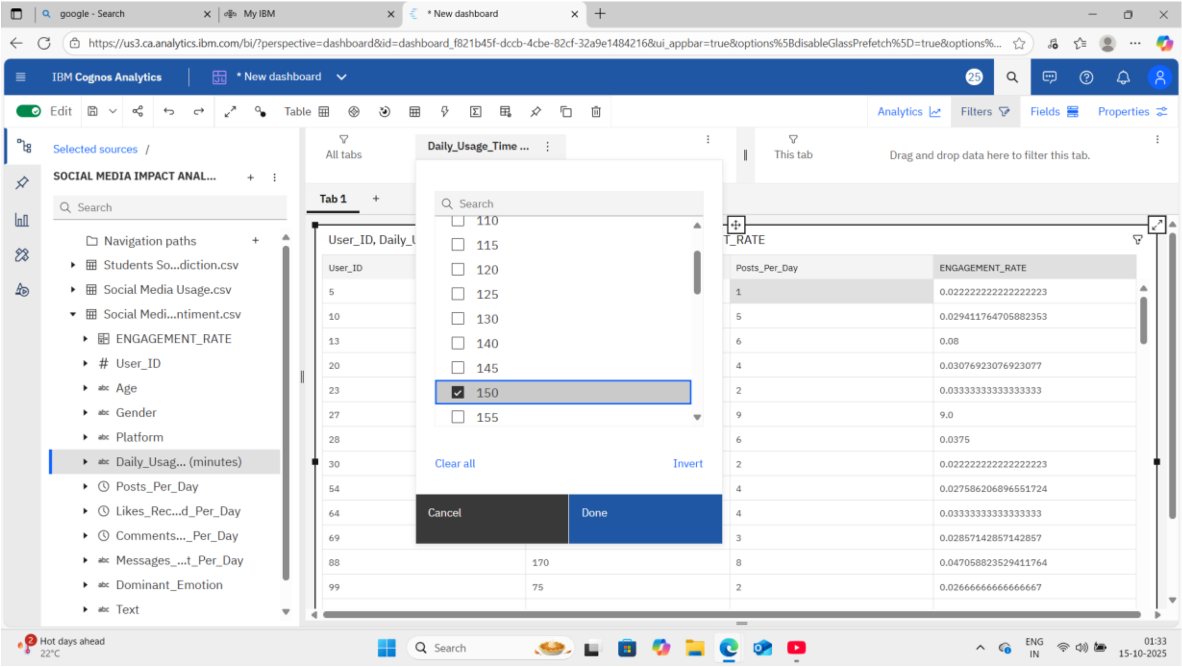


Step 9: Add Filters to find “problem users”

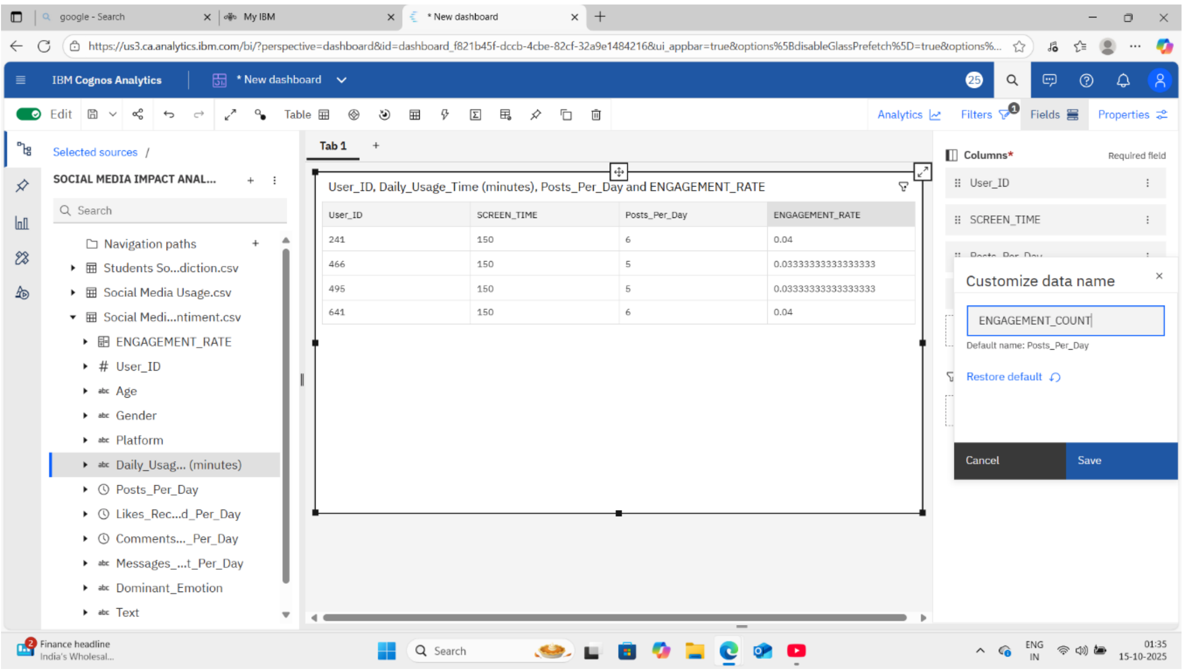


Page

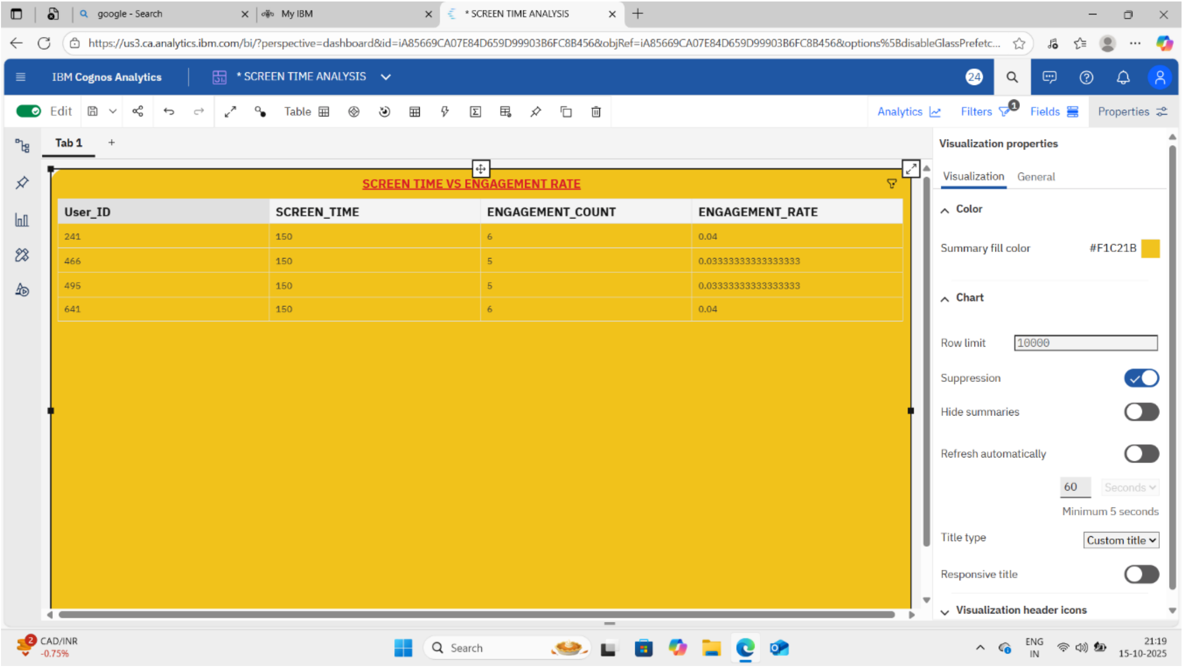
Step 11: ADD filter condition to find the users using more than 150 minutes.



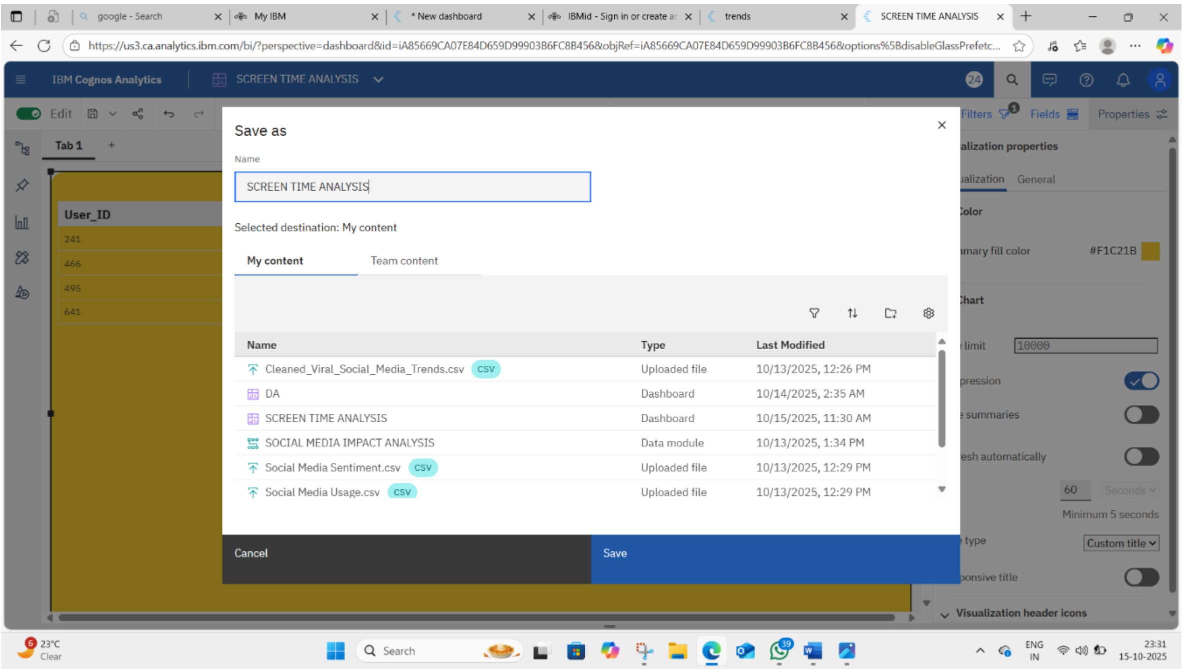
Step 12: Customise data name for better understanding



Step 13: Highlight them with color in properties icon



Step 14: Save the dashboard

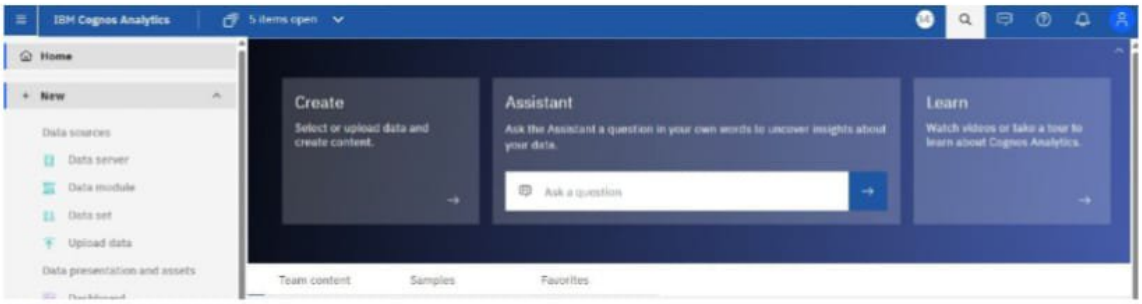


Project

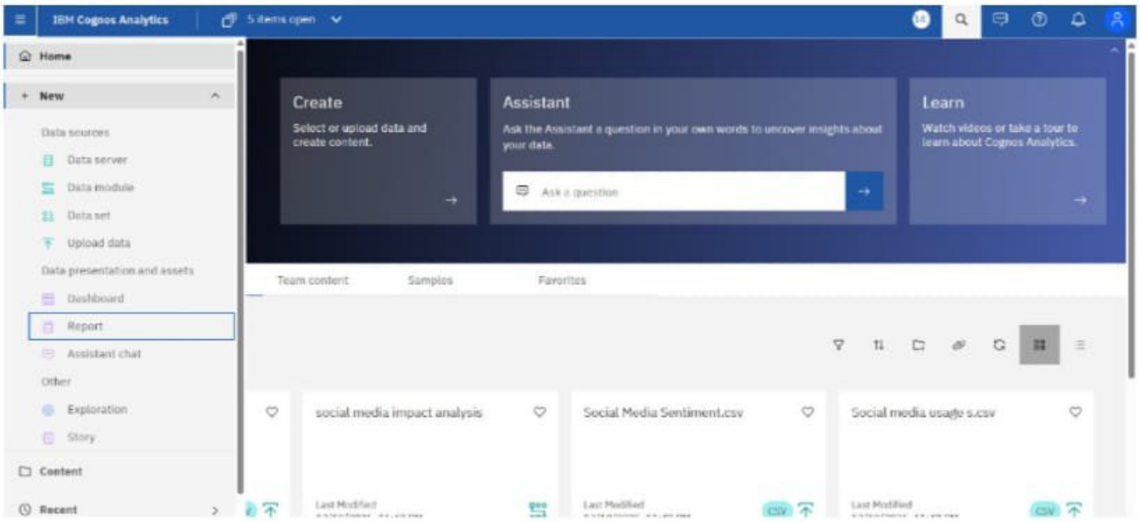
Identify the top 10 most engaging social media posts or trends

Solution: create a list using data module

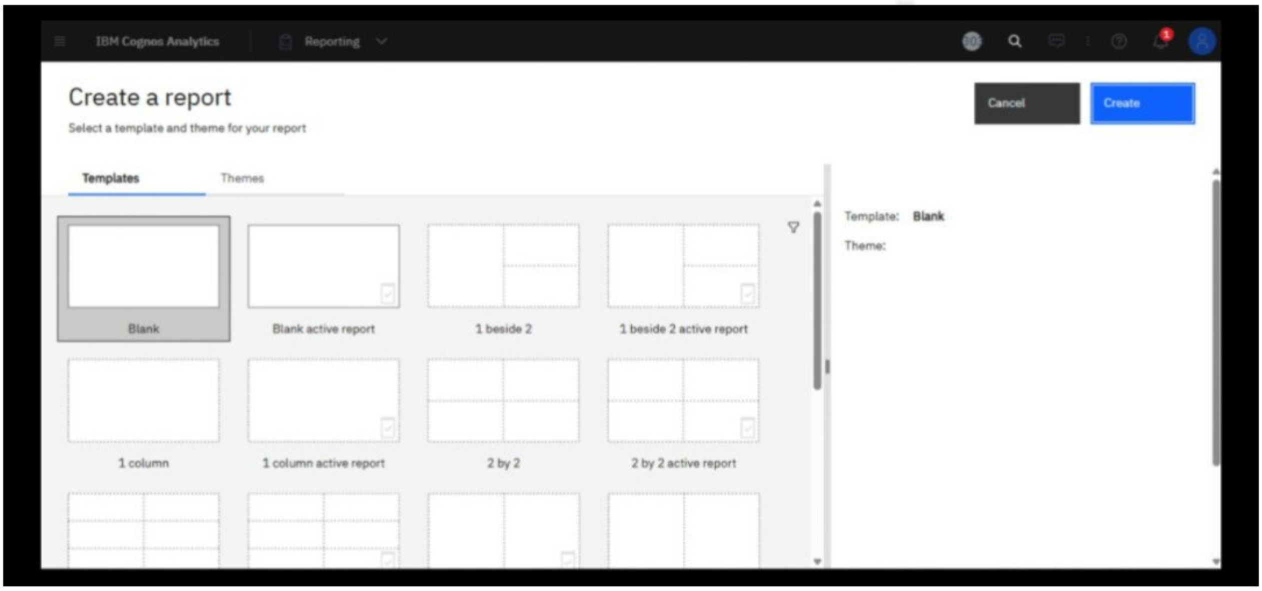
Step1 - Loging to IBM and followed interface will be opened



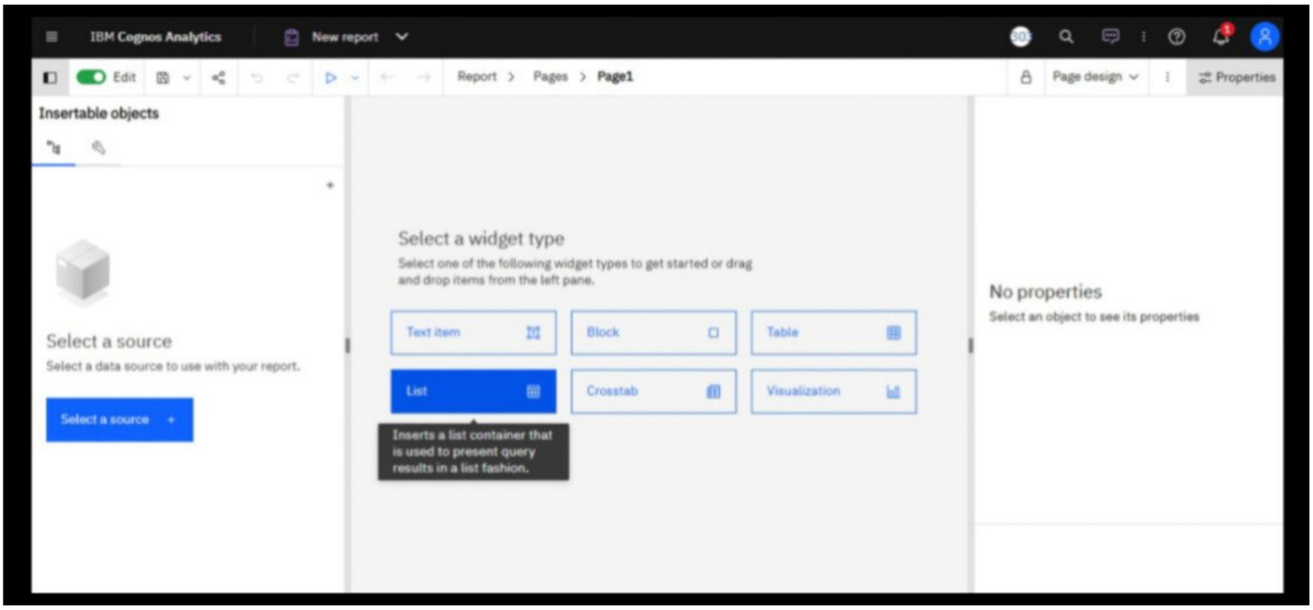
Step2 - Click three line icon and select new ->Report option.



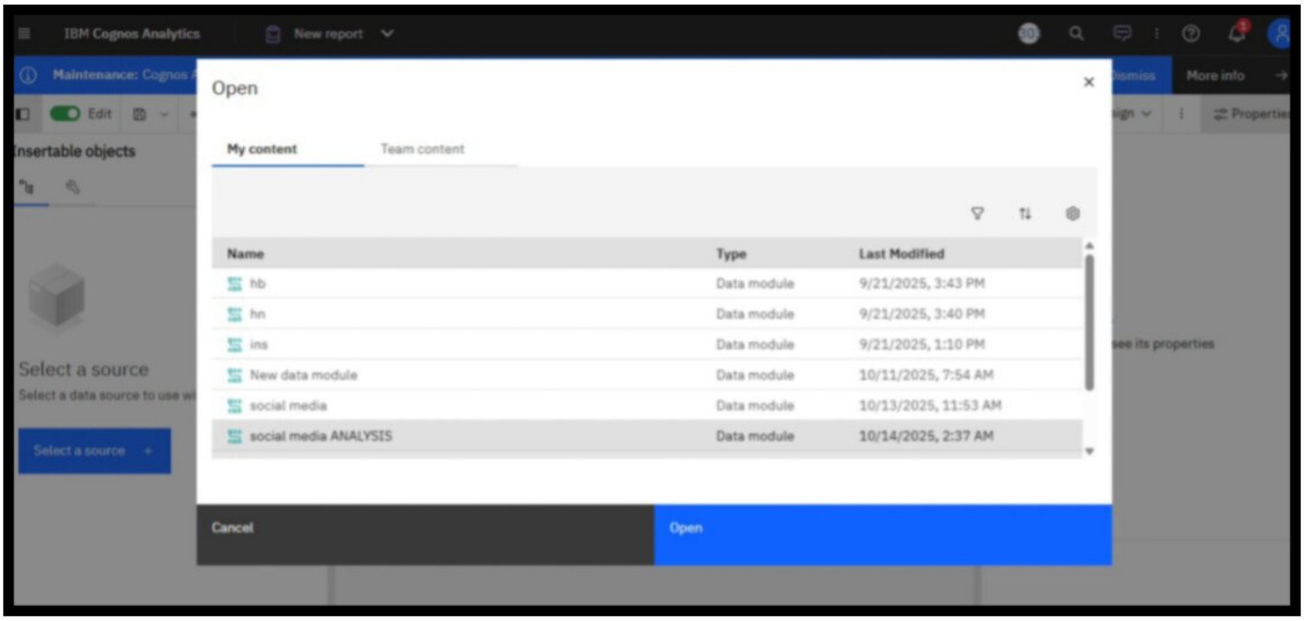
Step3- Click on Blank Report Option and then click on create option.



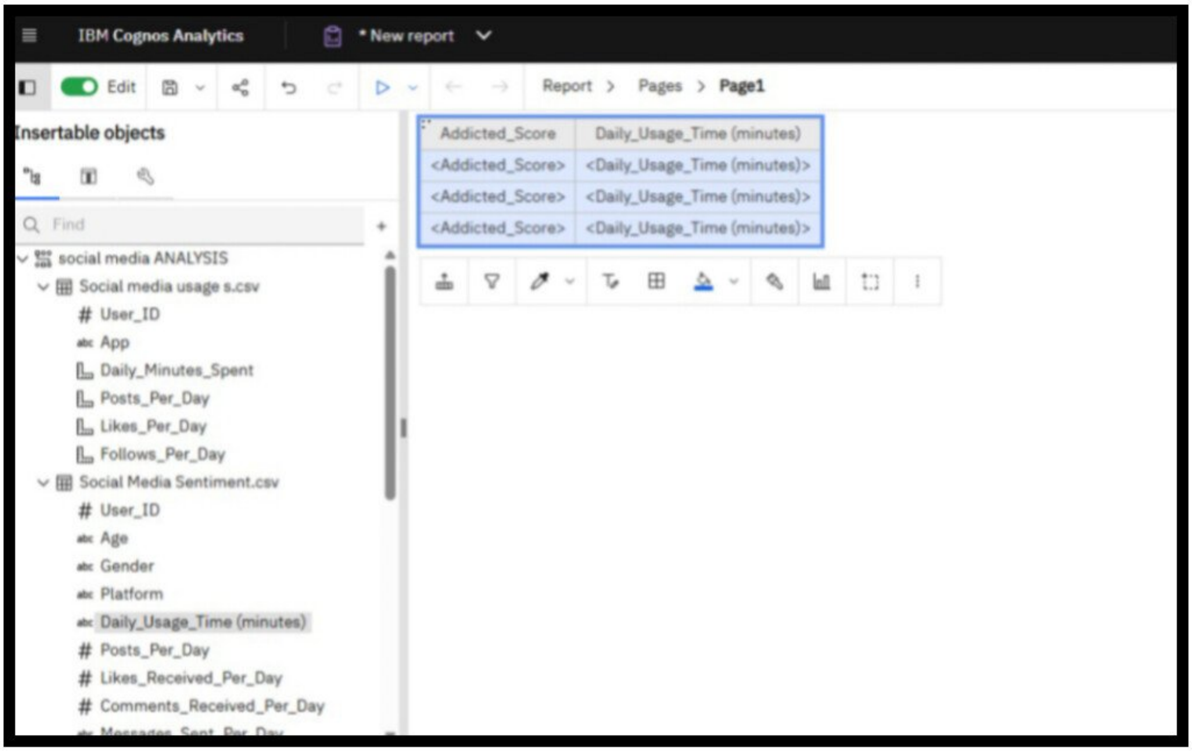
Step4- Select a List Report Option.



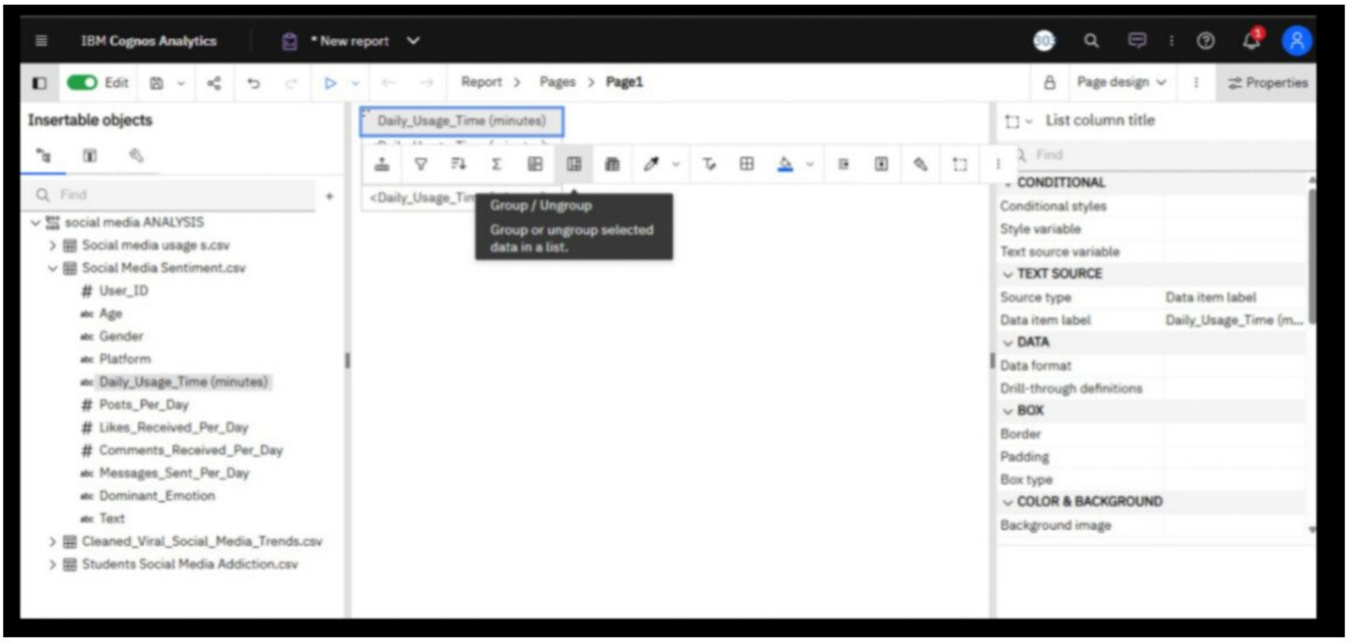
Step 5- Select Data Module from My Content option and Click an Open .



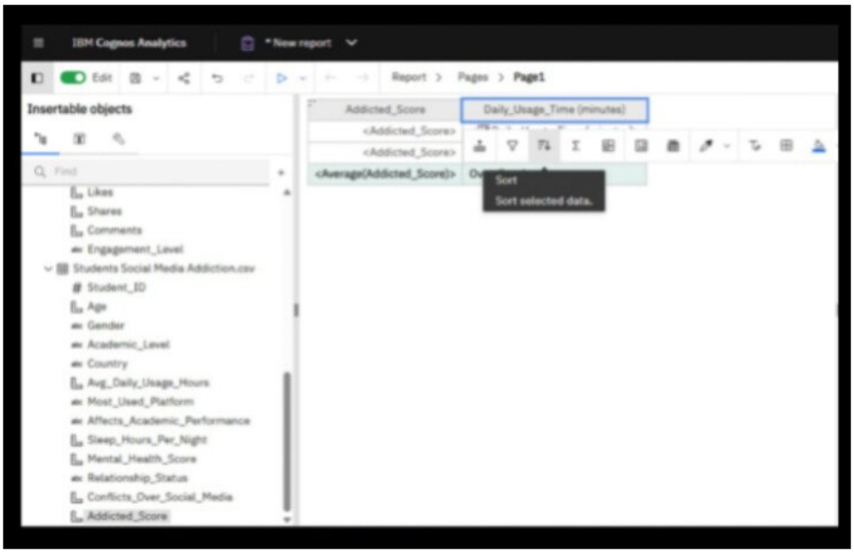
Step 6- Drag Daily Usage Time (minutes) and Addicted Score into the List Report.

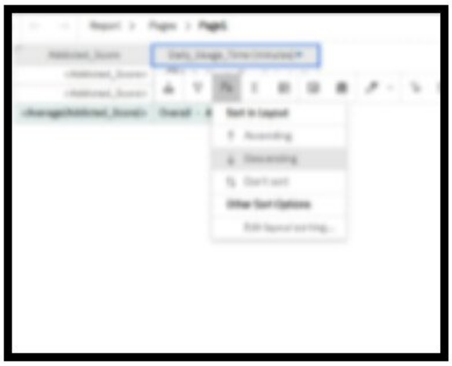


Step7- Click on Group Option .

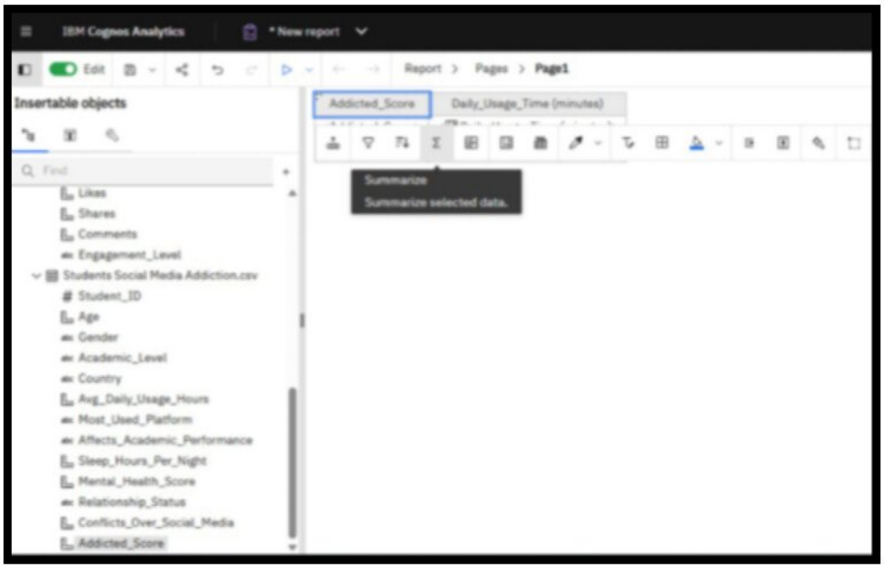


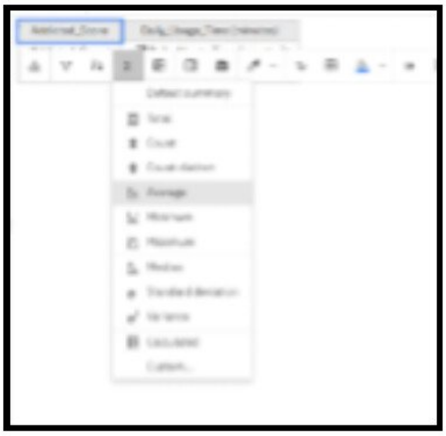
Step 8- Sort or Arrange the Data in Descending Order.



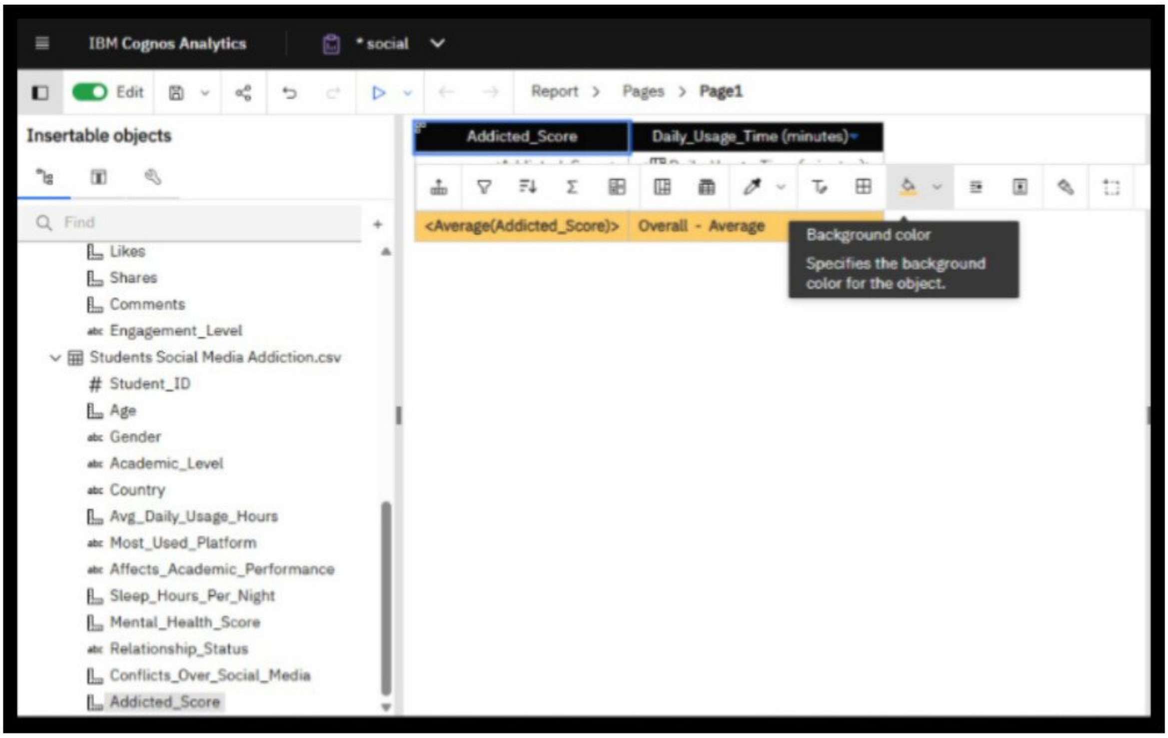


Step 9- Drag Addiction Score into the list .
Right Click the Addiction Score column-> Summarize-> Average .

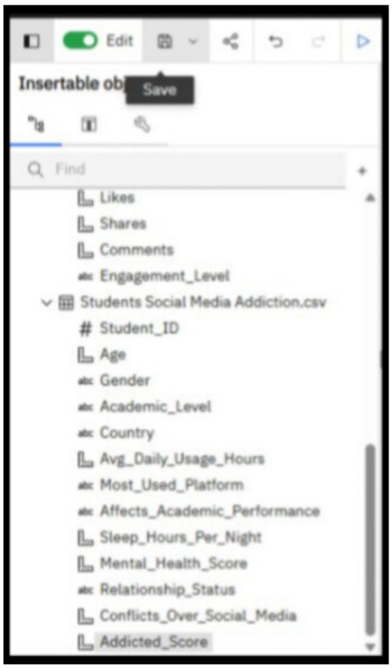


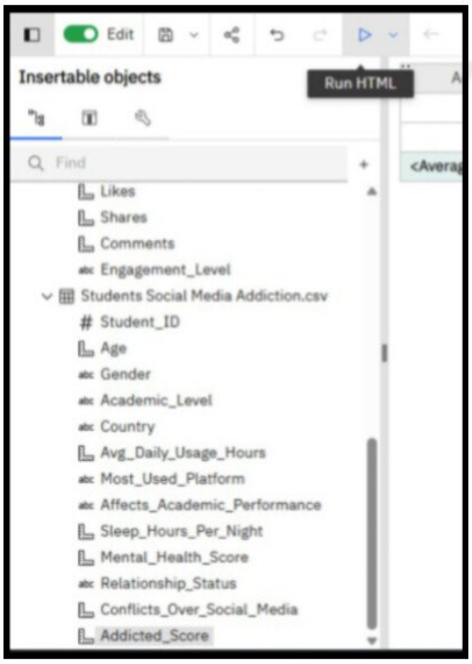


Step 10- Click on Background color for changing the color .



Step 11- Click Save -> Run





You'll see table like this ->

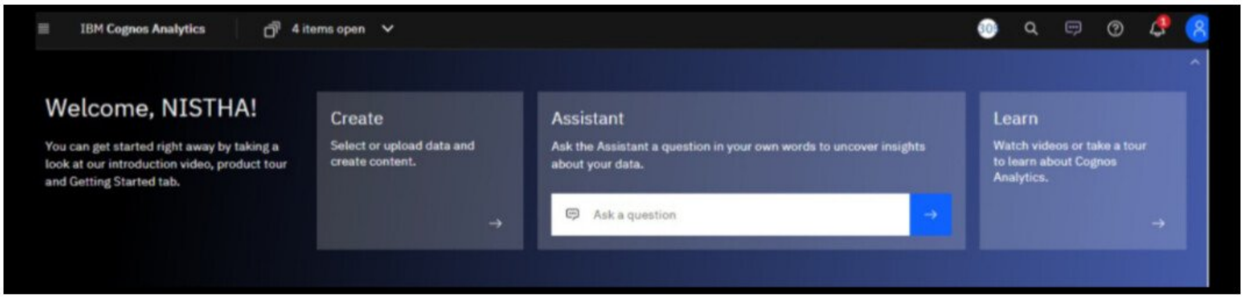
Addicted_Score	Daily_Usage_Time (minutes)
5.62202381	95
5.97415186	90
5.10434783	85
5.66666667	80
5.80184006	75
5.5	72
5.58436214	70
5.66666667	65
5.82252142	60
5.1959799	55
6.55421687	52
5.28802589	50
5.53773585	45
5.78481013	40
3.80952381	30
6.55421687	200
6.56590684	190
6.55421687	180
6.55421687	170
6.55421687	165

Project

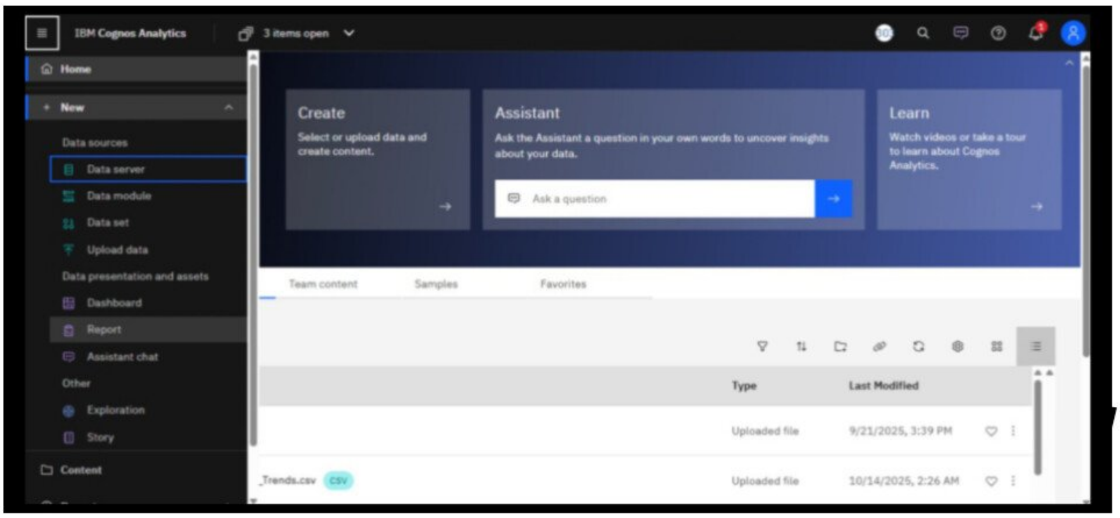
Problem statement: Examine the correlation between daily usage time and addiction score.

Solution: Created a List using Data module

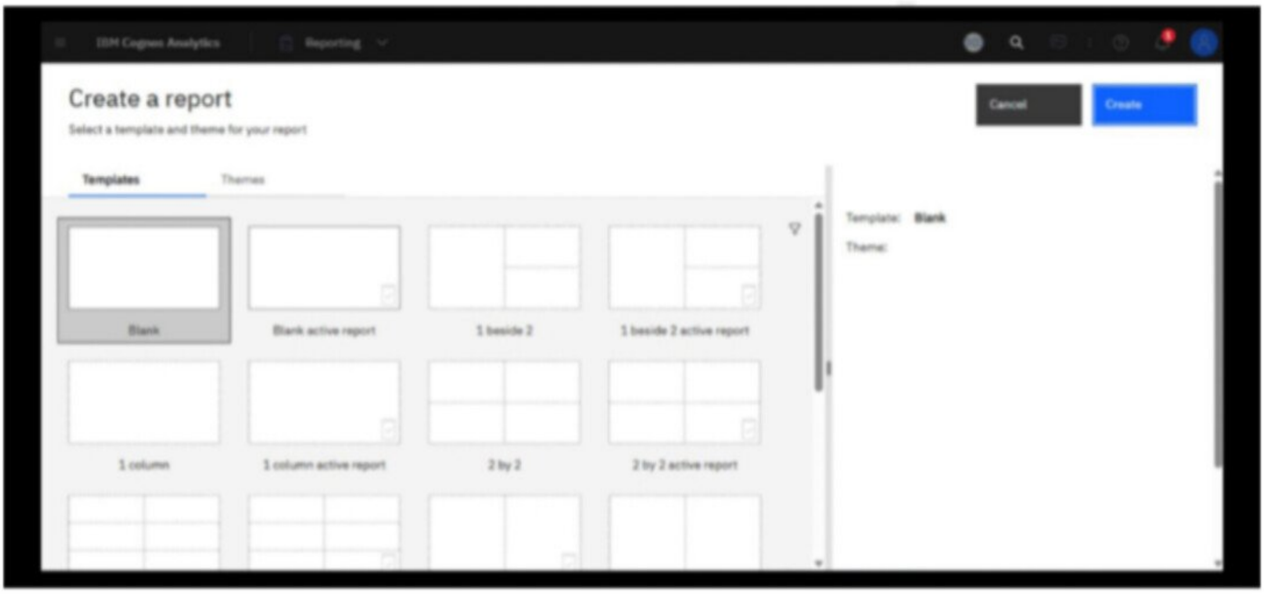
Step1- Login to IBM and followed interface will be opened .



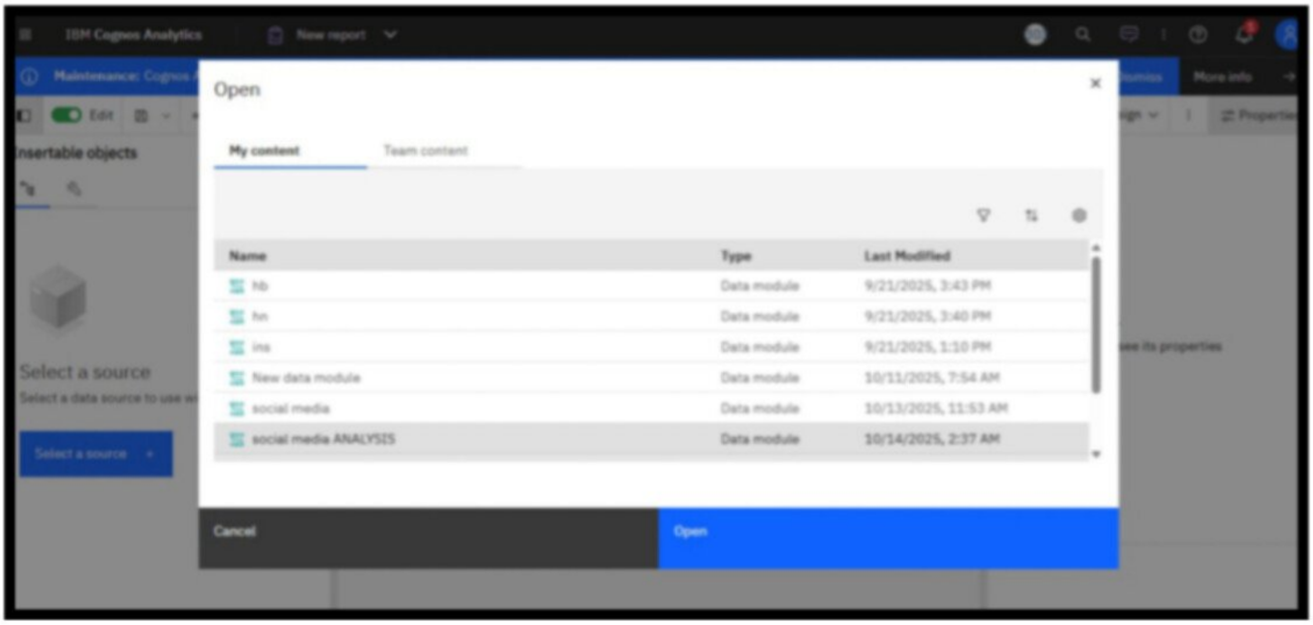
Step2- Click three-line icon and select New -> Report option.



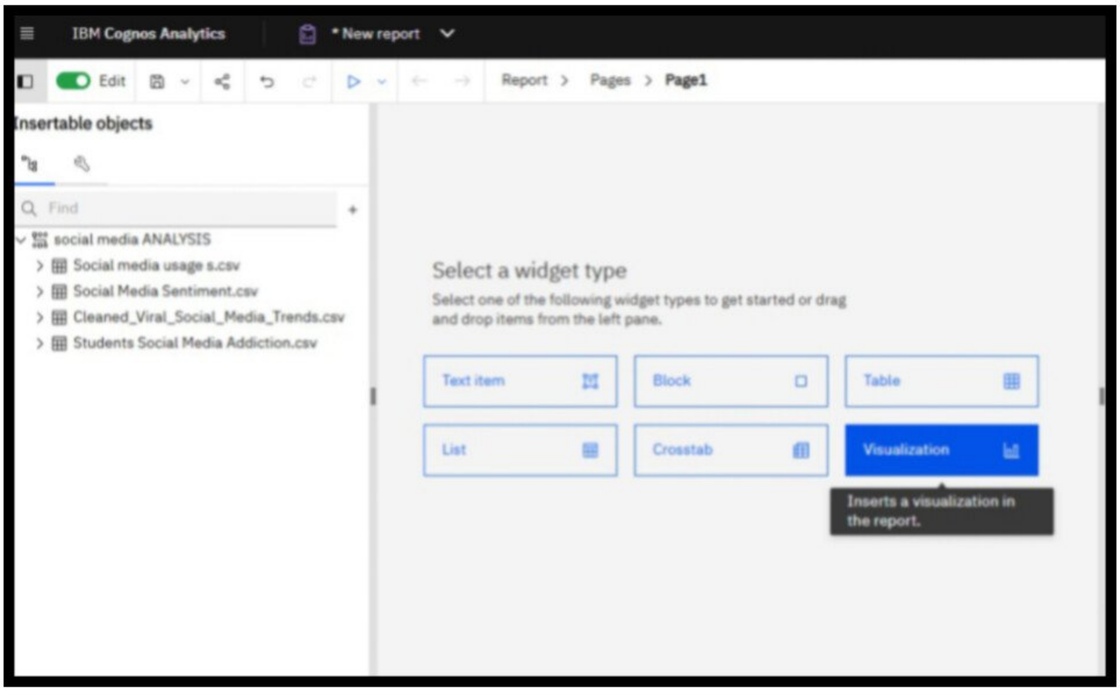
Step3- Click on Blank Report Option and then click on create option.



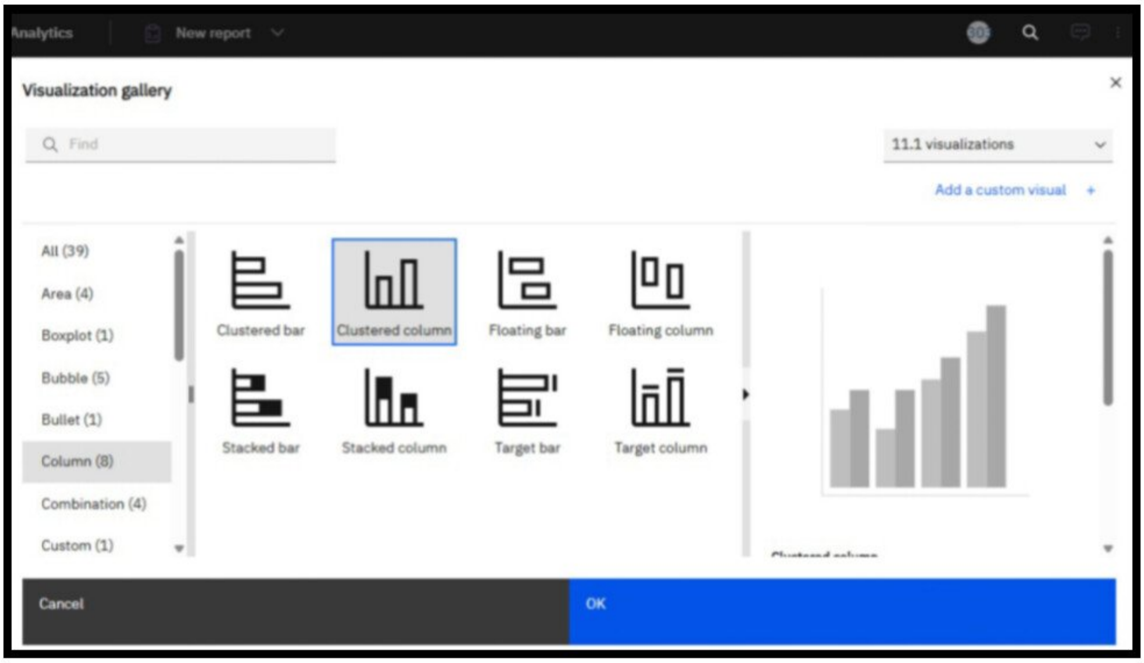
Step 4- Select Data Module from My Content and click an open



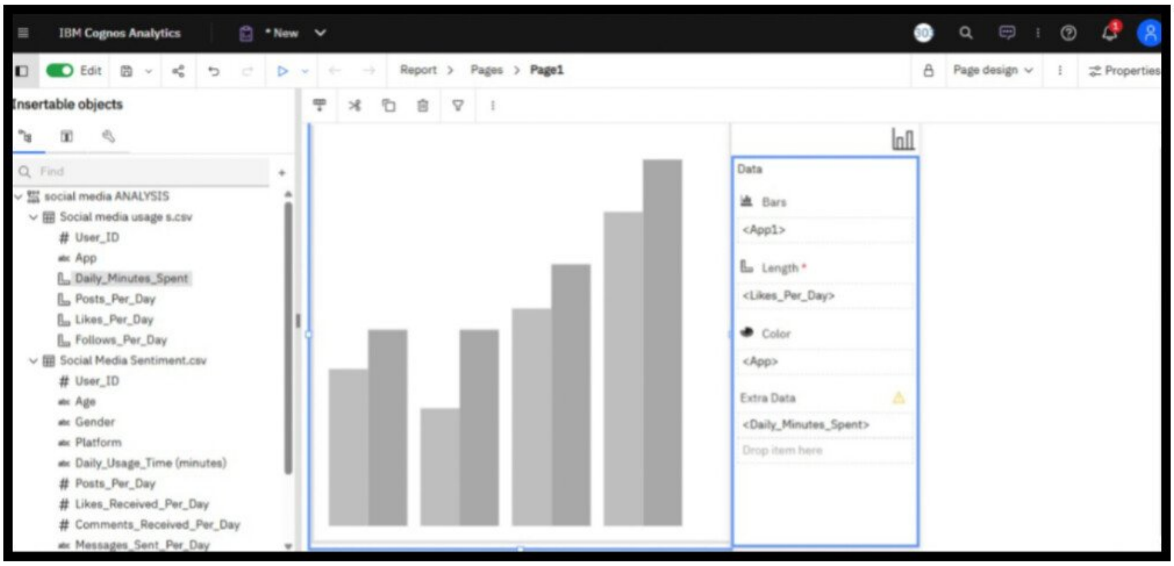
Step 5- Select Visualization widget from the given option.



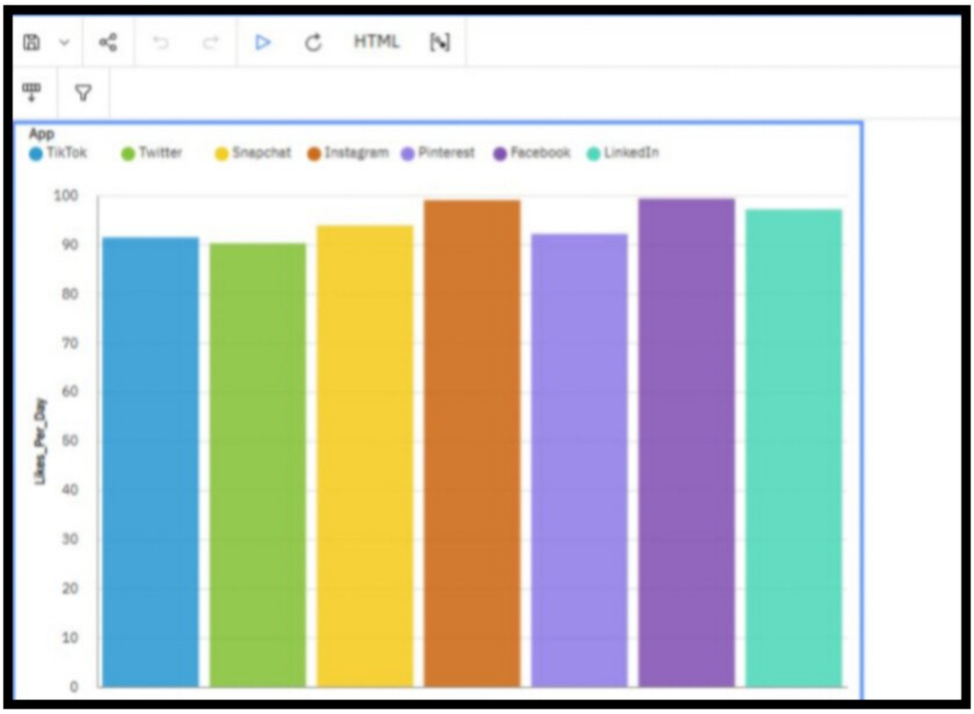
Step 6- Select a Clustered Columns .



Step 7- Drag 'App' from Social Media Usage into Bars and Color, and <Likes_Per_Day> in length.

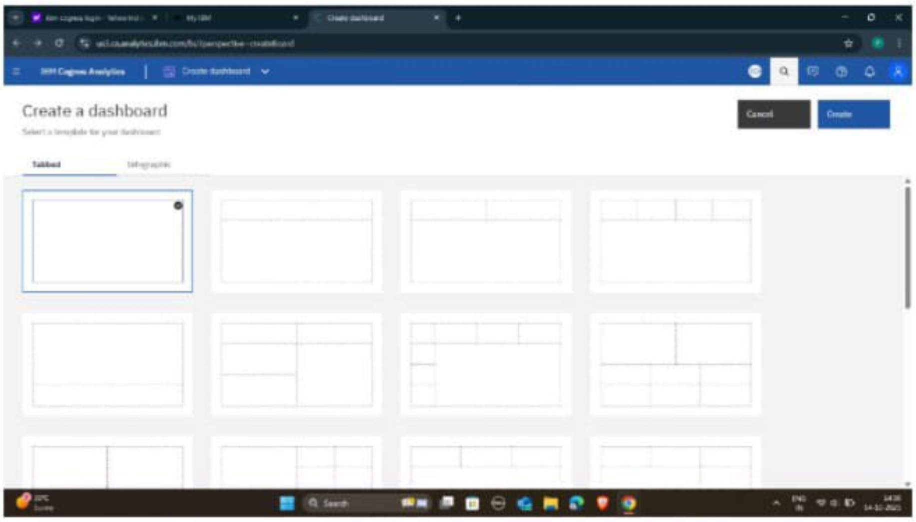


Step 8- Save and Run the report and following output will be generated.

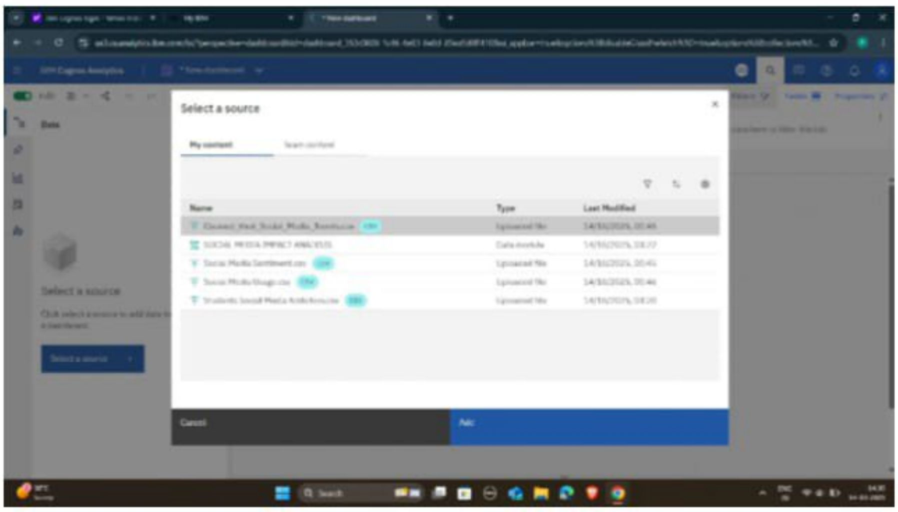


Find the most popular platform based on the number of viral trends the produce.

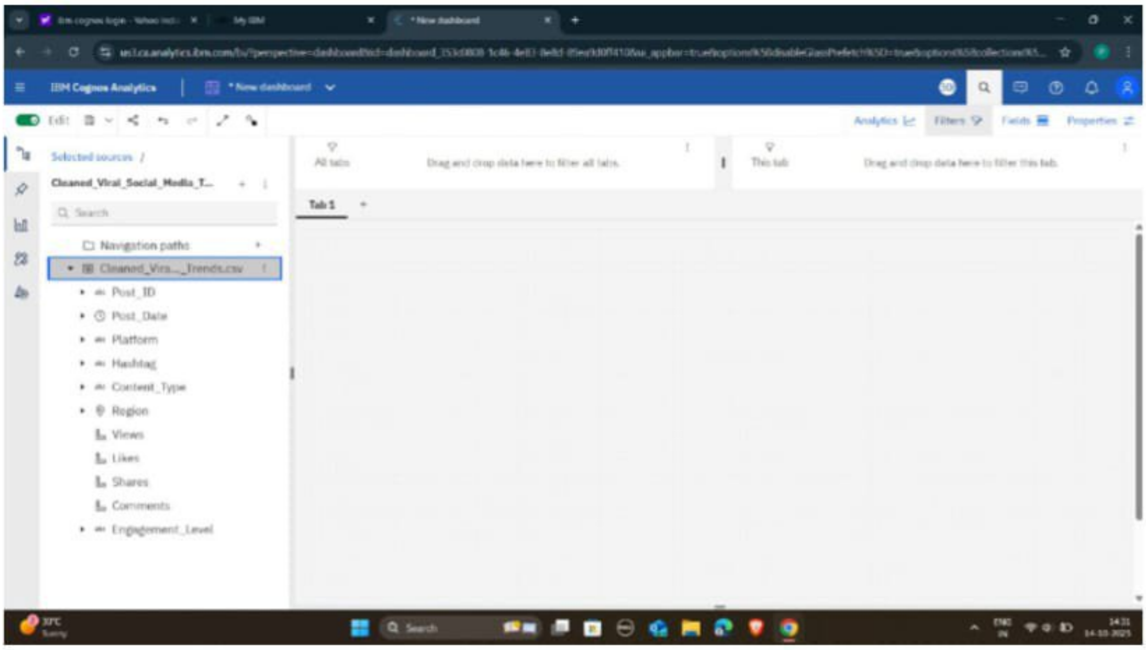
Step1 - Create New Dashboard: Log into IBM Cognos Analytics, select "Create a dashboard,"and choose a tabbed layout template



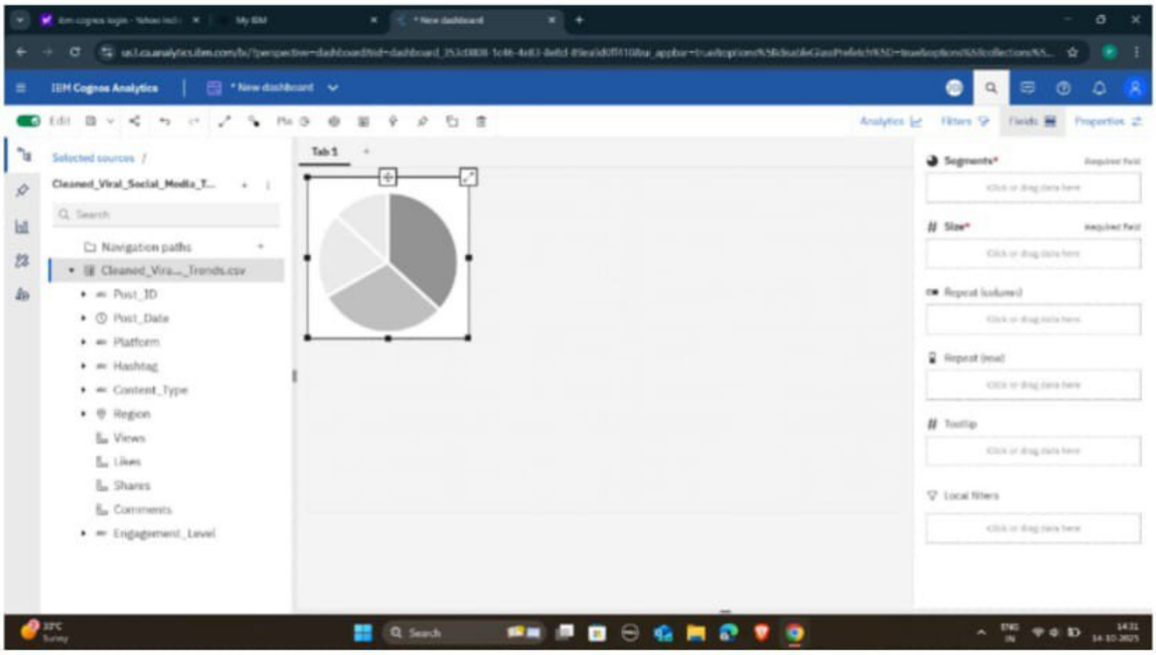
Select Data Source: Select and add the Cleaned_Viral_Social_Media_Trends.csv file as the data source for the dashboard



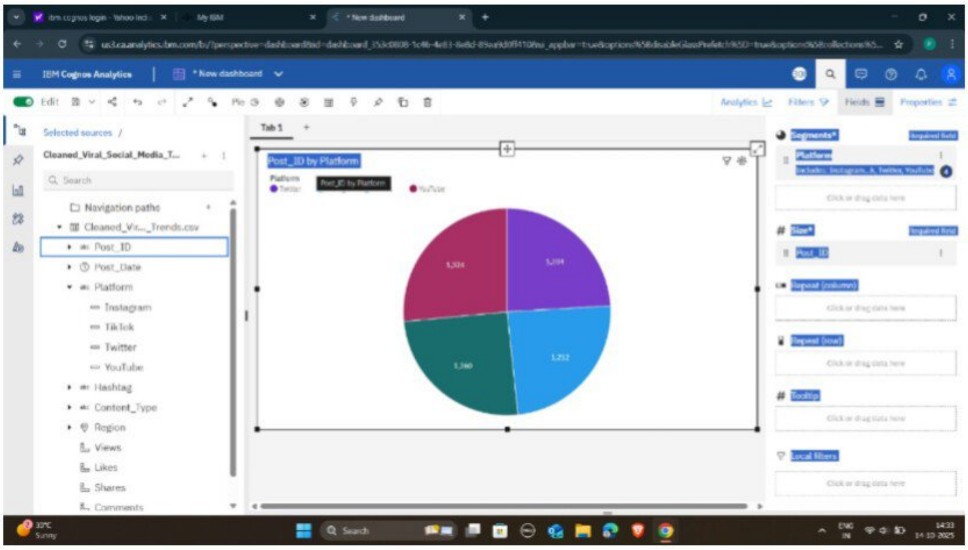
Step3: - Insert the Pie chart visualization onto the dashboard



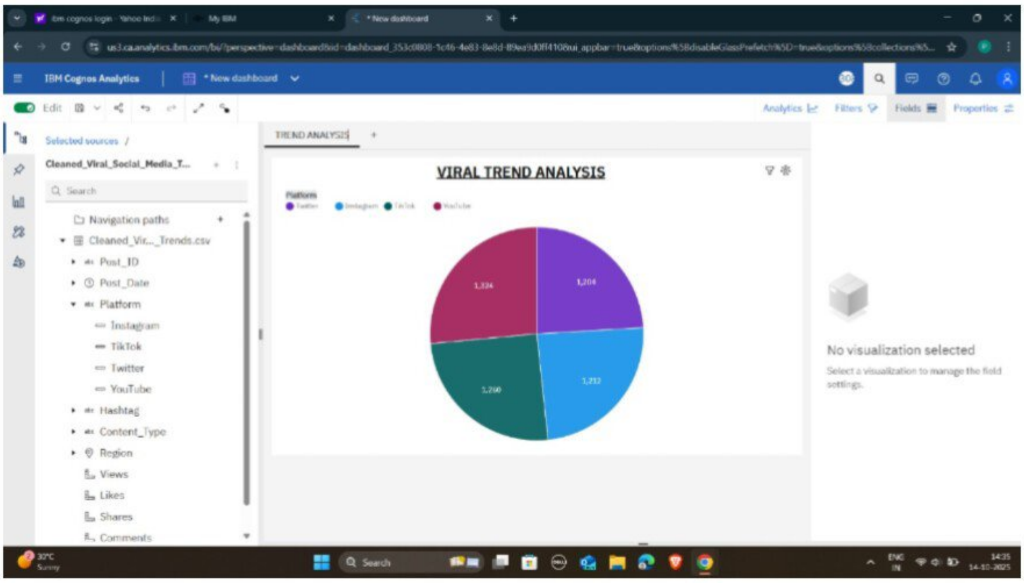
Step4: - Configure the chart by dragging Platform to Segments and Post_ID to Size (Count)



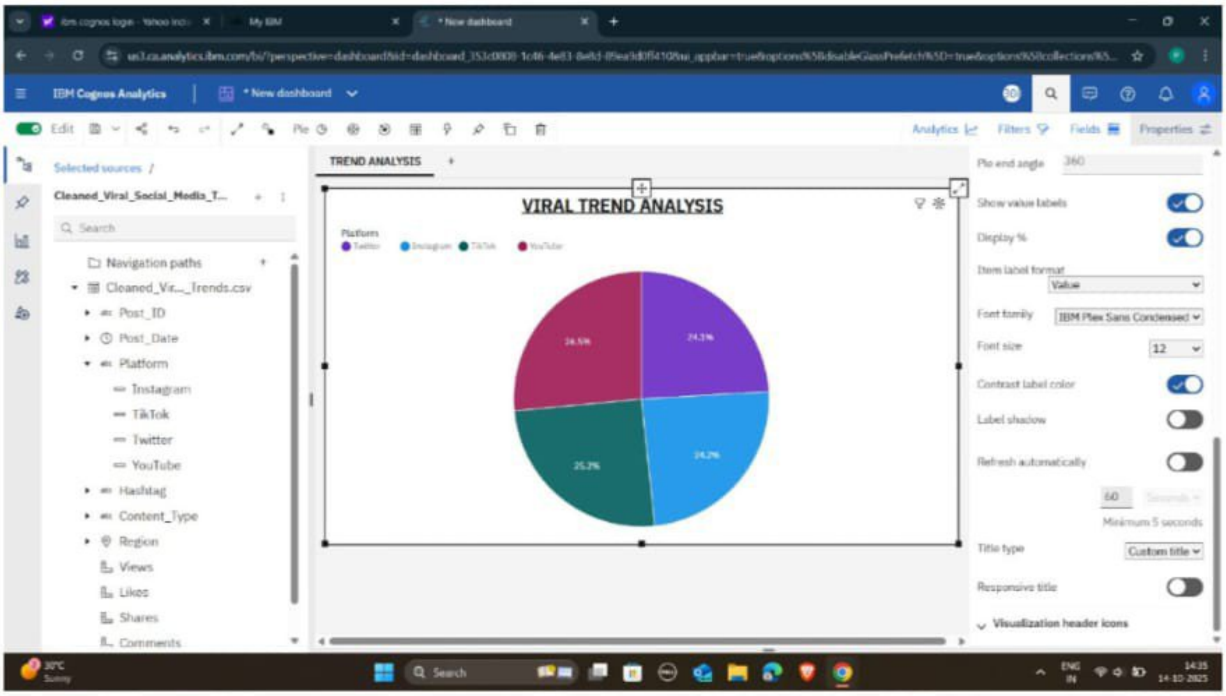
Step5: - Set the tab name to TREND ANALYSIS and the chart title to VIRAL TREND ANALYSIS



Step6: - Enable Display % in the properties pane l to show percentage values on the chart

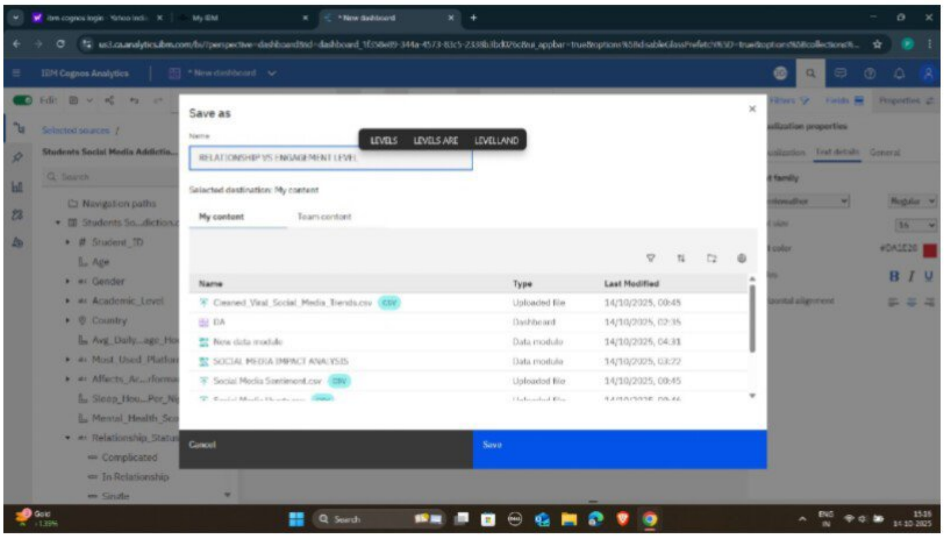


Step7: - Save the completed dashboard with the name TREND ANALYSIS to My content

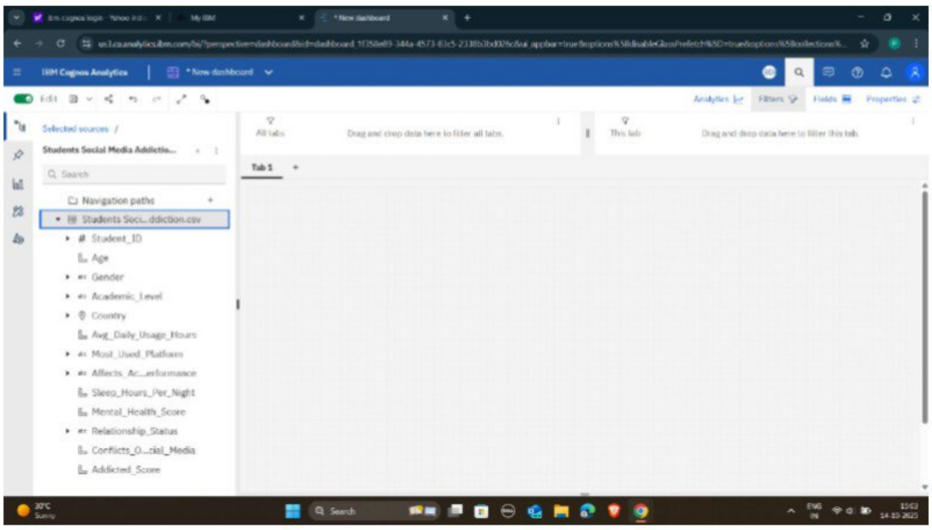


Study how relationship equality changes with increasing social media engagement.

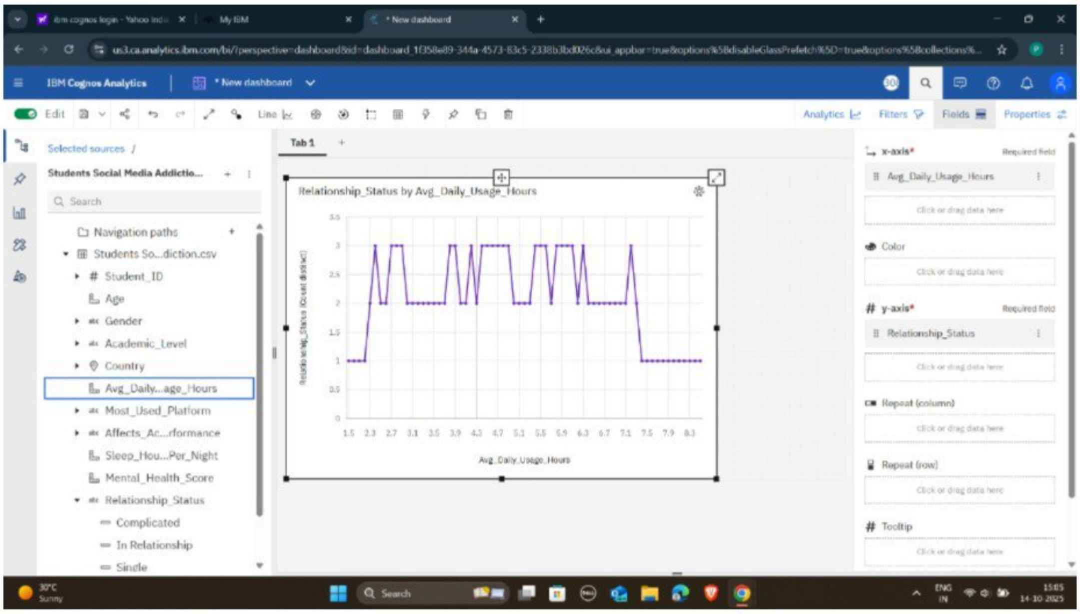
Step1:-Select Data: Load the Students_Social_Addiction.csv data file for analysi



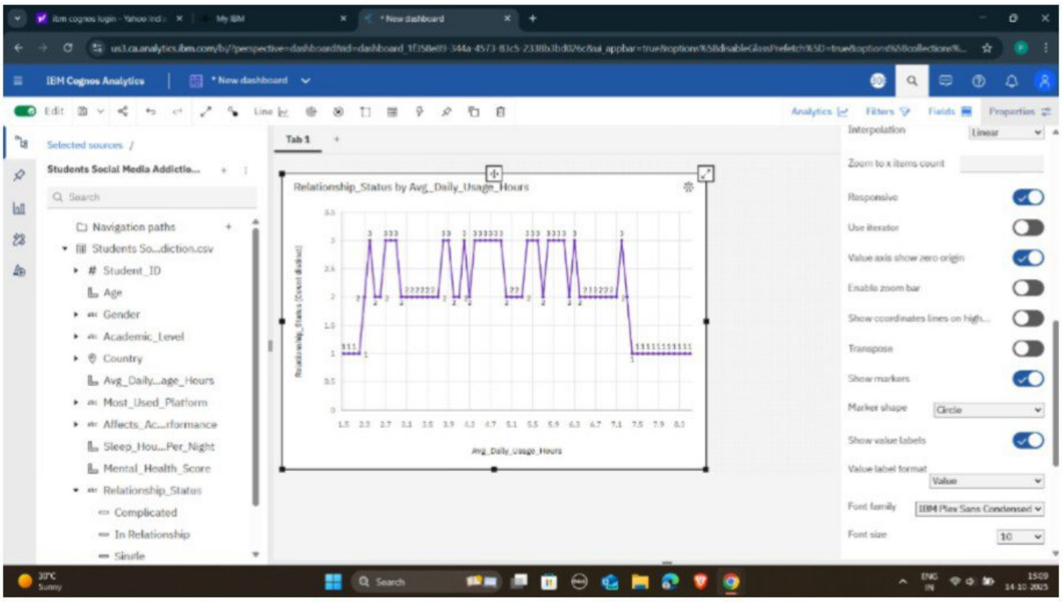
Step2:-Create Chart: Initialize a Line chart in a new dashboard.



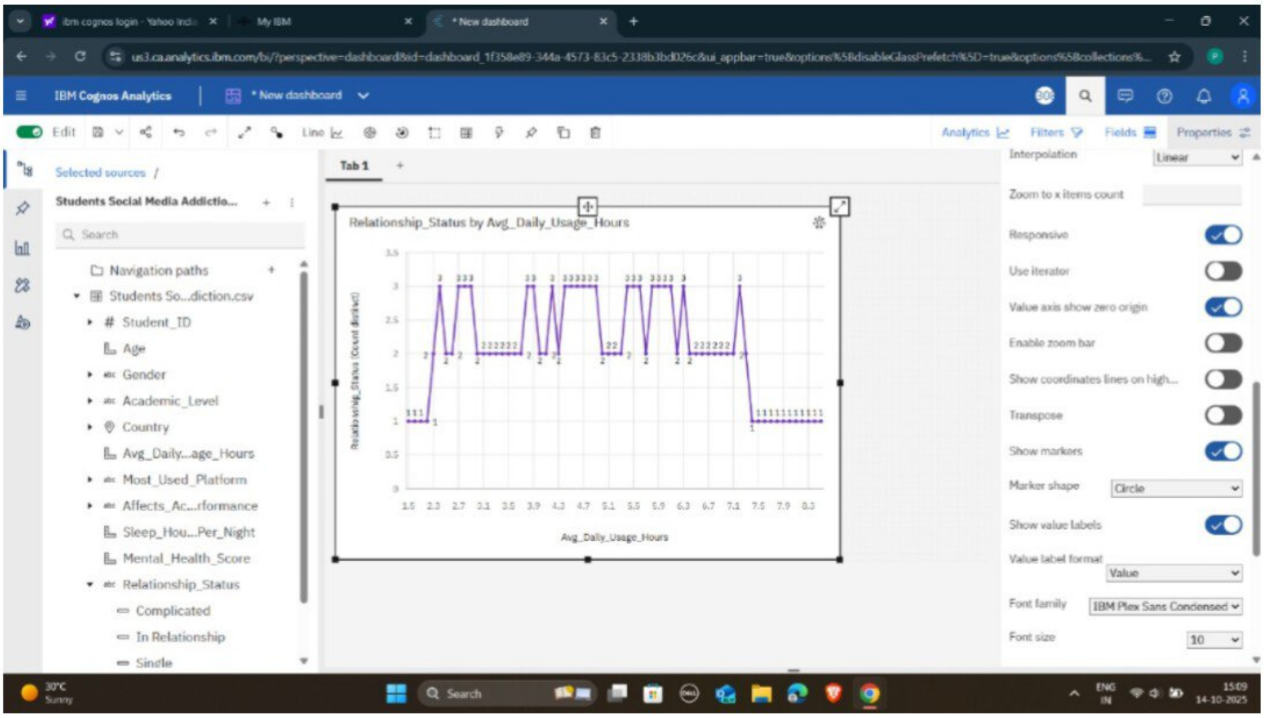
Step3:-Map Variables: Assign Avg_Daily_Usage_Hours to the X-axis and Relationship_Status (Count) to the Y-axis.



Step4:- Refine Visualization: Apply visualization properties such as showing value labels and setting the marker shape to 'Circle'.

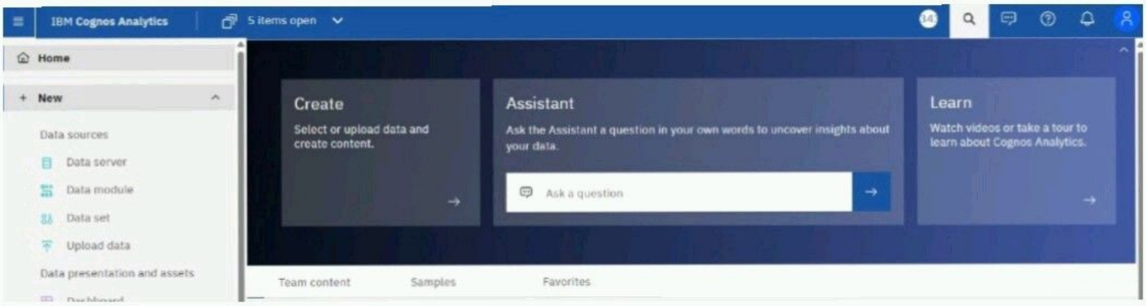


Save Dashboard: Save the final visualization with the title RELATIONSHIP VS ENGAGEMENT LEVEL.

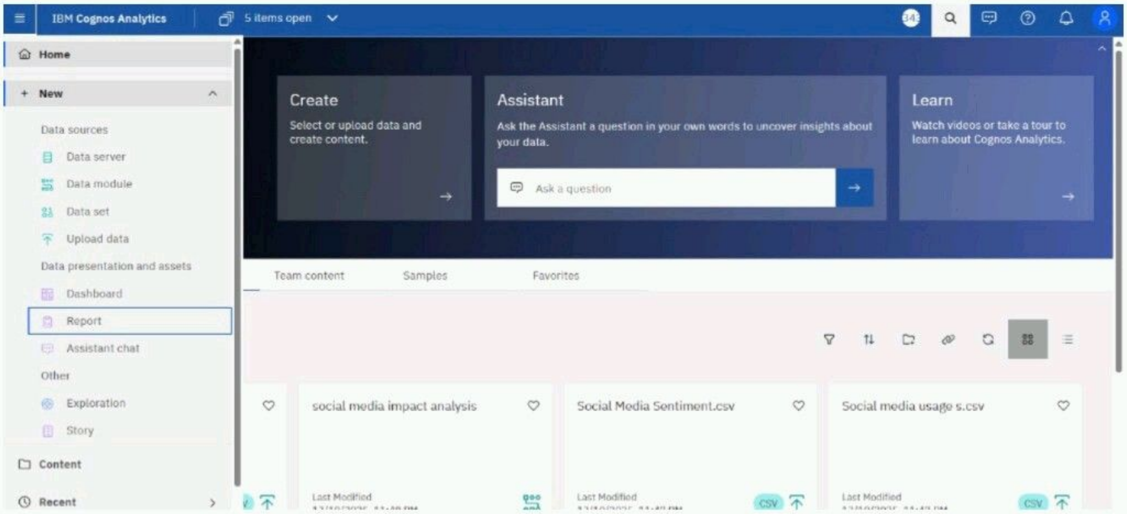


Determine the effect of social media addiction on emotional well-being scores.

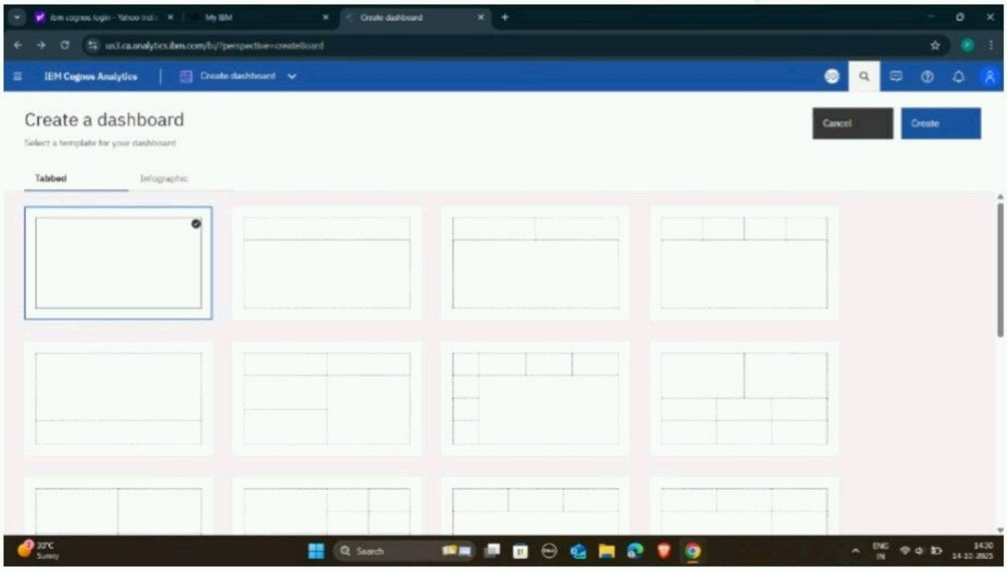
Step:1 - Loging to IBM and followed interface will be opened



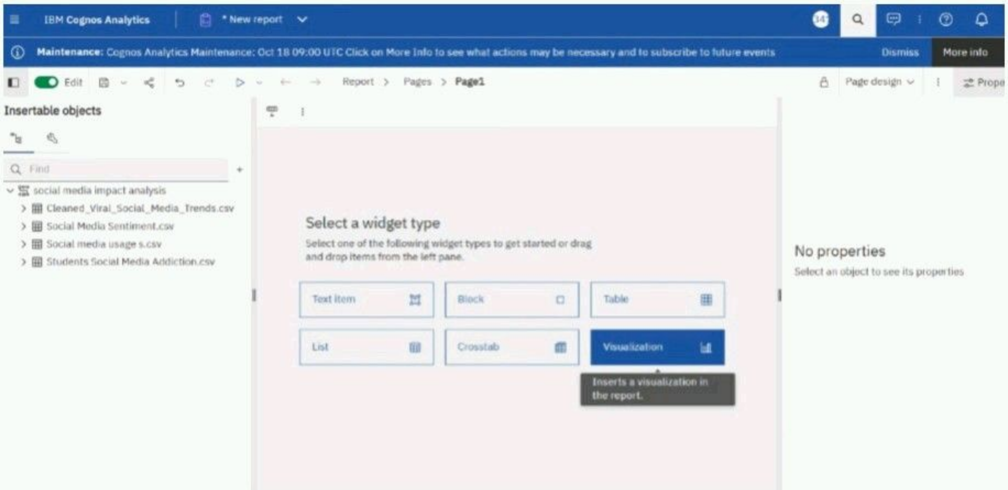
Step:2 - Click three line icon and select new -> Report option.



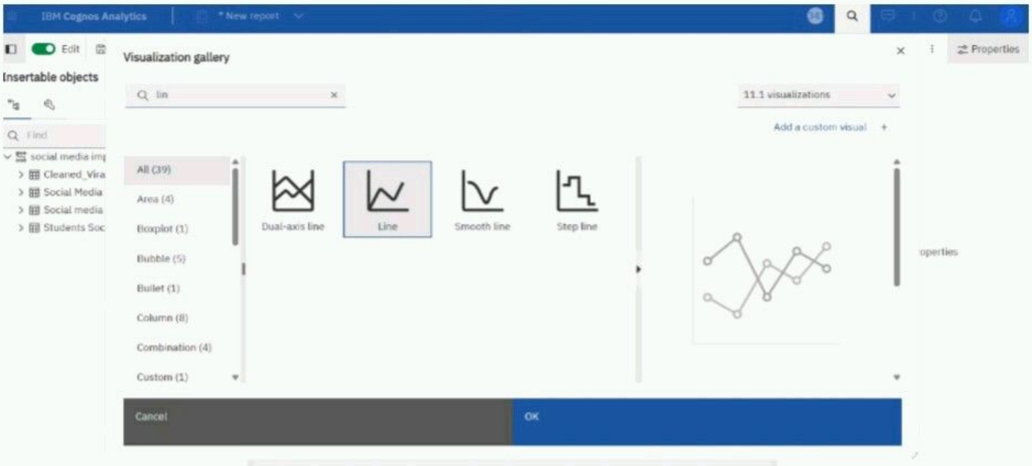
Step:3 - Click on Blank Report option and then click on create option.



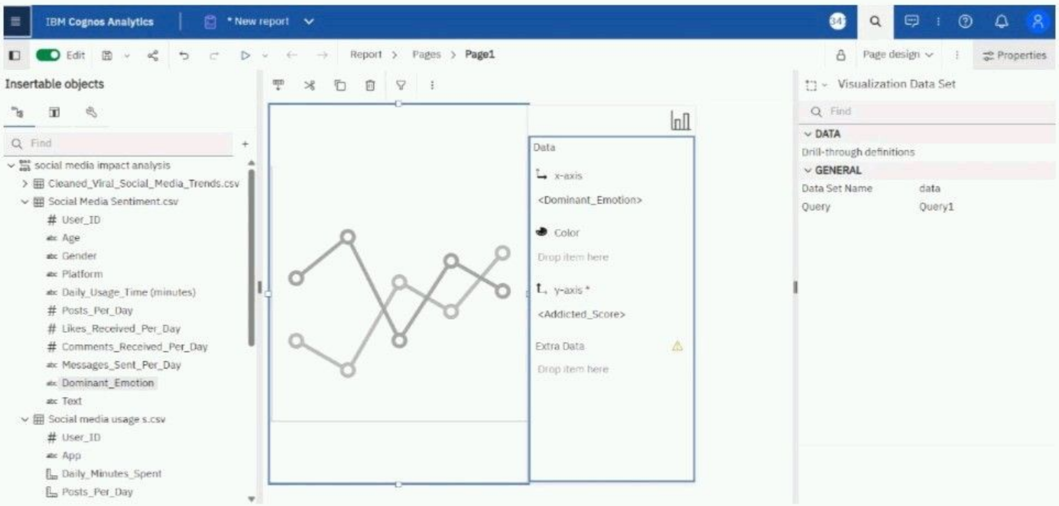
Step:4 - Select a visualization Report option.



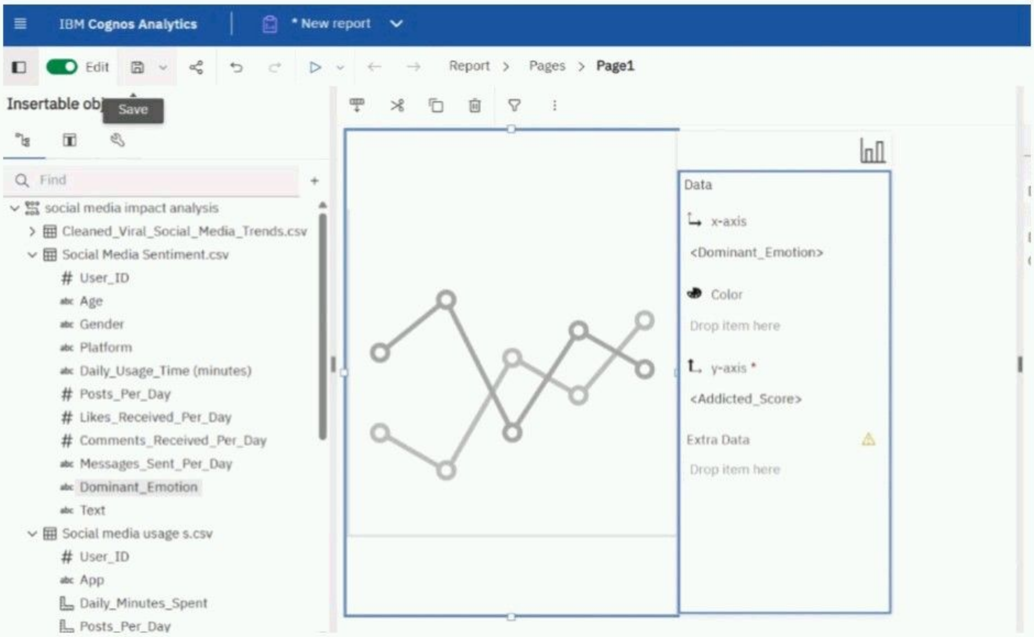
Step:5 - Select the Line chart from the Visualization gallery and click OK.



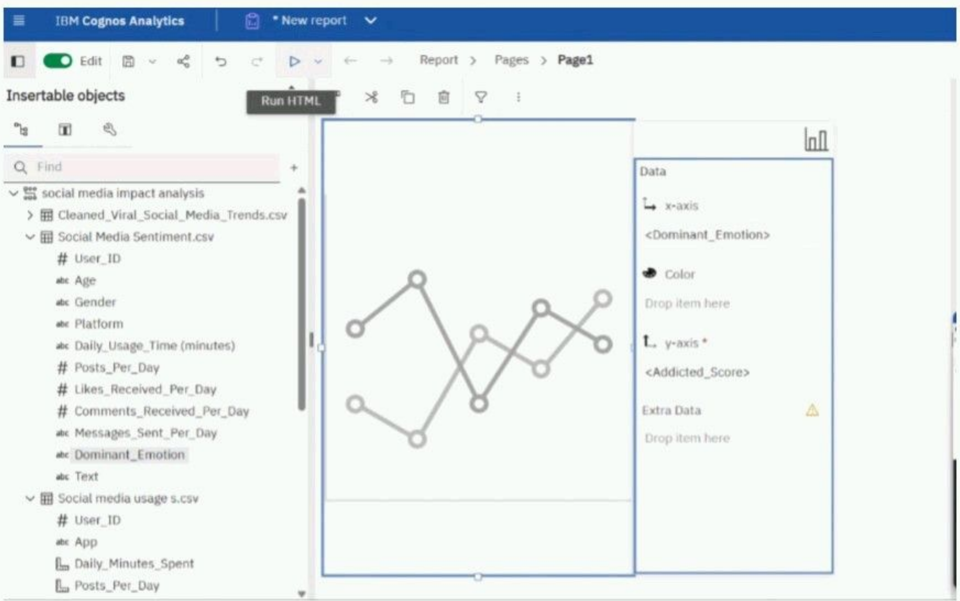
Step:6 - Drag <Dominant_Emotion> to the X-axis and <Addicted_Score> to the Y-axis* data slots.



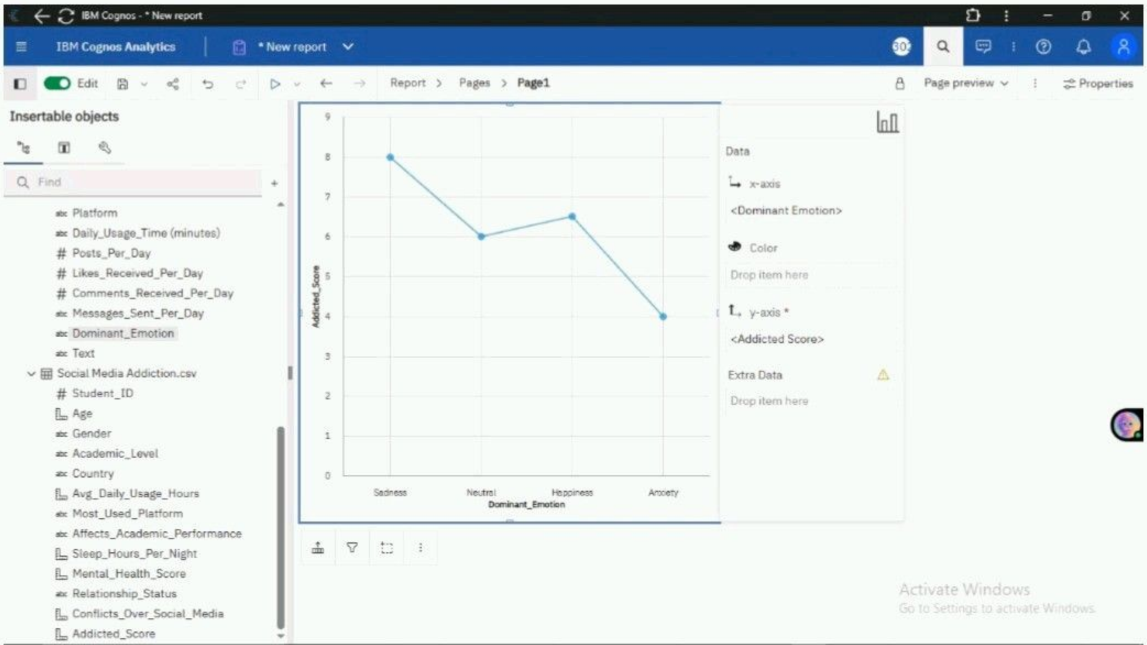
Step:7 - Click on the save to store the report.



Step:8 - Click on the Run to execute the Report.



Review the final line visualization showing the Addicted Score trend across different Dominant Emotions.

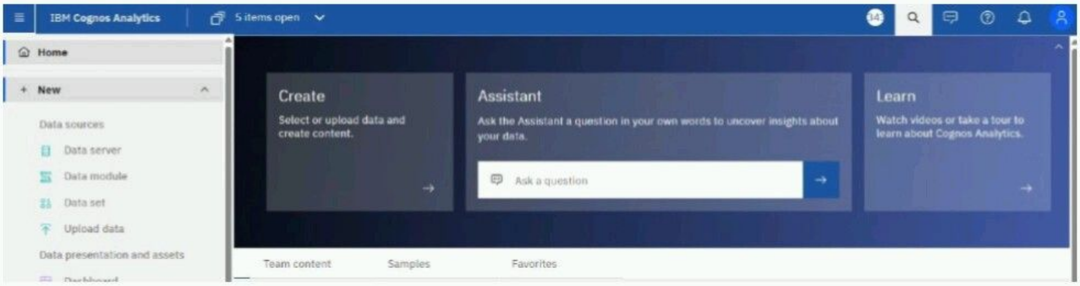


Project

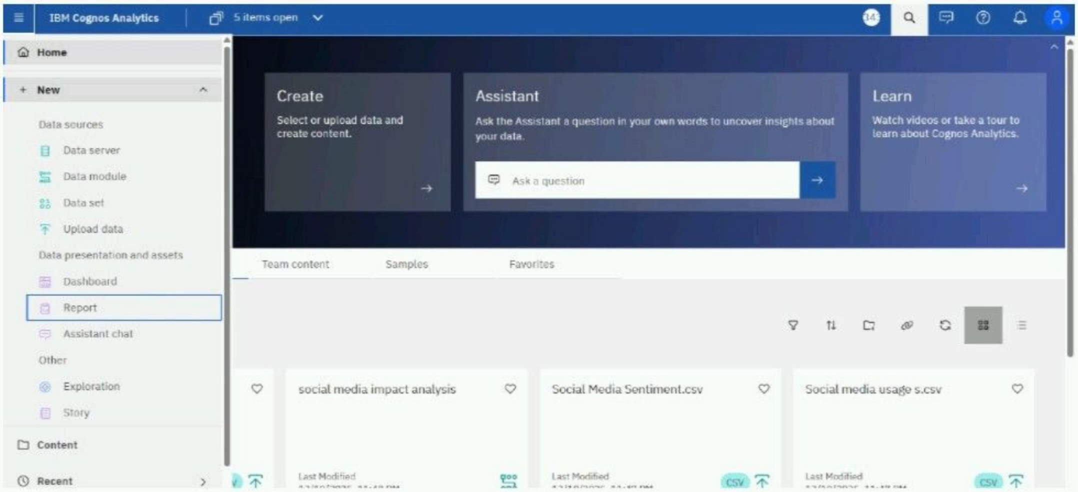
Identify the top 10 most engaging social media posts or trends

Solution: create a list using data module

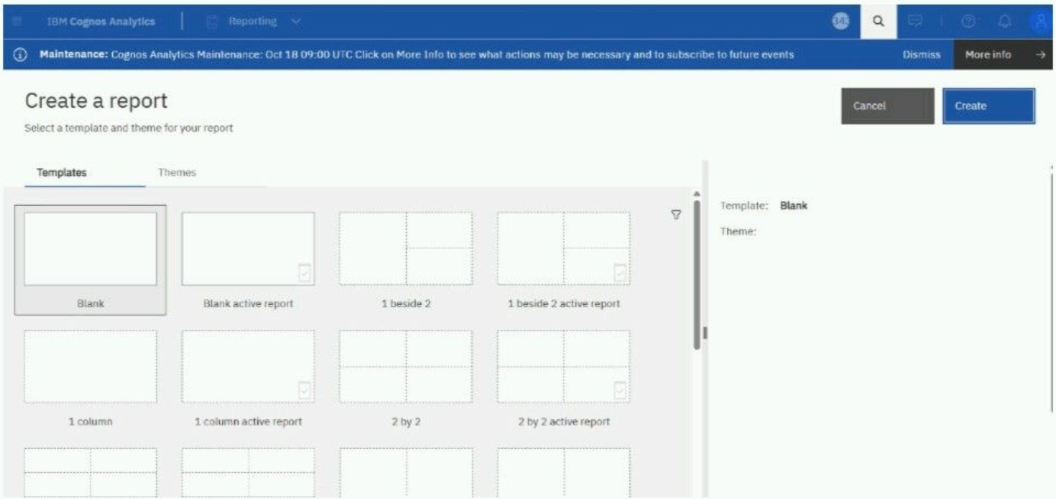
Step1 - Loging to IBM and followed interface will be opened



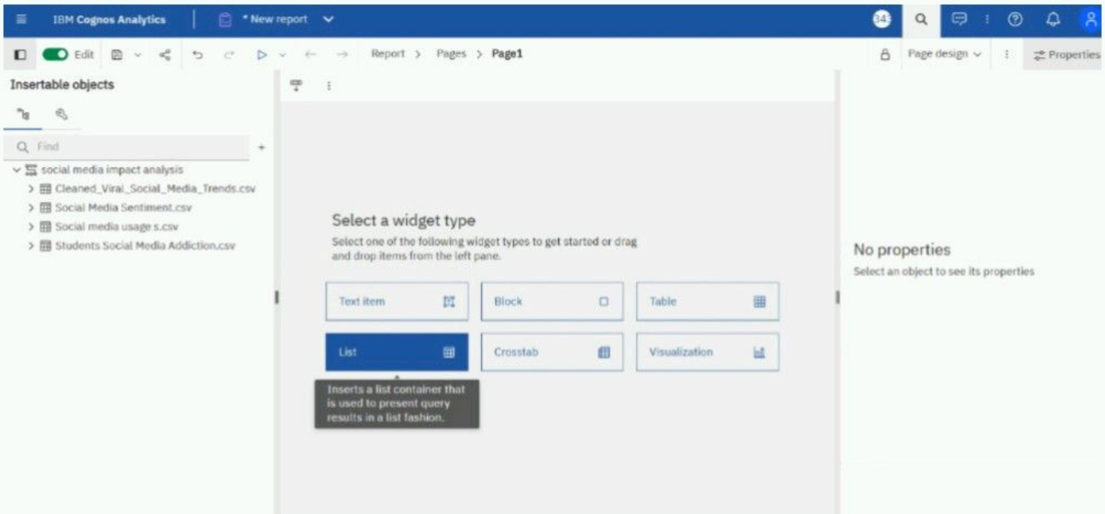
Step2 - Click three line icon and select new ->Report option.



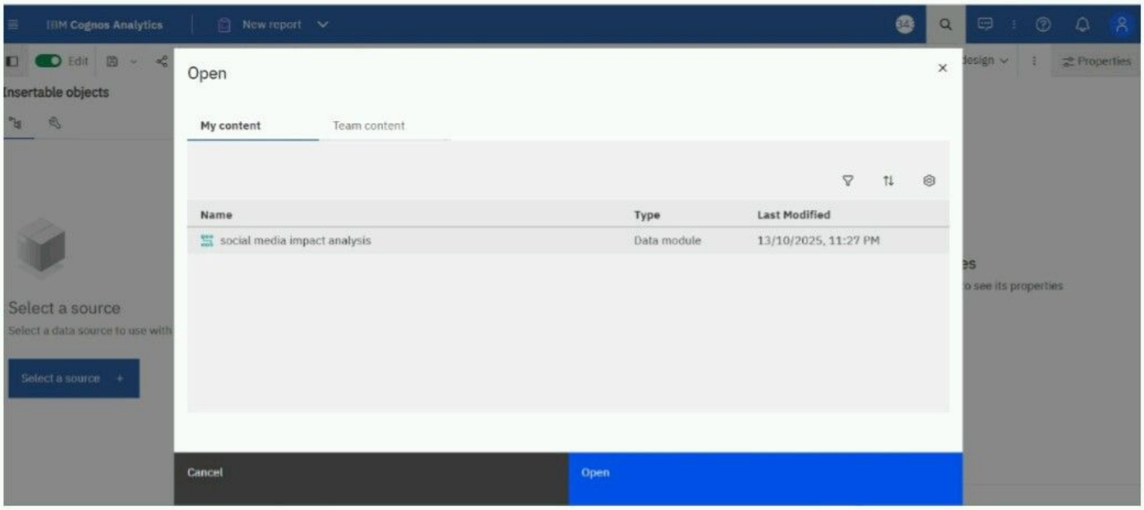
Step3 - Click on Blank Report option and then click on create option.



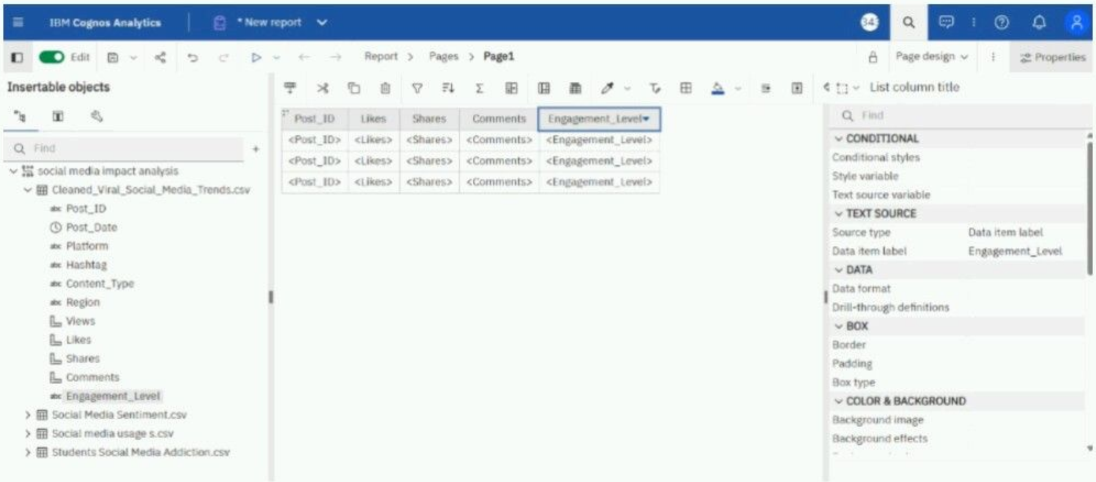
Step4 - Select a List Report Option



Step5 - Select Data module from my content option and click open.

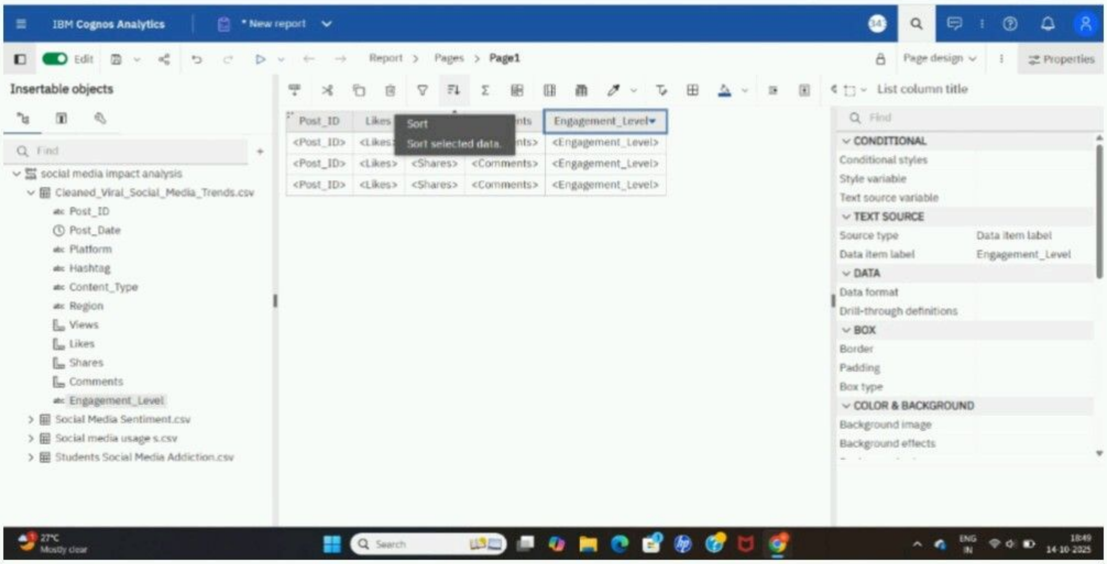


Step6 - Drag post_ID ,Likes, Share , Comments and Engagement _Level into the List Report.

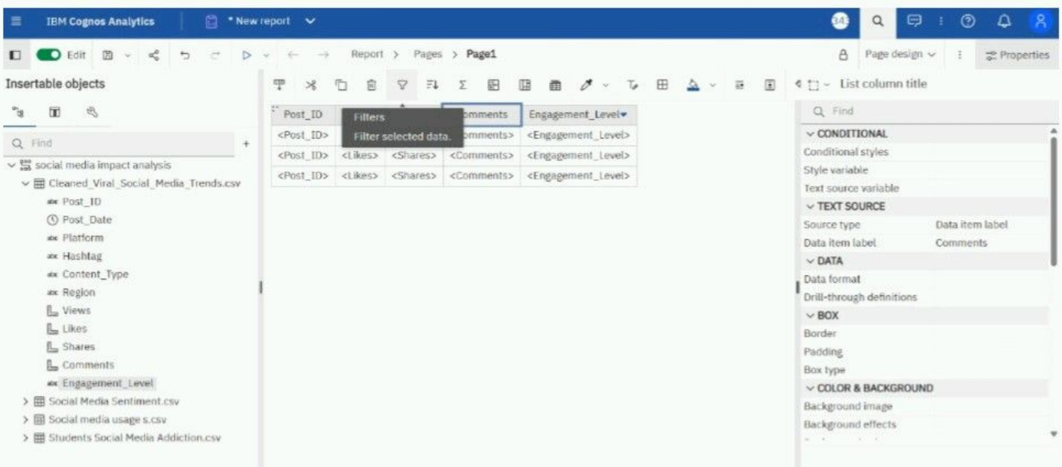


Step7 - Click on the column heading “Total Engagement”

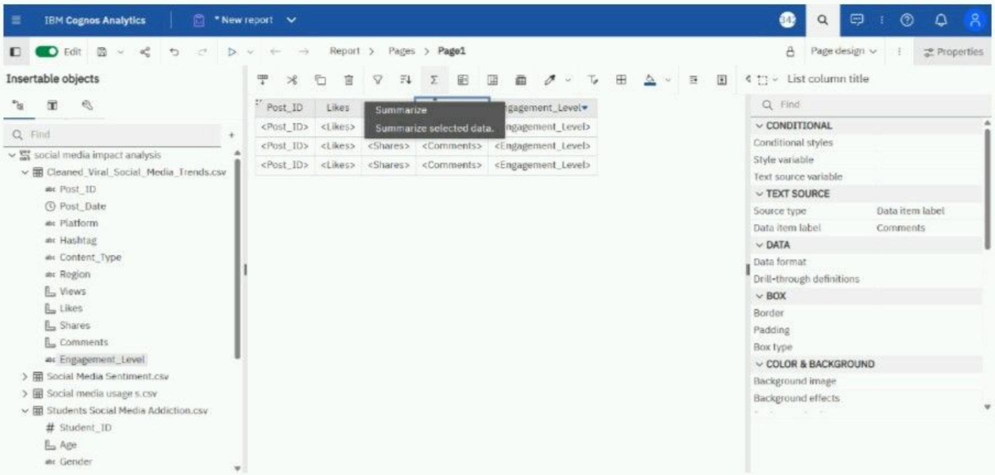
Choose “Sort → Descending” so that the highest engagement appears first.



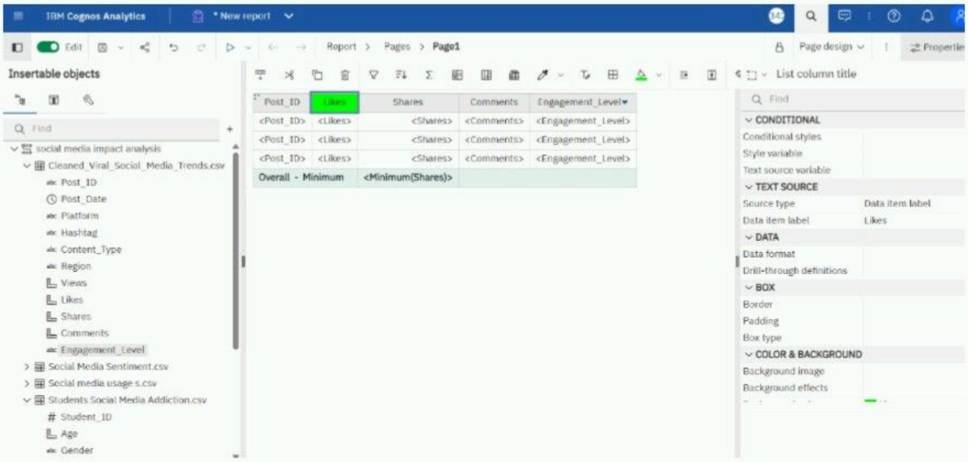
Step8 - Click on the Total Engagement column → choose “Filter” → “Top/Bottom”.



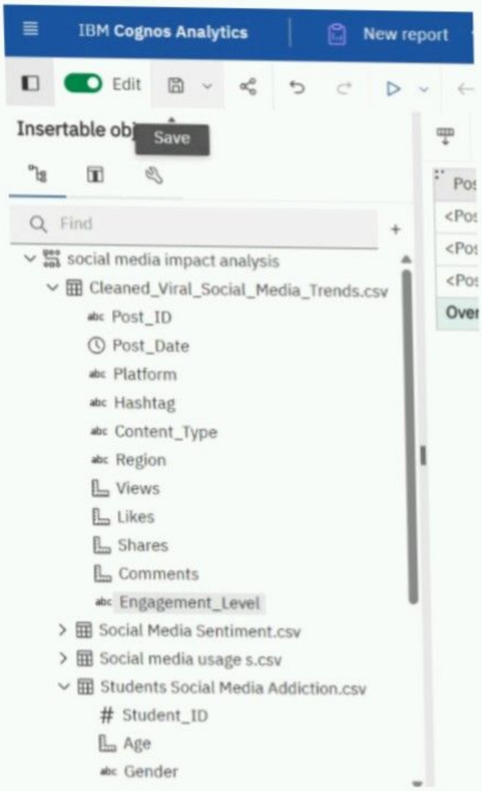
Step:9 - Select the field 'Share Data' and apply Summarize → Minimum to display the lowest (minimum) value of the shared data.

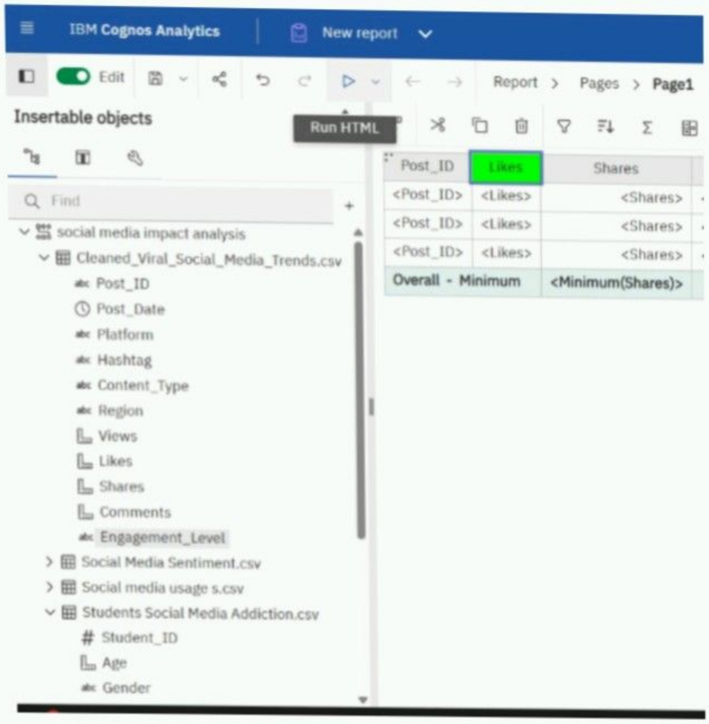


Step:10 - Applied color formatting to highlight the Like Data visually.

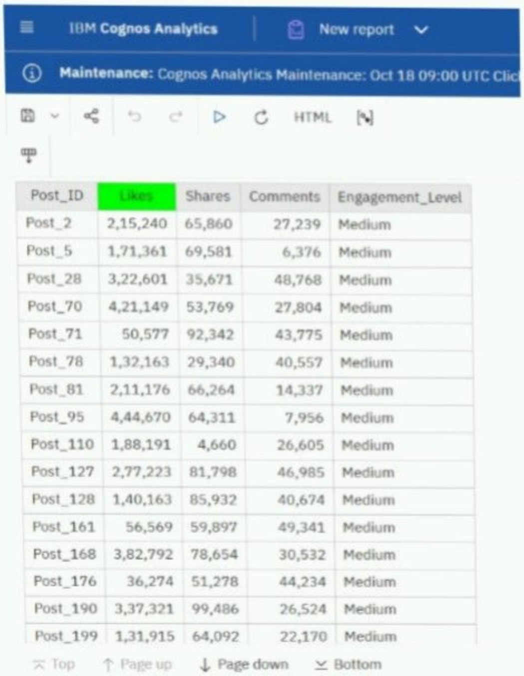


Step:11 - Click -> Run.





You'll see table like this ->



Post_ID	Likes	Shares	Comments	Engagement_Level
Post_2	2,15,240	65,860	27,239	Medium
Post_5	1,71,361	69,581	6,376	Medium
Post_28	3,22,601	35,671	48,768	Medium
Post_70	4,21,149	53,769	27,804	Medium
Post_71	50,577	92,342	43,775	Medium
Post_78	1,32,163	29,340	40,557	Medium
Post_81	2,11,176	66,264	14,337	Medium
Post_95	4,44,670	64,311	7,956	Medium
Post_110	1,88,191	4,660	26,605	Medium
Post_127	2,77,223	81,798	46,985	Medium
Post_128	1,40,163	85,932	40,674	Medium
Post_161	56,569	59,897	49,341	Medium
Post_168	3,82,792	78,654	30,532	Medium
Post_176	36,274	51,278	44,234	Medium
Post_190	3,37,321	99,486	26,524	Medium
Post_199	1,31,915	64,092	22,170	Medium